



Hierarchical Bayesian modelling of cognition

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 [@lei_zhang_lz](https://twitter.com/lei_zhang_lz)



UNIVERSITY OF
BIRMINGHAM



CHBH
CENTRE FOR HUMAN BRAIN HEALTH

My research:

- I ask people to **make decisions**

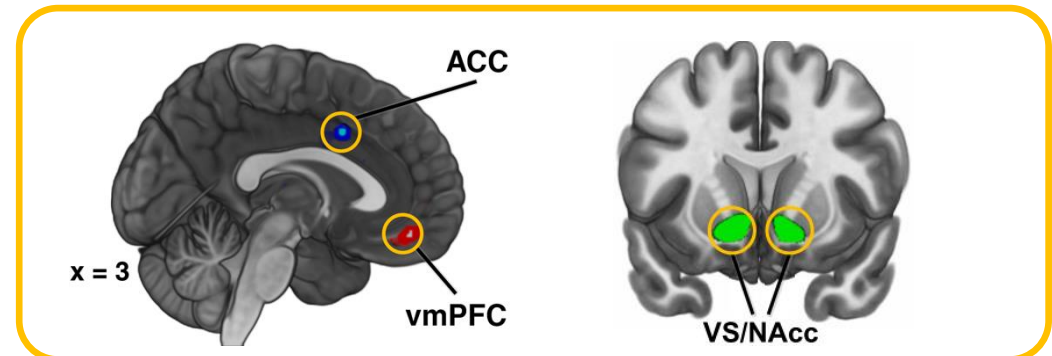
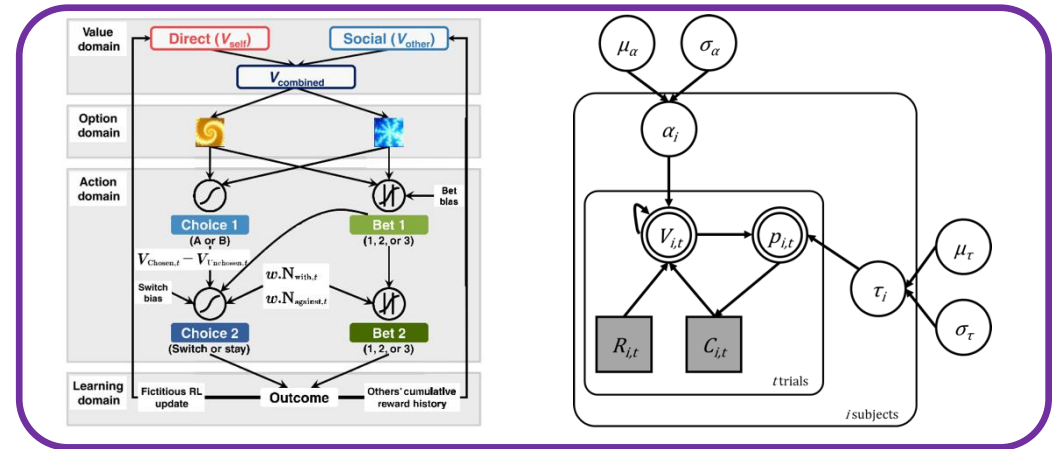
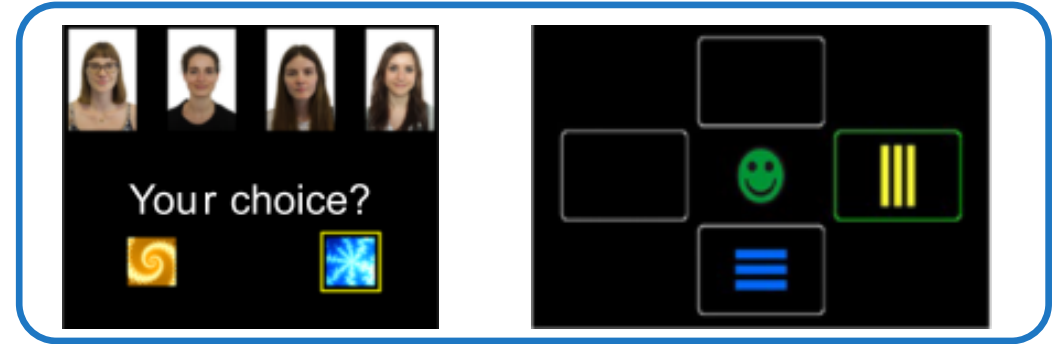
Computation

- I build **computational models**

Algorithm

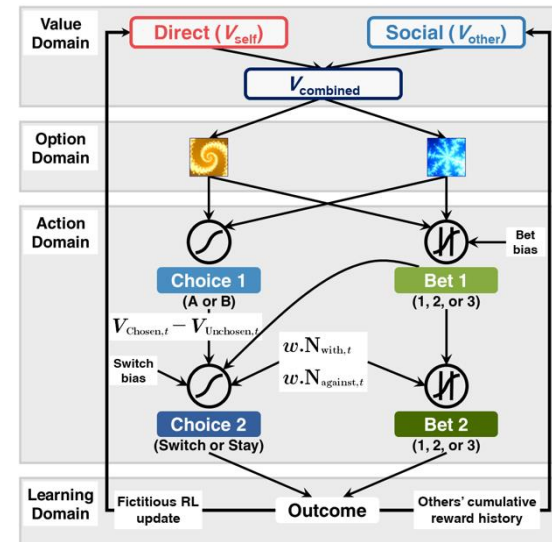
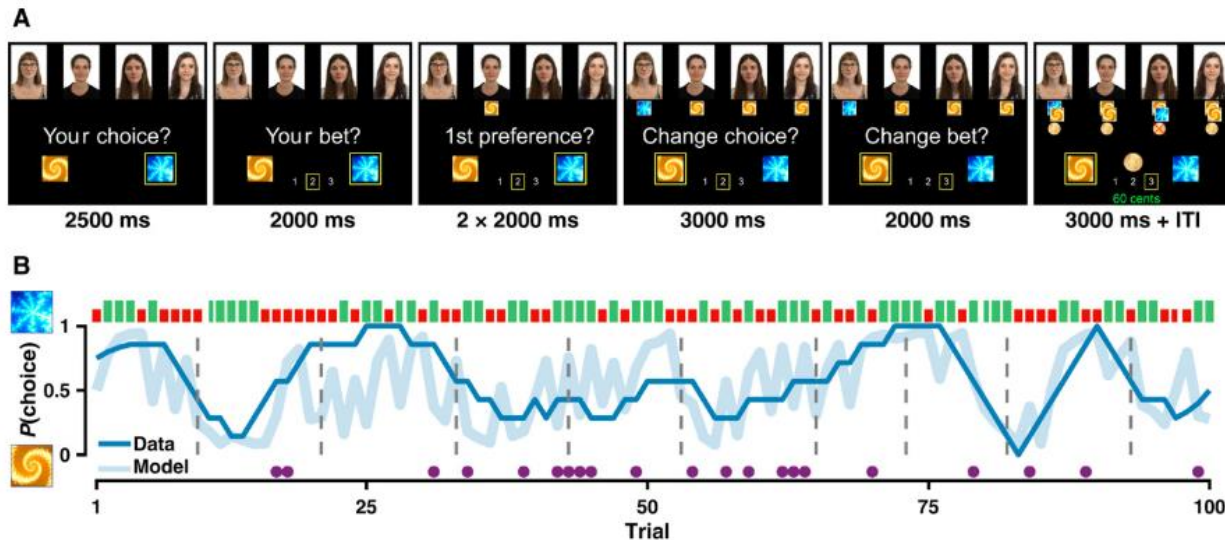
- I examine **neural mechanisms**

Implementation



About me

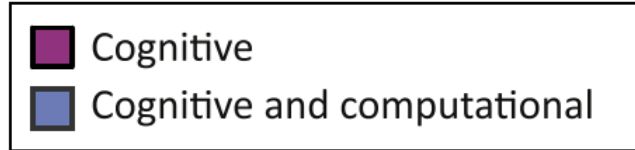
- My journey through computational modeling
 - Started with MLE (@fminsearch in Matlab)
 - Switched to Bayesian: first JAGS, then Stan
 - Why switching to Stan? Look at Fig. 2A and Table S3 in [Zhang & Gläscher \(2020\)](#)



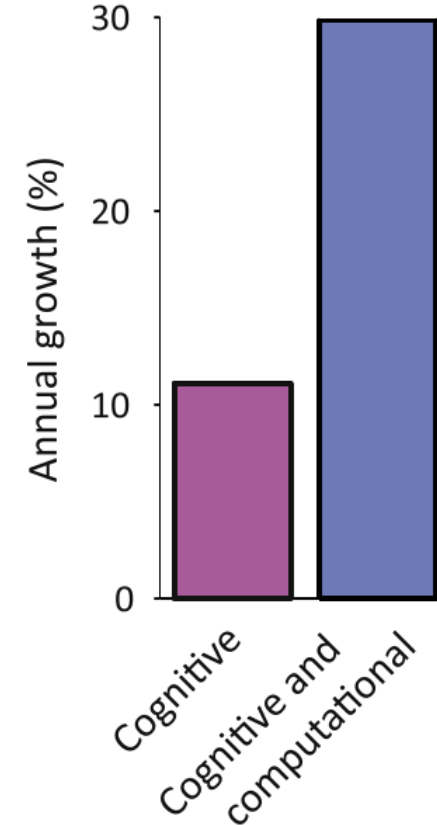
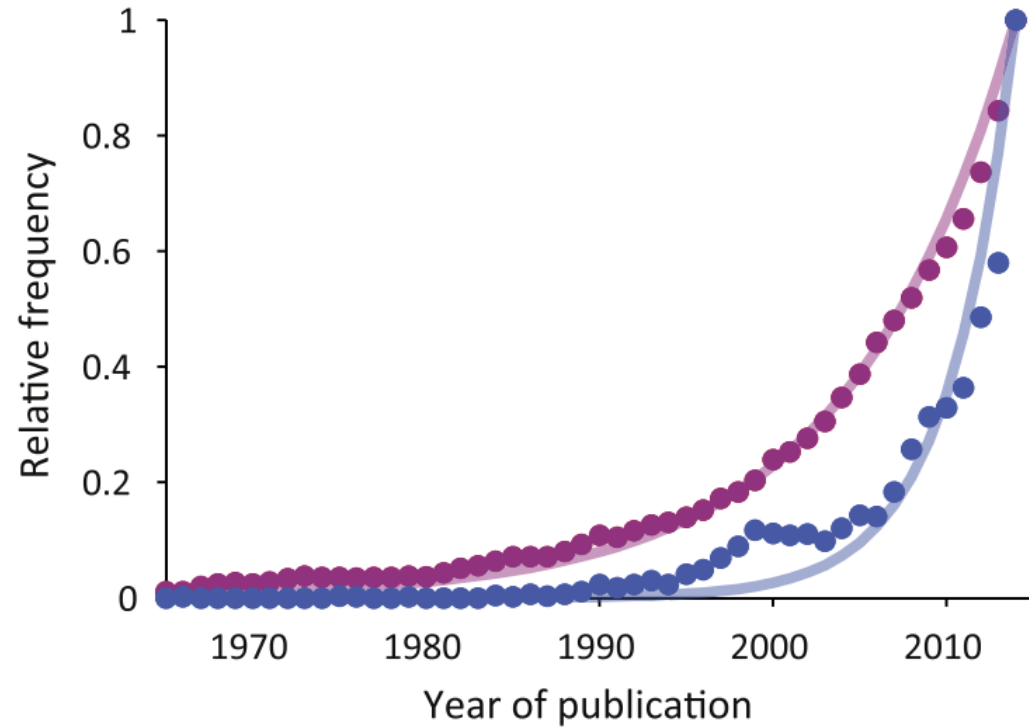
$$\begin{aligned}
 V_{self,t} &= [V_{self,t}(A), V_{self,t}(B)] \\
 V_{other,t} &= [V_{other,t}(A), V_{other,t}(B)] \\
 V_t &= \beta_{self} V_{self,t} + \beta_{other} V_{other,t} \\
 C1_t &\sim \text{Categorical}(\text{Softmax}(V_t)) \\
 U_{self,t} &= \beta_{self} V_{self,t} + \beta_{other} (V_{self,t} - V_{self,t,choice,C1,t}) \\
 B1_t &\sim \text{OrderedLogistic}(U_{self,t} | \theta) \\
 w.N_{opponent,t} &= \frac{\sum_{k=1}^K w_{k,t}}{\sum_{k=1}^K w_{k,t}}, K = 0, 1, \dots, 4 \\
 w.N_{with,t} &= \frac{\sum_{k=1}^K w_{k,t}}{\sum_{k=1}^K w_{k,t}} \\
 V_t(\text{switch}) &= \beta_{self} V_{self,t} + \beta_{other} (V_{self,t} - V_{self,t,choice,C1,t}) + \beta_{opponent} w.N_{opponent,t} \\
 C2 &\sim \text{Bernoulli}(V_t(\text{switch})) \\
 U_{self,t} &= U_{self,t} + \beta_{self} w.N_{with,t} + \beta_{opponent} w.N_{opponent,t}, \text{ if } C1 = C2 \\
 U_{self,t} &= U_{self,t} + \beta_{self} w.N_{with,t} + \beta_{opponent} w.N_{opponent,t}, \text{ if } C1 \neq C2 \\
 B2_t &\sim \text{OrderedLogistic}(U_{self,t} | \theta) \\
 \Phi(x) &= \frac{1}{1 + e^{-x}} \\
 \delta_{self,choice,C1,t} &= R_{self,t} - V_{self,t,choice,C1,t} \\
 \delta_{self,unchosen,C1,t} &= -R_{self,t} - V_{self,t,unchosen,C1,t} \\
 V_{self,t,choice,C1,t+1} &= V_{self,t,choice,C1,t} + \alpha \delta_{self,choice,C1,t} \\
 V_{self,t,unchosen,C1,t+1} &= V_{self,t,unchosen,C1,t} + \alpha \delta_{self,unchosen,C1,t}
 \end{aligned}$$

Boom in Computational Modeling

(A)



Source: PubMed



REPORT

Dopamine promotes cognitive effort by biasing the benefits versus costs of cognitive work

A. Westbrook^{1,2,3,*}, R. van den Bosch^{2,3}, J. I. Mänttä^{2,3}, L. Hofmans^{2,3}, D. Papadopetaki^{2,3}, R. Cools^{2,3,†}, M. J. Frank^{1,4,†}

* See all authors and affiliations

Science 20 Mar 2020:
Vol. 367, Issue 6484, pp. 1362-1366
DOI: 10.1126/science.aaz5891

Neuron

Available online 17 March 2020

In Press, Corrected Proof



Article

A Neuro-computational Account of Arbitration between Choice Imitation and Goal Emulation during Human Observational Learning

Caroline J. Charpentier^{1,2,✉}, Kiyohito Iigaya¹, John P. O'Doherty¹

3 out of 4 focused on Reinforcement Learning models!

nature reviews neuroscience

Review Article | Published: 12 March 2020

The neural and computational systems of social learning

Andreas Olsson[✉], Ewelina Knapska & Björn Lindström

Nature Reviews Neuroscience 21, 197–212(2020) | Cite this article

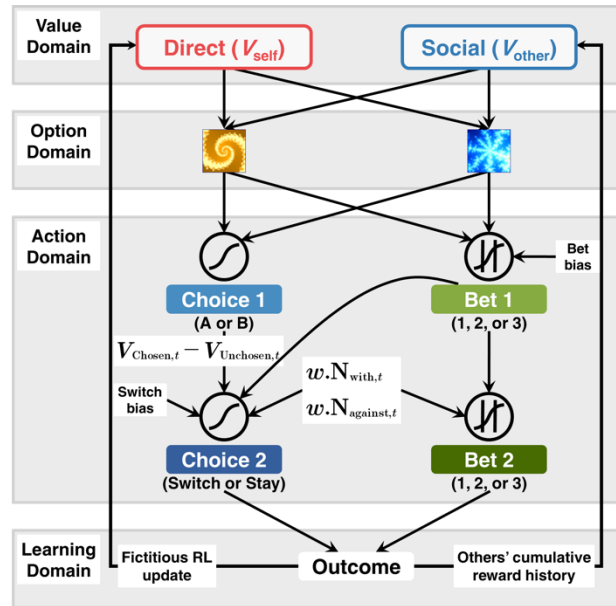
Translational Psychiatry

Article | Open Access | Published: 17 March 2020

Neurocomputational mechanisms underpinning aberrant social learning in young adults with low self-esteem

Geert-Jan Will[✉], Michael Moutoussis, Palee M. Womack, Edward T. Bullmore, Ian M. Goodyer, Peter Fonagy, Peter B. Jones, NSPN Consortium, Robb B. Rutledge & Raymond J. Dolan

hBayesDM package



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 V_{self,t} &= [V_{self,t}(A), V_{self,t}(B)] \\
 V_{other,t} &= [V_{other,t}(A), V_{other,t}(B)] \\
 V_t &= \beta_{self} V_{self,t} + \beta_{other} V_{other,t} \\
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 U_{self,t} &= \beta_{switch} + \beta_{diff_{C1}} (V_{chosen,C1,t} - V_{unchosen,C1,t}) \\
 B1_t &= \text{OrderedLogistic}(U_{self,t} | \theta) \\
 w.N_{against,t} &= \frac{\sum_{s=1}^K w_{s,t}}{\sum_{s=1}^K w_{s,t}}, K = 0, 1, \dots, 4 \\
 w.N_{with,t} &= \frac{\sum_{s=1}^{K-1} w_{s,t}}{\sum_{s=1}^{K-1} w_{s,t}} \\
 V_t(\text{switch}) &= \beta_{switch} + \beta_{diff_{C1}} (V_{chosen,C1,t} - V_{unchosen,C1,t}) + \beta_{against} w.N_{against,t} \\
 C2 &= \text{Bernoulli}(V_t(\text{switch})) \\
 U_{self,t} &= \begin{cases} U_{self,t} + \beta_{with_{C1}} w.N_{with,t} + \beta_{against_{C1}} w.N_{against,t}, & \text{if } C1 = C2 \\ U_{self,t} + \beta_{with_{C2}} w.N_{with,t} + \beta_{against_{C2}} w.N_{against,t}, & \text{if } C1 \neq C2 \end{cases} \\
 B2_t &= \text{OrderedLogistic}(U_{self,t} | \theta) \\
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 V_{self,unchosen,C2,t+1} &= V_{self,unchosen,C2,t} + \alpha \delta_{self,unchosen,C2,t}
 \end{aligned}$$



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

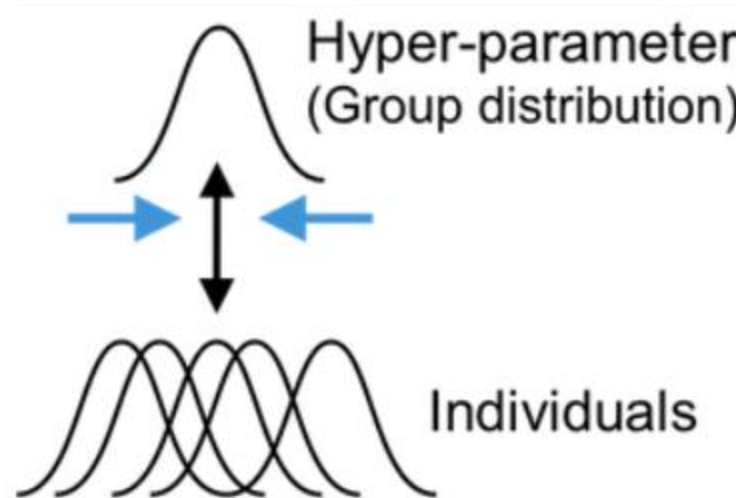
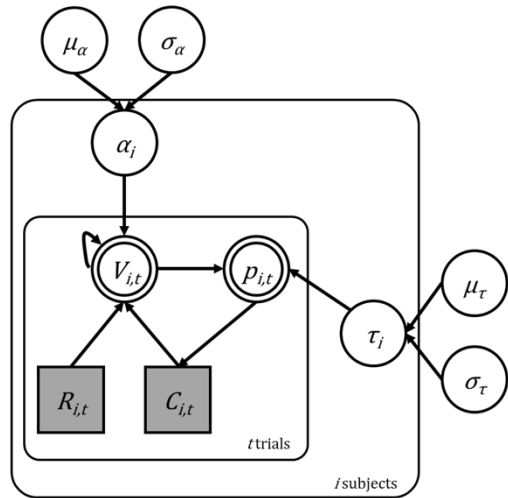
hBayesDM package

hBayesDM

repo status **Active** build **passing** CRAN **1.0.2 – 2019-11-13** downloads **33K**

DOI **10.1162/CPSY_a_00002**

hBayesDM (hierarchical Bayesian modeling of Decision-Making tasks) is a user-friendly package that offers hierarchical Bayesian analysis of various computational models on an array of decision-making tasks. hBayesDM uses **Stan** for Bayesian inference.



Bayesian Statistics and Hierarchical Bayesian Modeling for Psychological Science

Lecture 01

Lei Zhang

Social, Cognitive and Affective Neuroscience Unit (SCAN-Unit)
Department of Cognition, Emotion, and Methods in Psychology

https://github.com/lei-zhang/BayesCog_Wien

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Fakultät für Psychologie

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lei-zhang.net
@lei_zhang_lz

001 / 134:07

我的叔叔

Computational Modeling

我家世代代做中介调节模型。父亲是个路径分析员，整天在Mplus里调餐差相关，回家时衬衫上总是沾着不正定的矩阵。母亲负责调节效应，常常为找不到显著的交互项而叹气。我们住在SEM小区，对面fMRI大楼的邻居总用居高临下的眼神看我们——毕竟在他们眼里，没有生理信号的理论都是空中楼阁。

我的大姐嫁给了结构方程模型，二姐终于还是跟了网络分析——这事让父亲耿耿于怀，他说网络分析那小子“连个潜变量都没有，成何体统”。

但全家最大的希望，是我那位改行做计算建模的叔

叔。

“等他用漂移扩散模型发了顶刊，”母亲总望着窗外fMRI大楼的灯光说，“看那些做认知神经的还笑不笑话我们只会跑问卷数据。”

叔叔的故事在我们家像个传说。他本是家中最有天赋的中介分析员，却突然抛弃了传统的Bootstrap法，一头扎进计算建模的世界。父亲第一次看到他写的强化学习模型代码时气得发抖：“这算什么心理学？连问卷都不要做了！”母亲哭诉家门不幸，说计算建模“连因子载荷都算不出来”。

但当他开始在PsyArxiv网站上积累引用，家人的态度开始微妙地转变。

“他那个强化学习模型，”父亲悄悄对我说，“虽然看不懂，但据说PNAS在审稿。”

每当fMRI邻居炫耀他们的杏仁核激活时，母亲就会轻声说：“等我小叔子的计算模型在Nature Human

Behaviour上发表了……”

我们甚至开始想象那个场景：叔叔带着顶刊录用通知回来，fMRI邻居们排着队来请教如何将他们的脑电数据拟合到漂移扩散模型中——想到这些，连最保守的二姐都会脸红。

转机出现在全国心理学年会。姐姐托关系弄到门票，全家决定给叔叔一个惊喜——更重要的是，看看他是否真的像传闻中那样即将发顶刊。

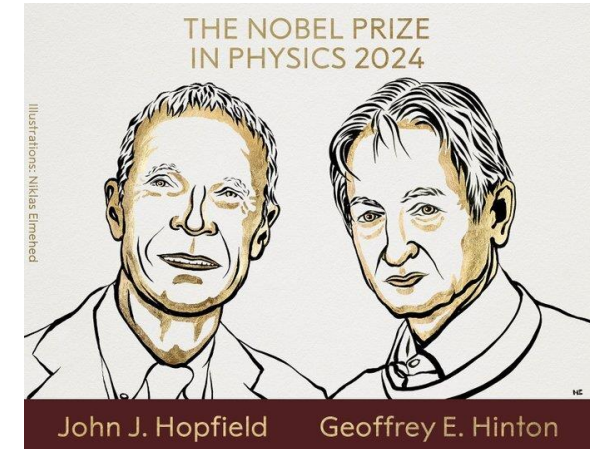
我们在会场找了整整一天，终于在分会场最后排看到了他。他正在向四个心不在焉的听众展示他的漂移扩散模型。

“这个模型……parameter recovery确实有点问题，”他头发凌乱，指着屏幕上混乱的后验分布，“现在审稿人都要我们和大模型对比，可DDM需要的试次数量……”

母亲手中的会议日程掉在地上。

Psychology jobs in the modern era

Hinton was educated at [Clifton College](#) in Bristol^[37] and the [University of Cambridge](#) as an undergraduate student of [King's College, Cambridge](#). After repeatedly changing his degree between different subjects like [natural sciences](#), [history of art](#), and [philosophy](#), he eventually graduated with a BA degree in [experimental psychology](#) in 1970.^[8] He



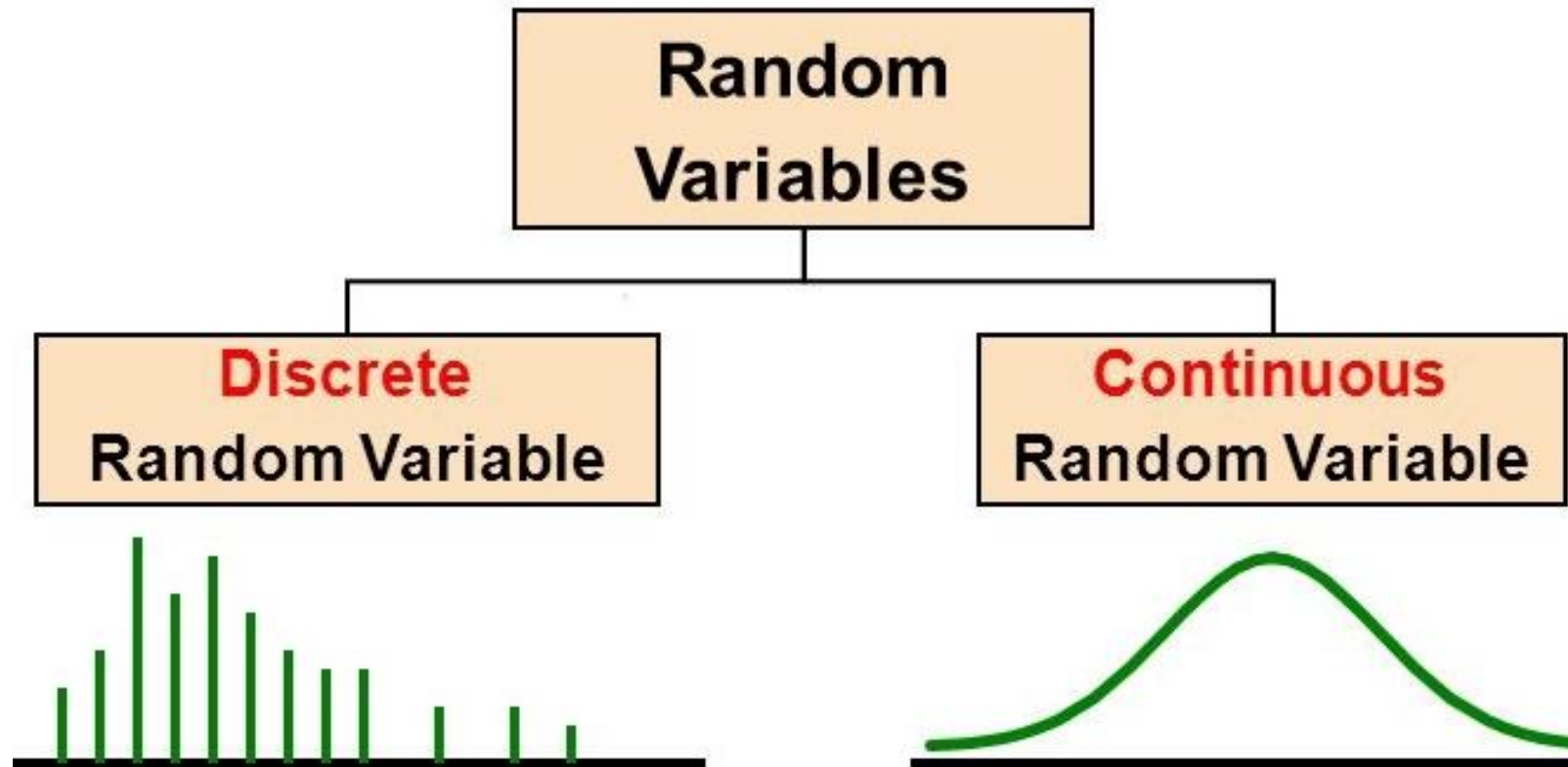
Outline

- Recap on Bayesian modeling
- Why Stan?
- What is (computational) cognitive modeling?
- The idea of the simple Q-Learning model
- Implementing Q-Learning model for one subject in Stan
- Fitting multiple subjects with the hBayesDM package
- Q&A

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Probability Functions

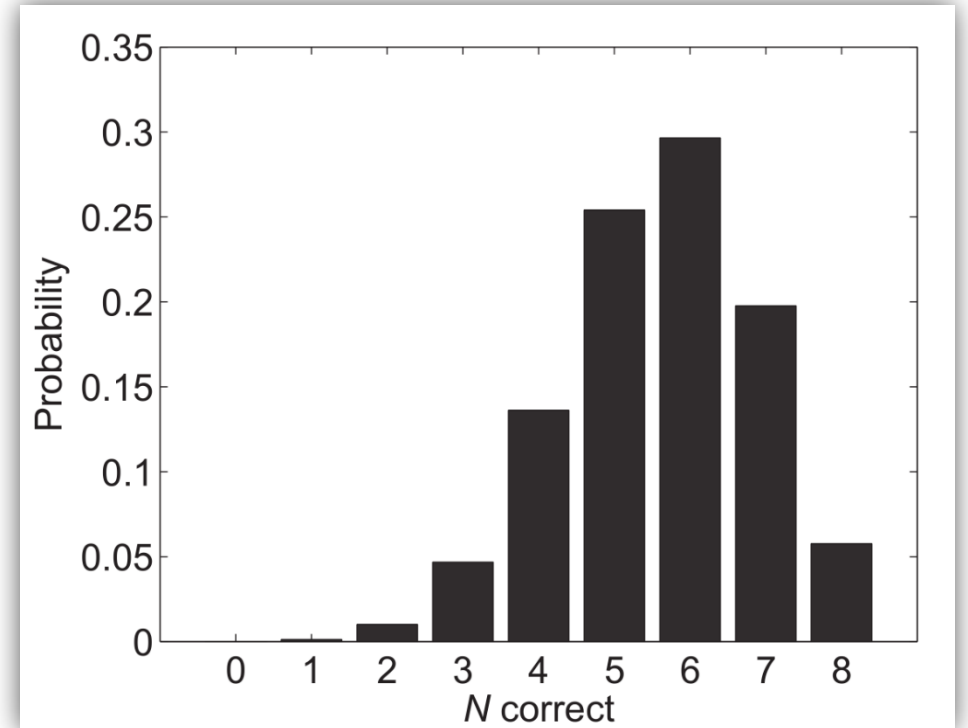




Probability Functions

discrete variable – we talk about **mass**

- Run a test of 8 questions, and record each student's correct responses
- Count and plot the # of correct answers
- Then divide those counts by the total number of students
- The lengths of these bars sum up to 1



Probability Functions

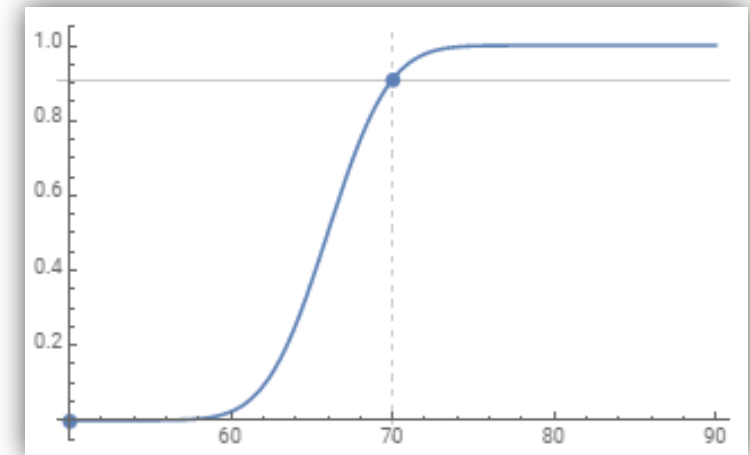
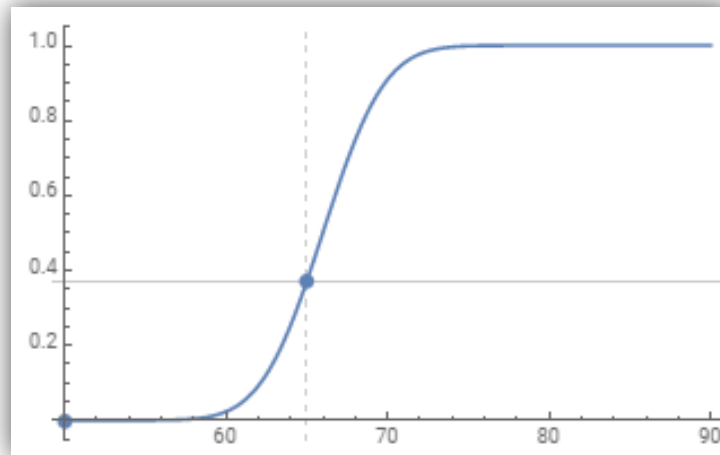
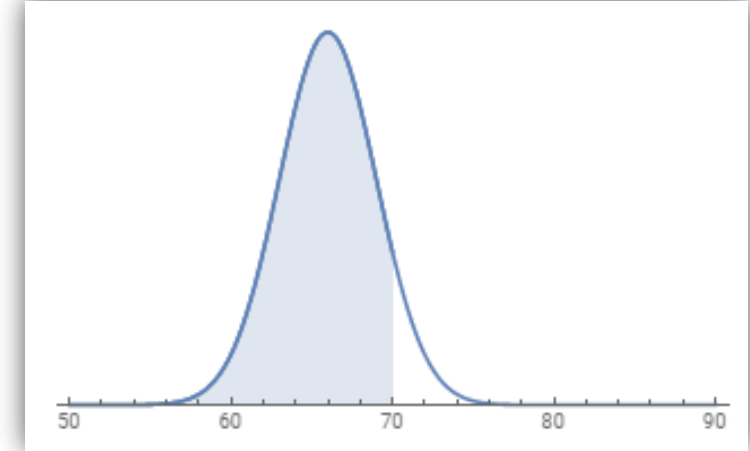
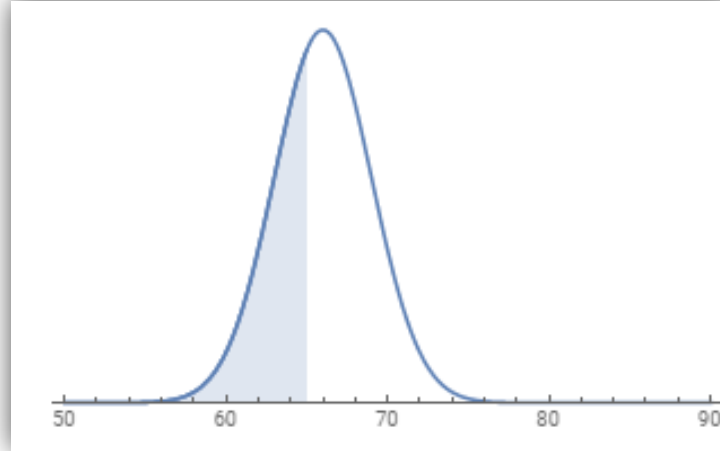
continuous variable – we talk about **density**

probability density function
(PDF)

The entire area under the curve
of PDF is 1.



cumulative distribution function
(CDF)



Joint, Marginal and Conditional Probability

Joint Probability

$$p(A, B) = p(B, A)$$

- e.g., $p(\text{rain, cold})$: $p(\text{rain})$ AND $p(\text{cold})$

Marginal Probability

$p(A)$ – ‘ p of A irrespective of B ’

- e.g., $p(\text{rain})$: $p(\text{rain, cold}) + p(\text{rain, not cold})$

Conditional Probability

$p(A|B)$ – ‘ p of A given B ’ – event B is fixed, not uncertainty

$$p(A, B) = p(A|B)p(B)$$

- e.g., $p(\text{rain, cold}) = p(\text{rain}|\text{cold})p(\text{cold})$

Example: discrete

Joint probability :

$$P(X = 0, Y = 1) =$$

$$\sum_{x,y} P(X = x, Y = y) = 1$$

		snow	
		X	
cold	Y	1	0
	1	0.5	0.1
	0	0.1	0.3

Marginal probability :

$$P(Y = 1) =$$

$$P(X = 0) =$$

$$P(X = x) = \sum_y P(X = x, Y = y)$$

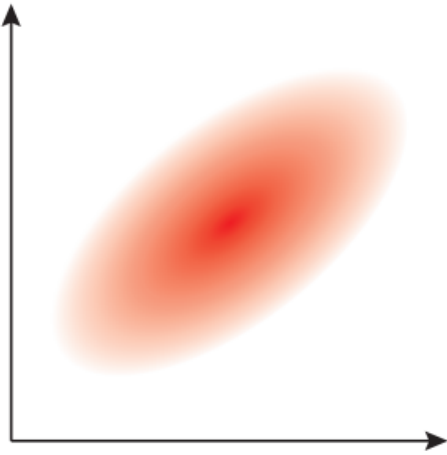
Conditional probability :

$$P(X = 1|Y = 1) =$$

$$\begin{aligned} P(X = x|Y = y) &= \frac{P(X = x, Y = y)}{P(Y = y)} \\ &= \frac{P(X = x, Y = y)}{\sum_x P(X = x, Y = y)} \end{aligned}$$

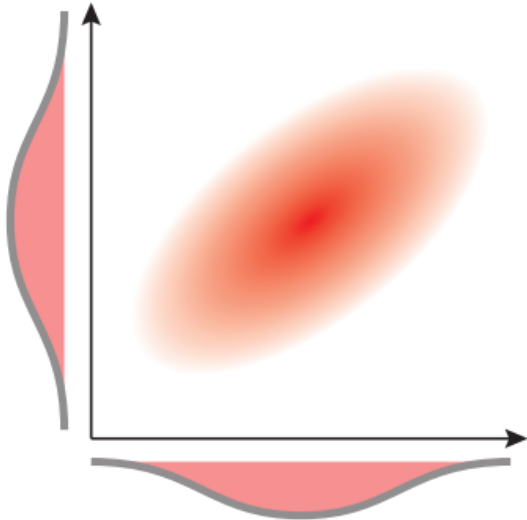
Example: continuous

joint distribution



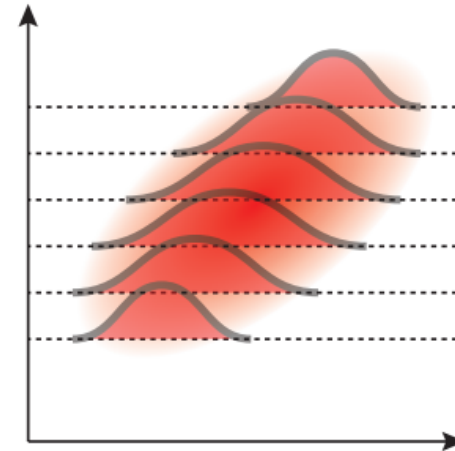
The "co-distribution" of x and y .

marginal distribution



The density of x - (or y -) values,
without knowing the other's value.

conditional distribution



The probability distribution of x ,
given that we know the value of y .

Bayes' theorem

$$p(A,B) = p(B,A)$$

$$p(A,B) = p(A|B)p(B)$$

$$p(B,A) = p(B|A)p(A)$$

$$p(A|B)p(B) = p(B|A)p(A)$$

$$p(A | B) = \frac{p(B | A) p(A)}{p(B)}$$

Linking Data and Parameter



The diagram shows two blue arrows originating from the expression $p(A|B)$ in the equation below. One arrow points to the symbol θ , and the other points to the symbol D .

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Linking Data and Parameter

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}$$

Linking Data and Parameter

Likelihood

How plausible is the data given our parameter is true?

Prior

How plausible is our parameter before observing the data?

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}$$

Posterior

How plausible is our parameter given the observed data?

Evidence

How plausible is the data under all possible parameters?

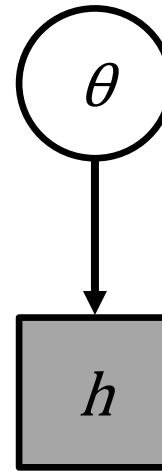
Bernoulli Model

- You are interested in if a coin is biased.
- You will flip the coin.
- You will record whether it comes up a head (h) or a tail (t).
- You might observe 15 heads out of 20 flips.
- What is your degree of belief about the biased parameter ϑ ?



Bernoulli Model

$$p(h \mid \theta) = \theta^h (1 - \theta)^{1-h}$$

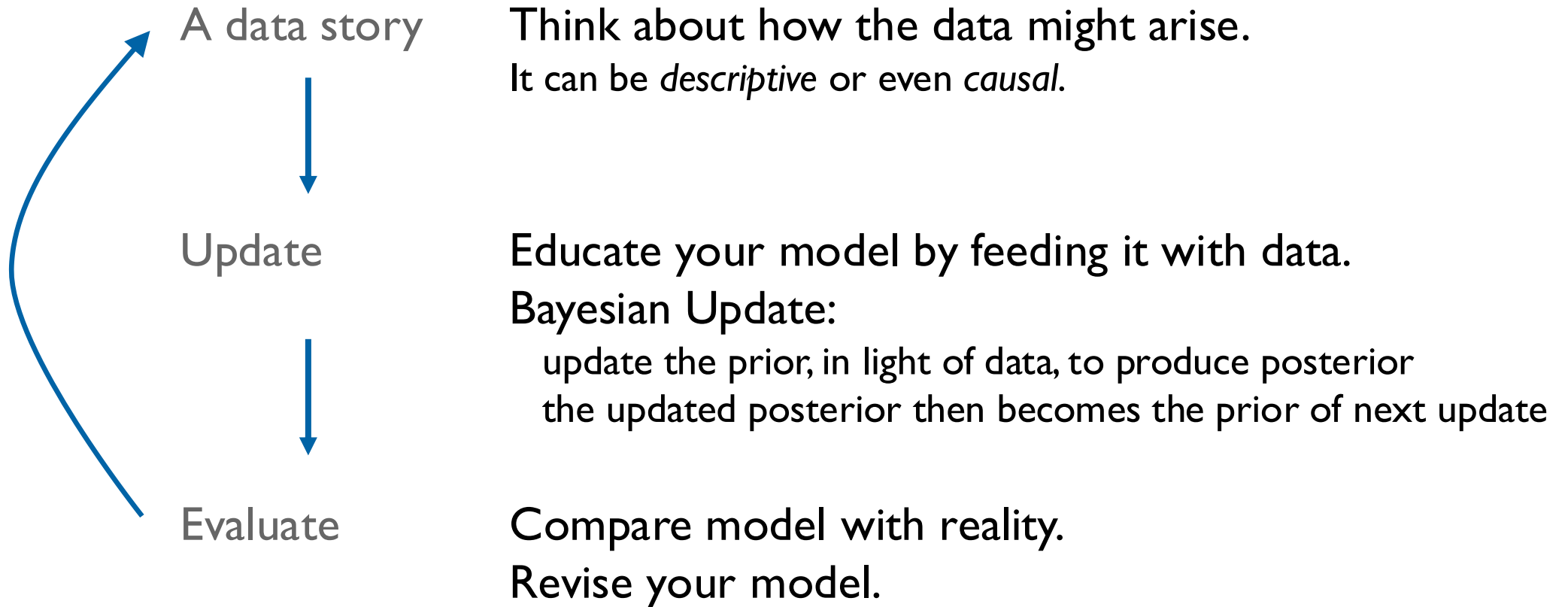


$\theta \sim \text{Uniform}(0, 1)$

$h \sim \text{Bernoulli}(\theta)$

	continuous	discrete
unobserved	θ	δ
observed	y	N

Steps of (Bayesian) Modeling?



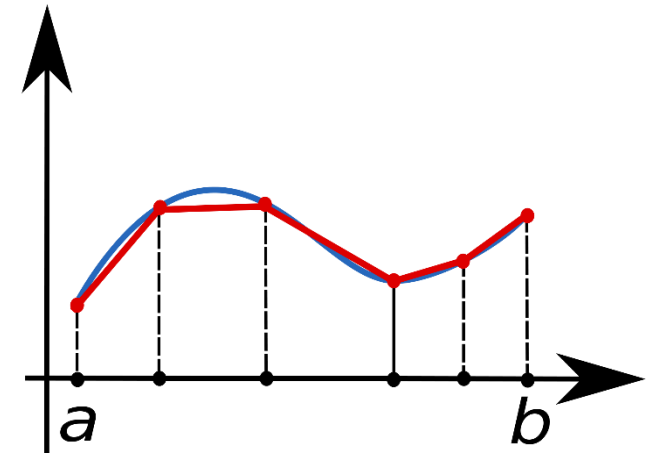
Solve it by **Grid** Approximation

discrete parameters

$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{\sum_{\theta^*} p(D | \theta^*) p(\theta^*)}$$

continuous parameters

$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{\int p(D | \theta^*) p(\theta^*) d\theta^*}$$



What if I have multiple parameters?

grid approximation for
2 parameters?
5 parameters?
10 parameters?

$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{\int p(D | \theta^*) p(\theta^*) d\theta^*}$$

$$p(data) = \int_{\text{All } \theta_1} \int_{\text{All } \theta_2} p(data, \theta_1, \theta_2) d\theta_1 d\theta_2$$

$$p(data) = \int_{\mu_1} \int_{\sigma_1} \dots \int_{\mu_{100}} \int_{\sigma_{100}} \underbrace{p(data | \mu_1, \sigma_1, \dots, \mu_{100}, \sigma_{100})}_{\text{likelihood}} \times \underbrace{p(\mu_1, \sigma_1, \dots, \mu_{100}, \sigma_{100})}_{\text{prior}} d\mu_1 d\sigma_1 \dots d\mu_{100} d\sigma_{100}$$

- Analytical solutions (often does not exist)
- Grid approximation (takes too long)
- solution: **Markov Chain Monte Carlo**

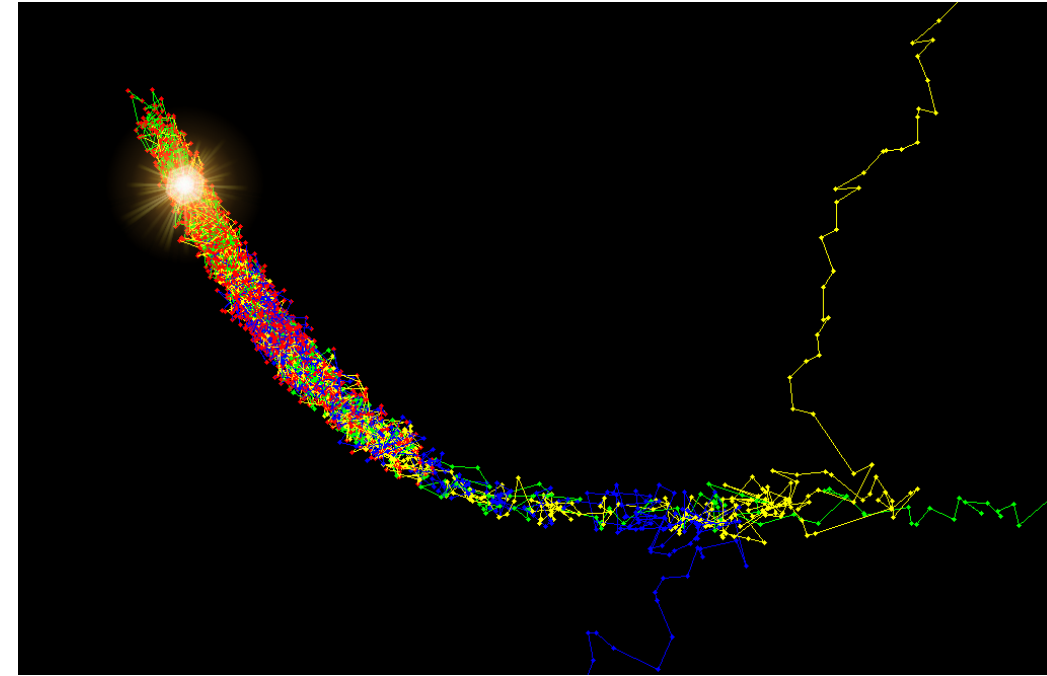
$$p(\theta | D) \propto p(D | \theta) p(\theta)$$

MCMC Sampling Algorithms

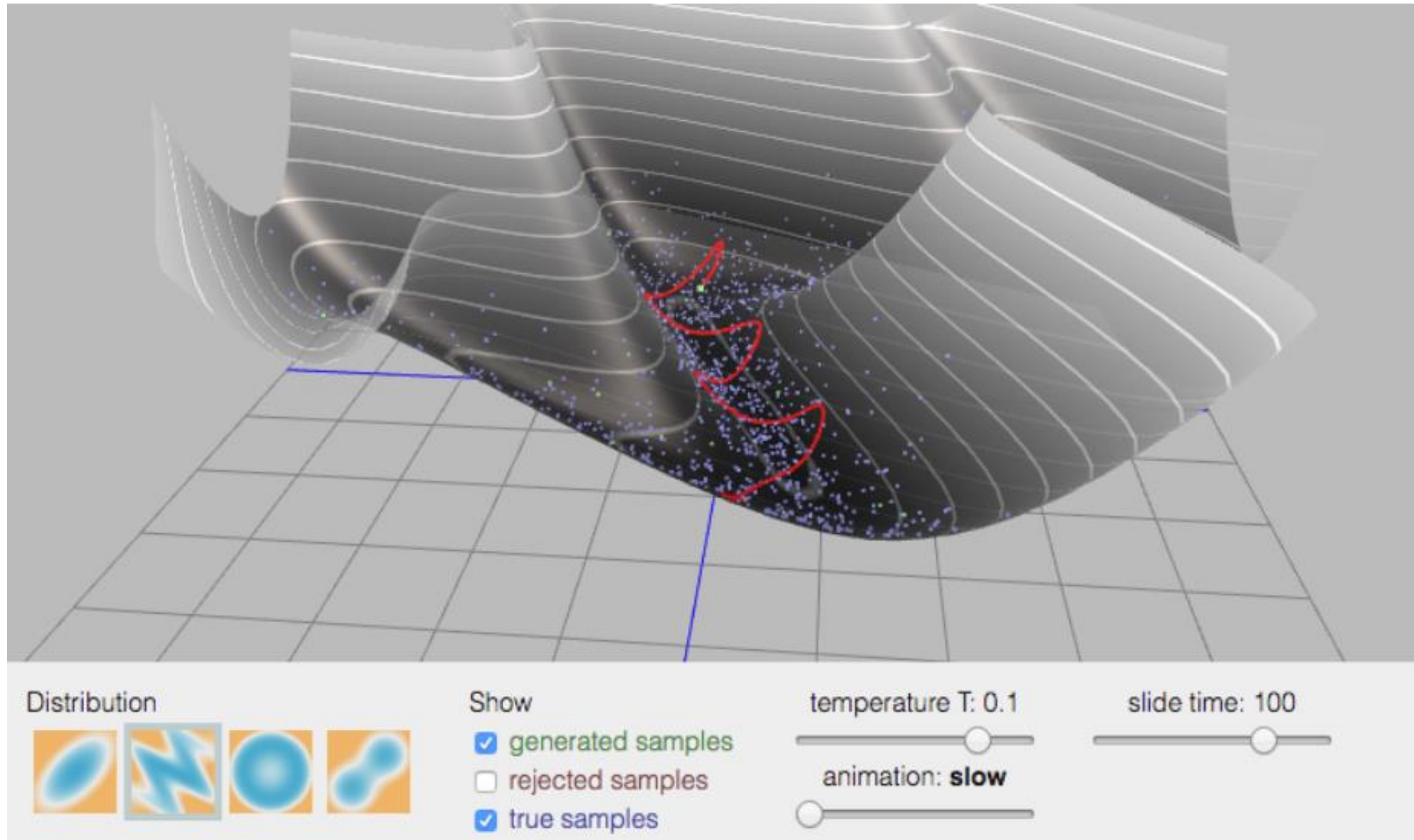
- Rejection sampling
- Importance sampling
- Metropolis algorithm
- Gibbs sampling → JAGS
- HMC sampling*



Stan!



build some intuition



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Is Stan popular?

2021 Summer

Stan: A Probabilistic Programming Language

Carpenter, Bob; Gelman, Andrew; Hoffman, Matthew D.; Lee, Daniel; Goodrich, Ben; Betancourt, Michael; Brubaker, Marcus A.; Guo, Jiqiang; Li, Peter; Riddell, Allen

Grantee Submission, Journal of Statistical Software v76 n1 p1-32 Jan 2017

Stan is a probabilistic programming language for specifying statistical models. A Stan program imperatively defines a log probability function over parameters conditioned on specified data and constants. As of version 2.14.0, Stan provides full Bayesian inference for continuous-variable models through Markov chain Monte Carlo methods such as the No-U-Turn sampler, an adaptive form of Hamiltonian Monte Carlo sampling. Penalized maximum likelihood estimates are calculated using optimization methods such as the limited memory Broyden-Fletcher-Goldfarb-Shanno algorithm. Stan is also a platform for computing log densities and their gradients and Hessians, which can be used in alternative algorithms such as variational Bayes, expectation propagation, and marginal inference using approximate integration. To this end, Stan is set up so that the densities, gradients, and Hessians, along with intermediate quantities of the algorithm such as acceptance probabilities, are easily accessible. Stan can be called from the command line using the "cmdstan" package, through R using the "rstan" package, and through Python using the "pystan" package. All three interfaces support sampling and optimization-based inference with diagnostics and posterior analysis. "rstan" and "pystan" also provide access to log probabilities, gradients, Hessians, parameter transforms, and specialized plotting.

Descriptors: [Programming Languages](#), [Probability](#), [Bayesian Statistics](#), [Monte Carlo Methods](#), [Statistical Inference](#), [Maximum Likelihood Statistics](#), [Computation](#), [Statistical Distributions](#), [Computer Software](#)



brms: An R package for Bayesian multilevel models using Stan

[PC Bürkner](#) - Journal of statistical software, 2017 - jstatsoft.org

... The **brms** package does not fit **models** itself but uses Stan on the back-end. Accordingly, all samplers implemented in Stan can be used to fit **brms models**. Currently, these are the static ...

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Cited by 3989



[PDF] **JAGS**: A program for analysis of Bayesian graphical models using Gibbs sampling

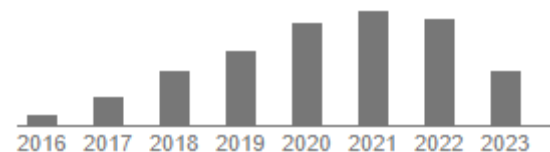
[M Plummer](#) - Proceedings of the 3rd international workshop on ..., 2003 - ci.tuwien.ac.at

JAGS is a program for Bayesian Graphical modelling which aims for compatibility with Classic BUGS. The program could eventually be developed as an R package. This article explains the motivations for this program, briefly describes the architecture and then ...

☆ 99 Cited by 4543 Related articles All 8 versions 99

2023 Summer

Cited by 6594



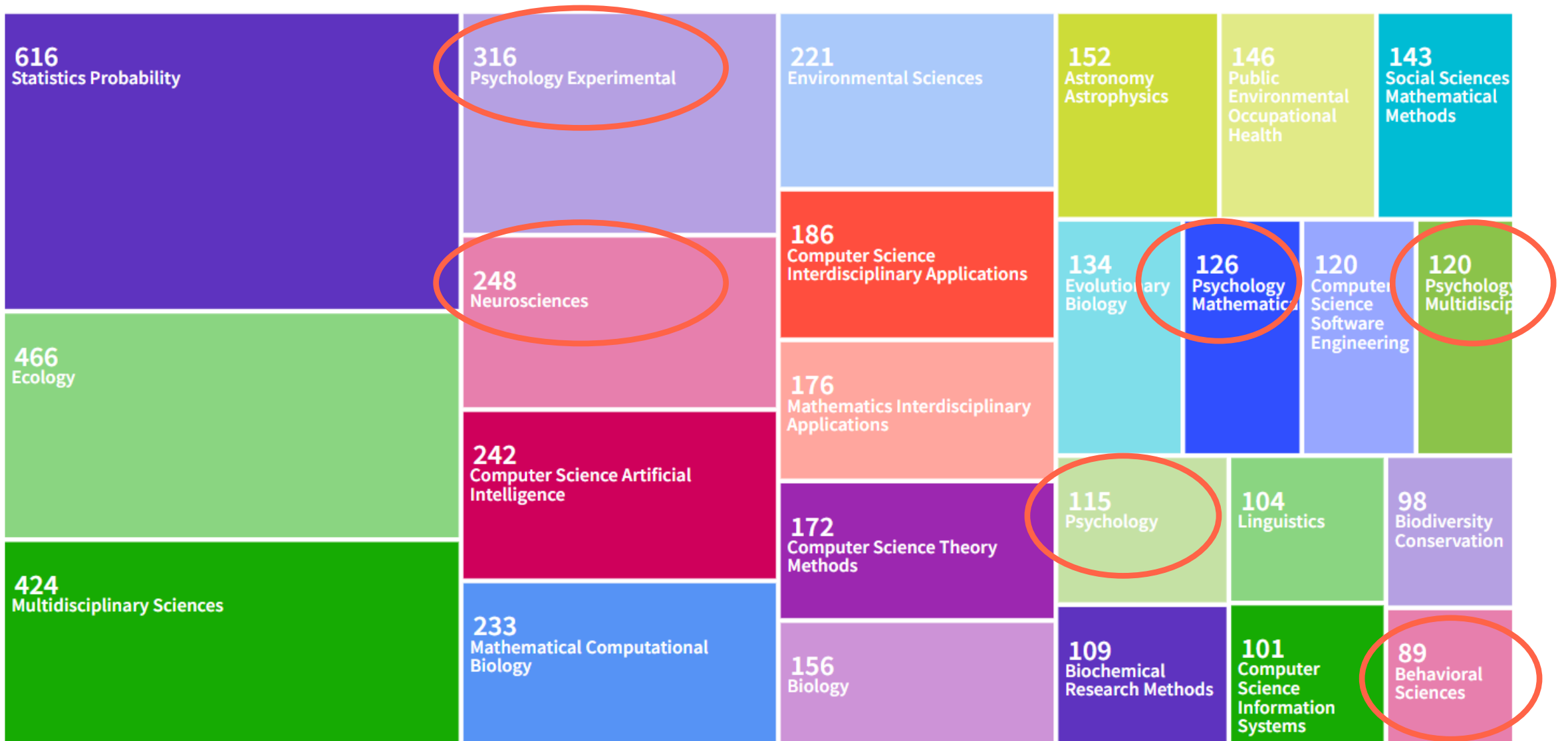
[PDF] **JAGS**: A program for analysis of Bayesian graphical models using Gibbs sampling

[M Plummer](#) - Proceedings of the 3rd international workshop on ..., 2003 - r-project.org

... **JAGS** is a program for Bayesian Graphical modelling which aims for compatibility ... **JAGS** and so avoid having to write a new program for each application. A second motivation for **JAGS** ...

☆ Save 99 Cite Cited by 6277 Related articles All 7 versions 99

Who are using Stan?



Why Use Stan?



vs. BUGS / JAGS

- Less spatial correlation → effective samples
- Time to converge and per effective sample size:
 - | - ∞ times faster
- Memory usage: 1–10%
- Language features
 - variable overwrite: `a = 4`, then `a = 5`
 - formal control flow (same as R)
 - full support of vectorizing & matrix calculation



Krzysztof Sakrejda
@sakrejda

I keep getting asked why people should use [@mcmc_stan](#) so I wrote an answer:

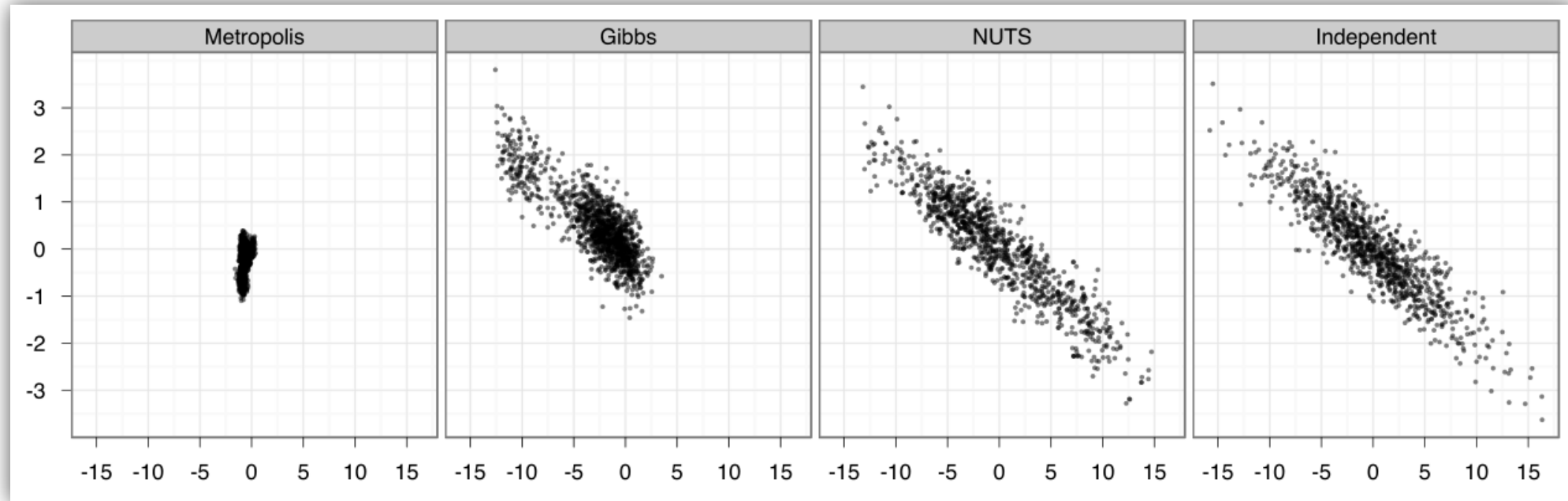


"Selling" Stan
discourse.mc-stan.org

27.03.18, 16:01

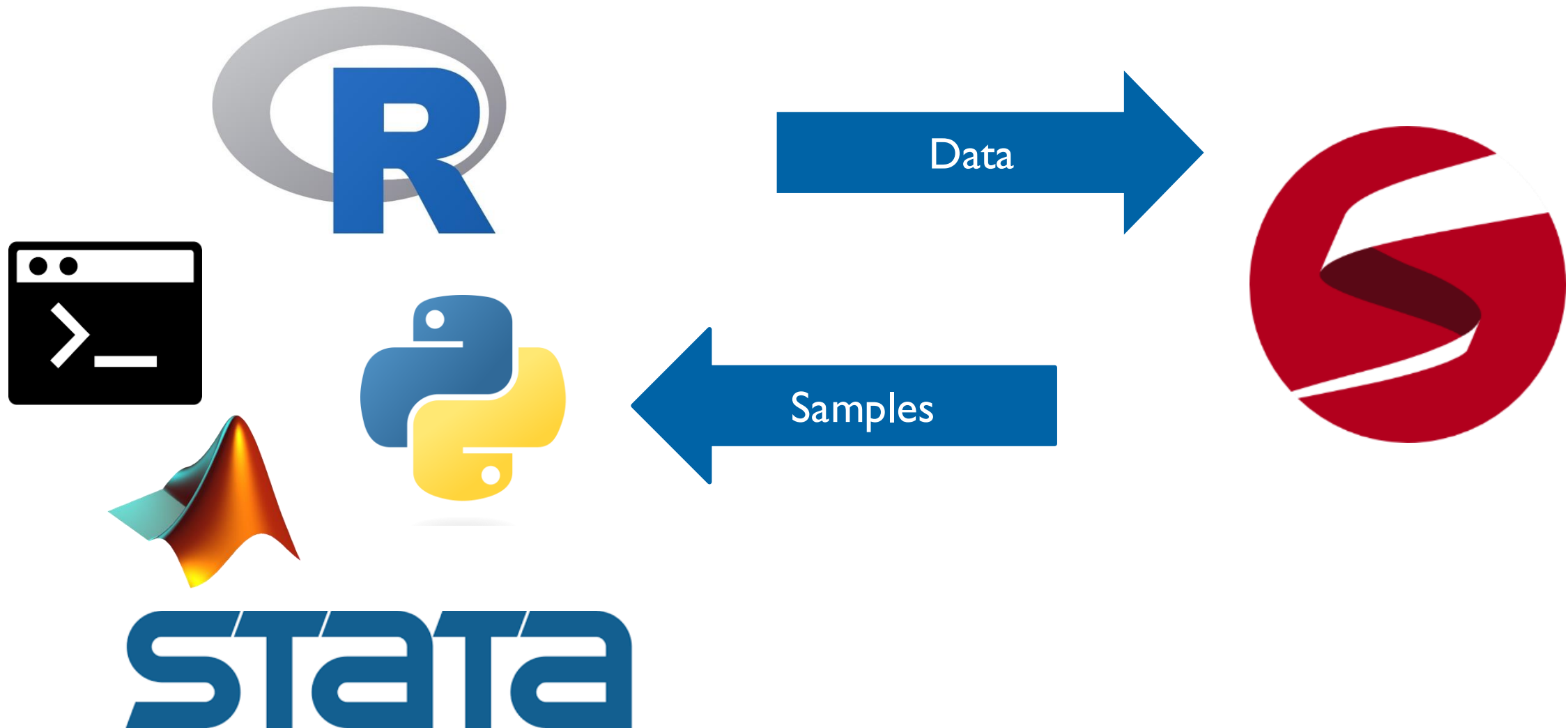
HMC vs. Gibbs and Metropolis

Hamilton MC (HMC) implements No-U-Turn Sampler (NUTS)



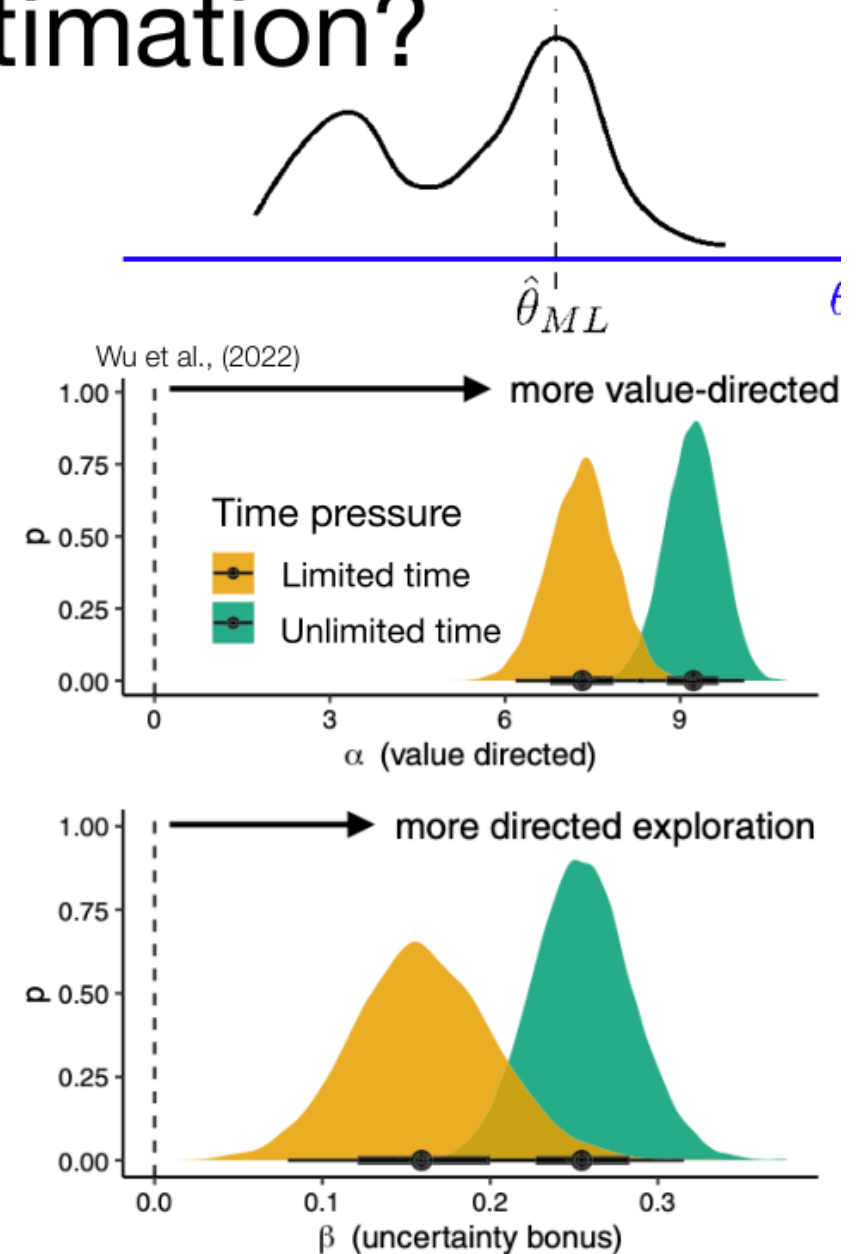
- Two dimensions of highly correlated 250-dim normal
- 1,000,000 draws from Metropolis and Gibbs (thin to 1000)
- **1,000** draws from NUTS; 1000 independent draws

Stan and Other Platforms



Why Bayesian model estimation?

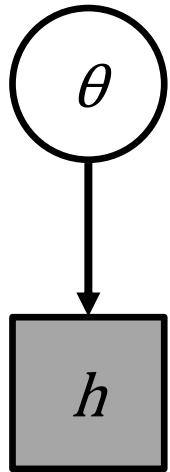
1. Not just a point estimate, but an entire **probability distribution over parameters**
2. Rather than only assuming participants are independent samples, we can model **hierarchical relationships**
3. Naturally avoid overfitting through **Bayesian Occam's Razor**, since we evaluate the model across the entire range of parameters



But, any cons?

- HMC provides huge improvements in computational efficiency, but **mathematical foundations** are more difficult to follow, at least sometimes.
- Stan cannot sample from the posterior distribution of **discrete parameters** (e.g., [1, 2]).
 - → with additional effort, it can be achieved through marginalisation, see the Stan User Manual.
- There are a few unwritten **tips and tricks**.
 - Practise makes perfect and follow discussions on the forum.
- **Block-based** language → might seem rigid in the first place.

A quick look at Stan



$\theta \sim \text{Uniform}(0, 1)$

$h \sim \text{Bernoulli}(\theta)$

```
data {  
  int<lower=0> N;  
  int<lower=0,upper=1> flip[N];  
}  
  
parameters {  
  real<lower=0,upper=1> theta;  
}  
  
model {  
  flip ~ bernoulli(theta);  
}
```

Stan Language

```
data {  
  //... read in external data...  
}
```

```
transformed data {  
  //... pre-processing of data ...  
}
```

```
parameters {  
  //... parameters to be sampled by HMC ...  
}
```

```
transformed parameters {  
  //... pre-processing of parameters ...  
}
```

```
model {  
  //... statistical/cognitive model ...  
}
```

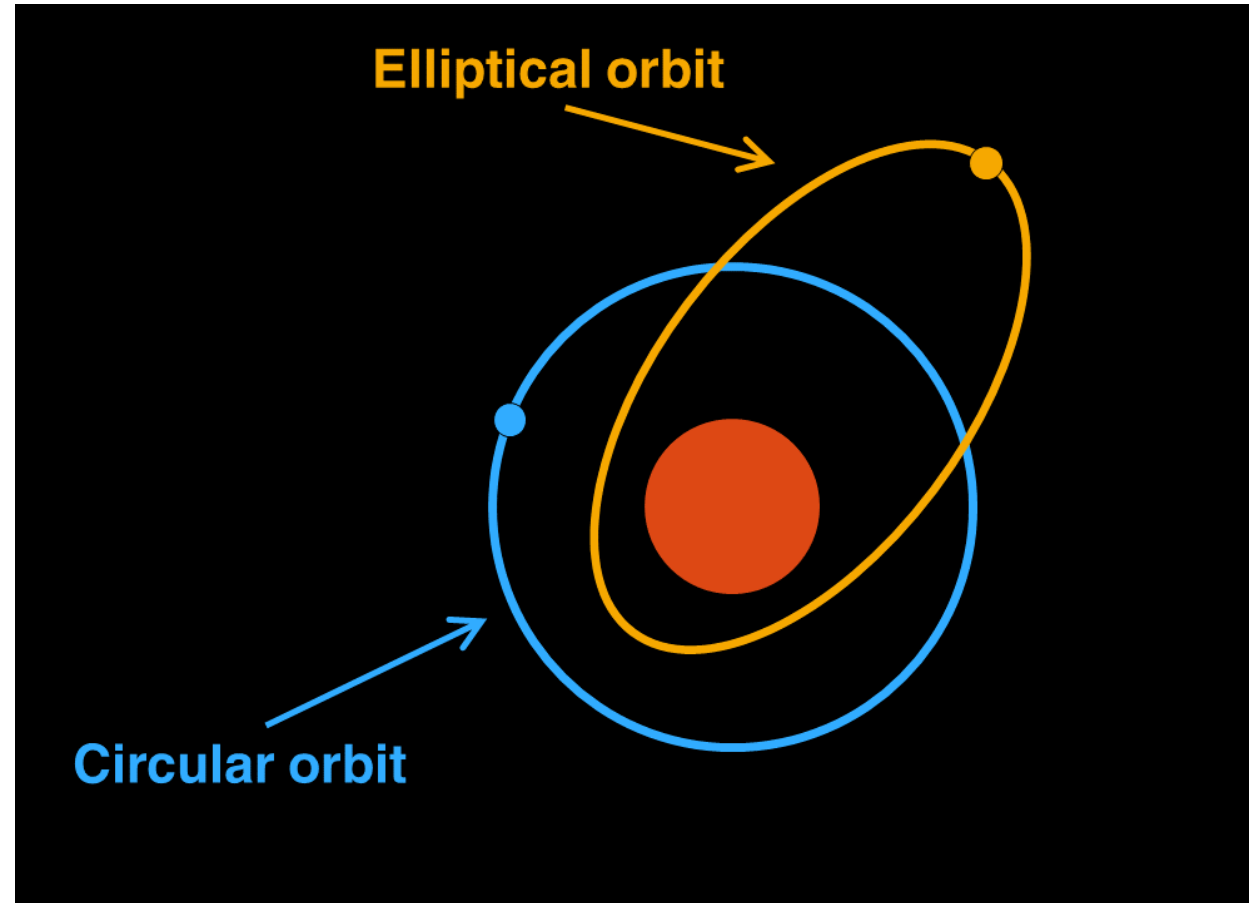
```
generated quantities {  
  //... post-processing of the model ...  
}
```

Outline

- Recap on Bayesian modeling
- Why Stan?
- **What is (computational) cognitive modeling?**
- The idea of the simple Q-Learning model
- Implementing Q-Learning model for one subject in Stan
- Fitting multiple subjects with the hBayesDM package
- Q&A

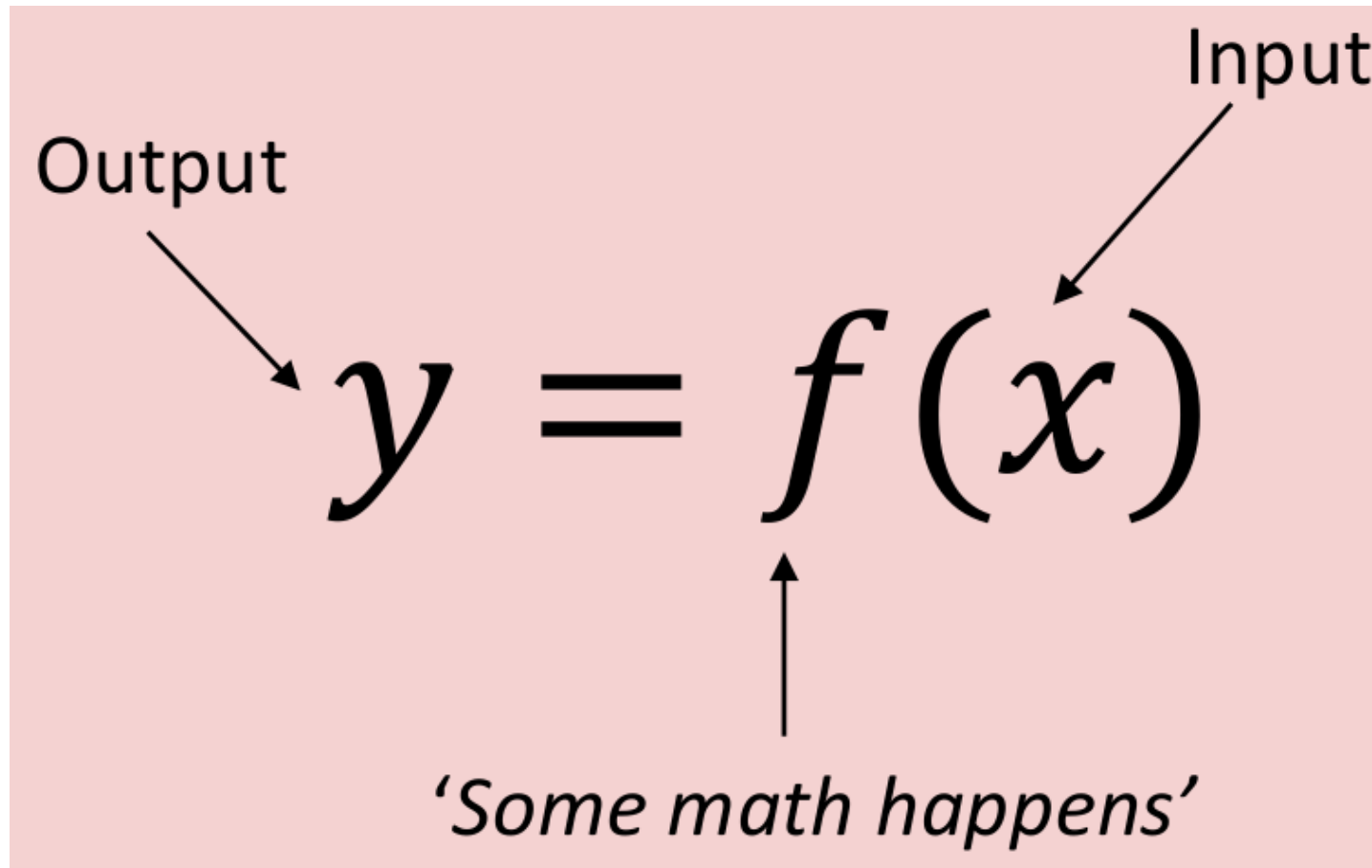
The idea of **computational modeling** is never new

Scientists use mathematical models to **approximate certain processes** (physical or mental), in order to explain and to predict.



Computational modeling

Cognition as information processing



Computational modeling

Cognition as information processing

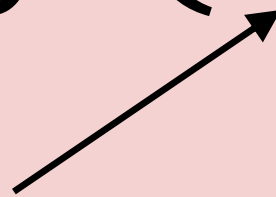


Kurt Lewin, (1936)

Behavior



$$B = f(P, E)$$



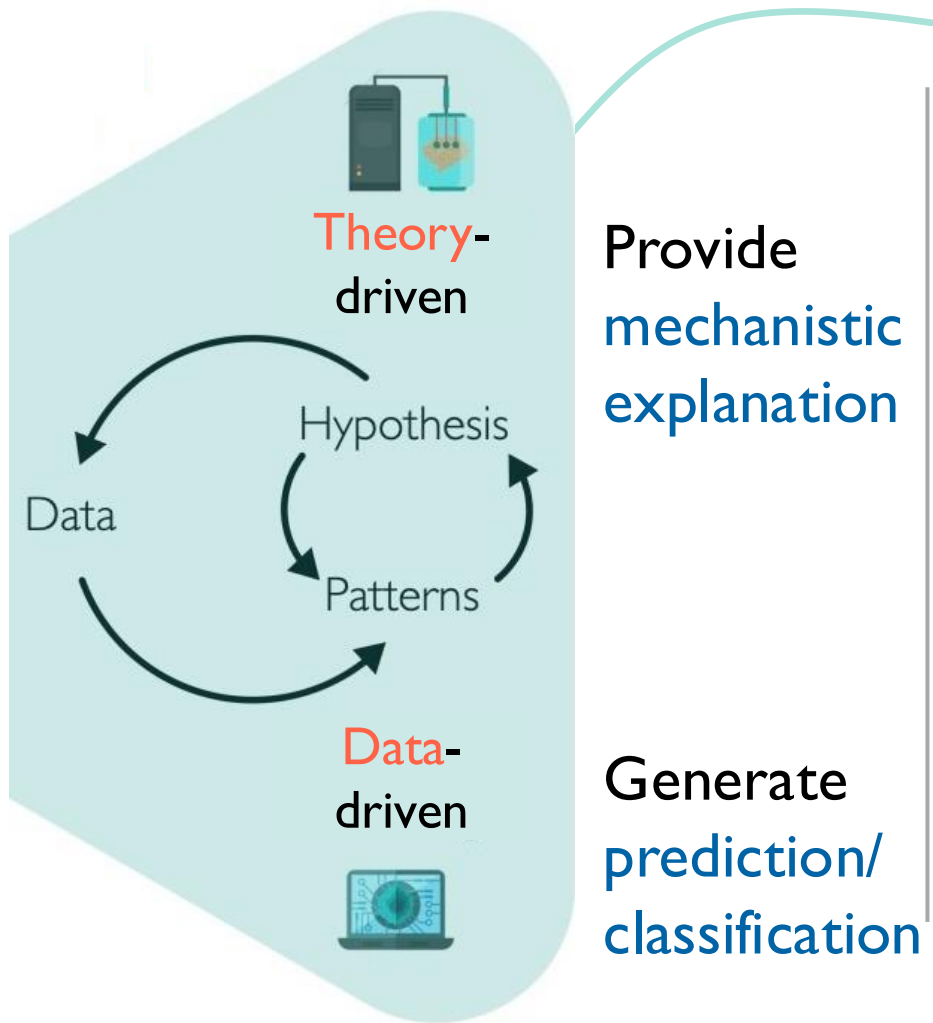
Person



Environment



Two flavours of computational modeling



Construct



- Reward,
- Learning,
- Self v. Other,
- etc.

Computational modeling

On the Role of Theory and Modeling in Neuroscience

Daniel Levenstein,¹ Veronica A. Alvarez,² Asohan Amarasingham,³ Habiba Azab,⁴ Zhe S. Chen,⁵ Richard C. Gerkin,⁶ Andrea Hasenstaub,⁷ Ramakrishnan Iyer,⁸ Renaud B. Jolivet,⁹ Sarah Marzen,¹⁰ Joseph D. Monaco,¹¹ Astrid A. Prinz,¹² Salma Quraishi,¹³ Fidel Santamaria,¹³ Sabyasachi Shivkumar,¹⁴ Matthew F. Singh,¹⁵ Roger Traub,¹⁶ Farzan Nadim,^{1,7*} Horacio G. Rotstein,^{1,7*} and A. David Redish^{18*}

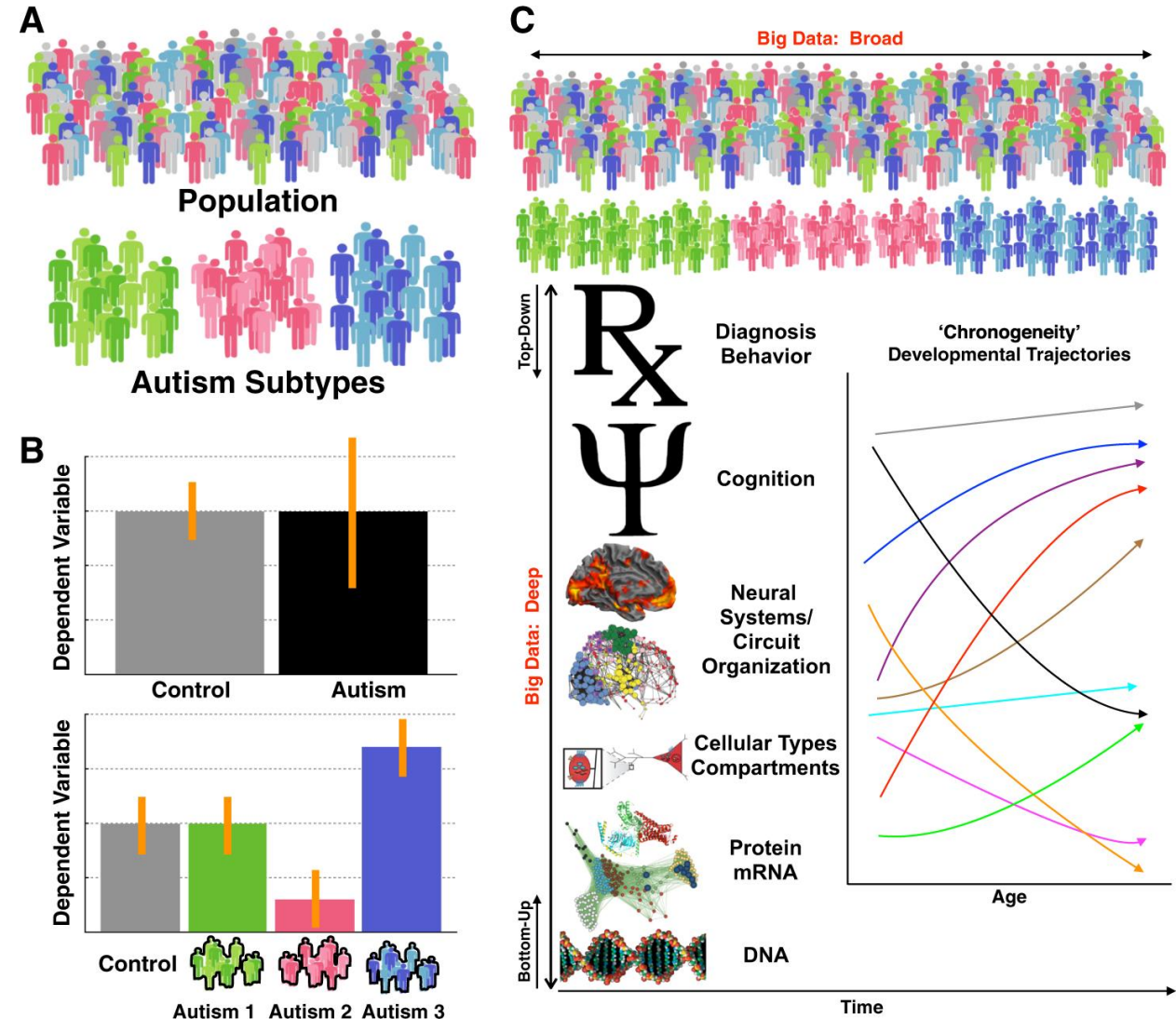
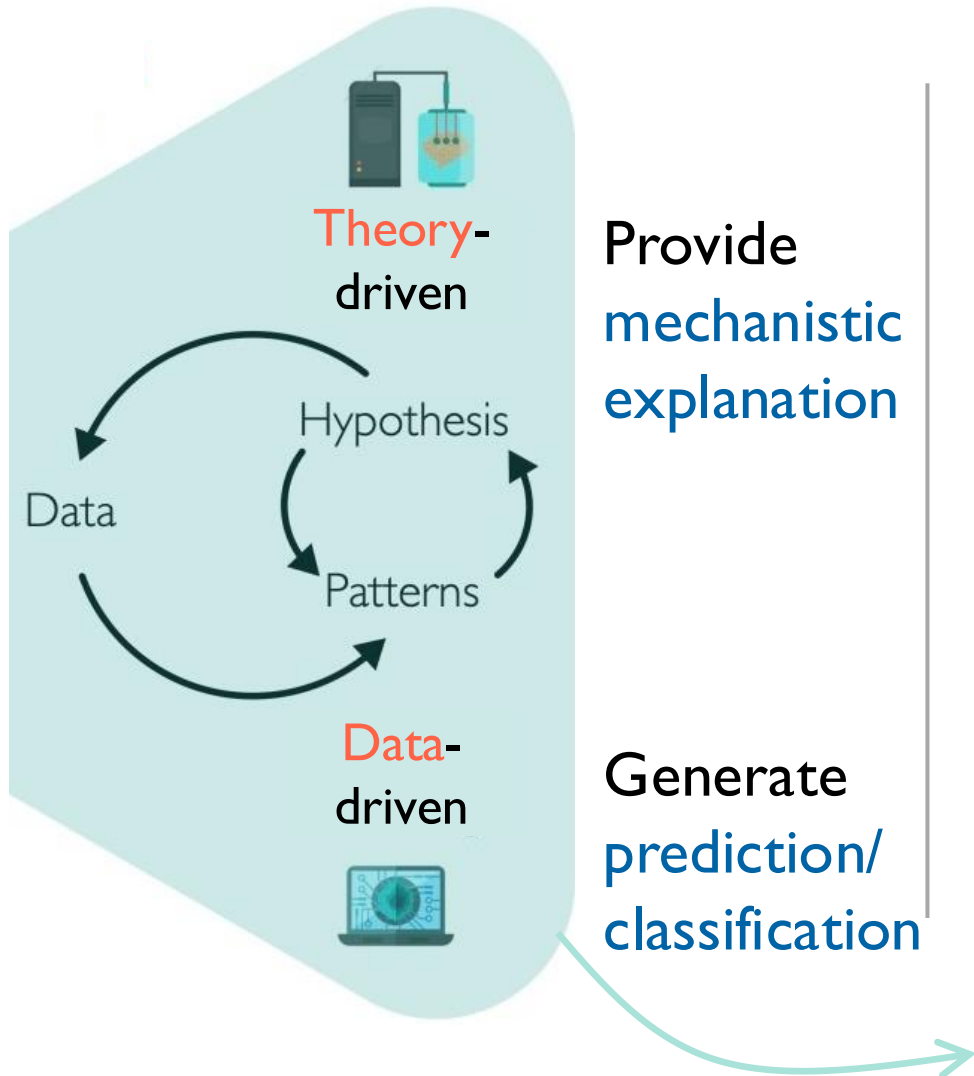
How Computational Modeling Can Force Theory Building in Psychological Science

Olivia Guest¹, Andrea E. Martin

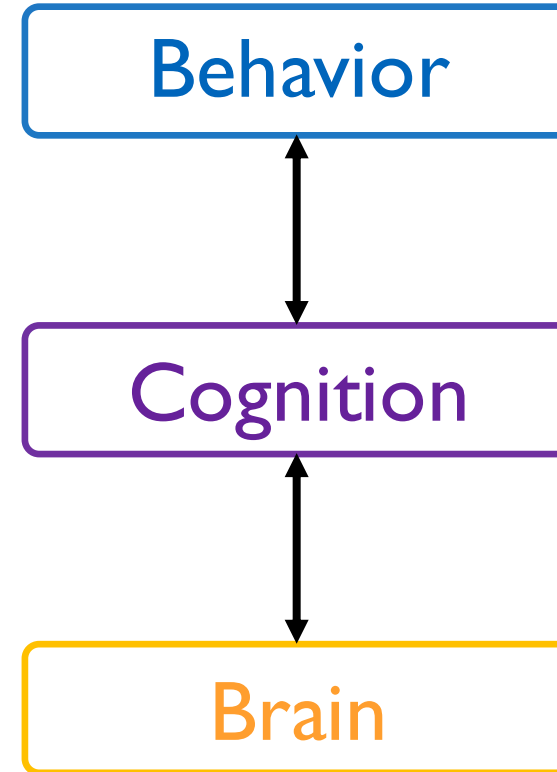
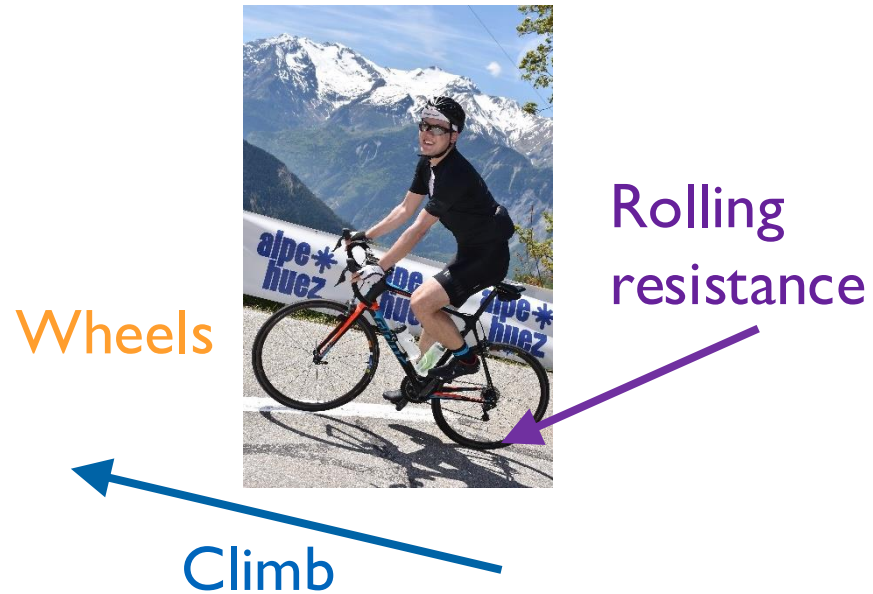
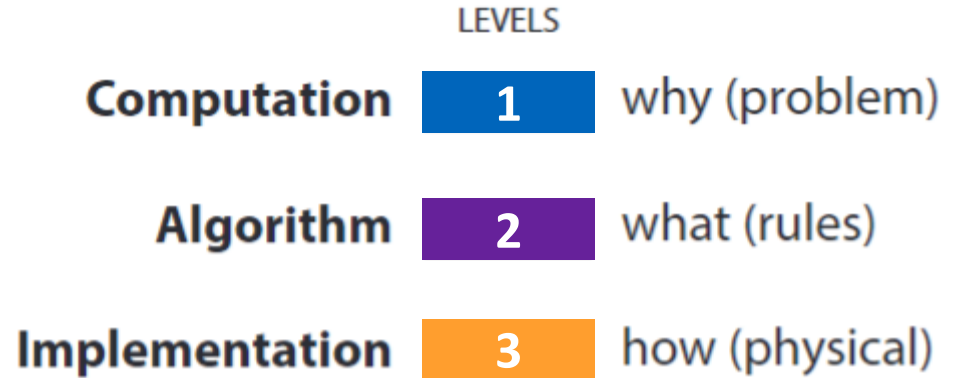
Building better theories

Clare Press^{1, 2}, Daniel Yon¹, Cecilia Heyes^{3, 4}

Two flavours of computational modeling



Influential perspective: Marr's 3 levels of analysis

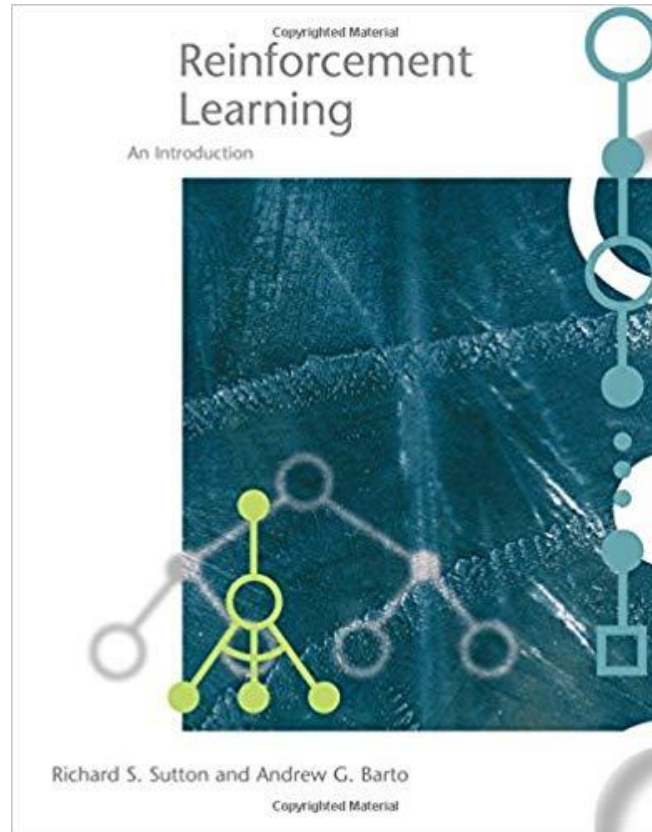


David Marr, (1982)

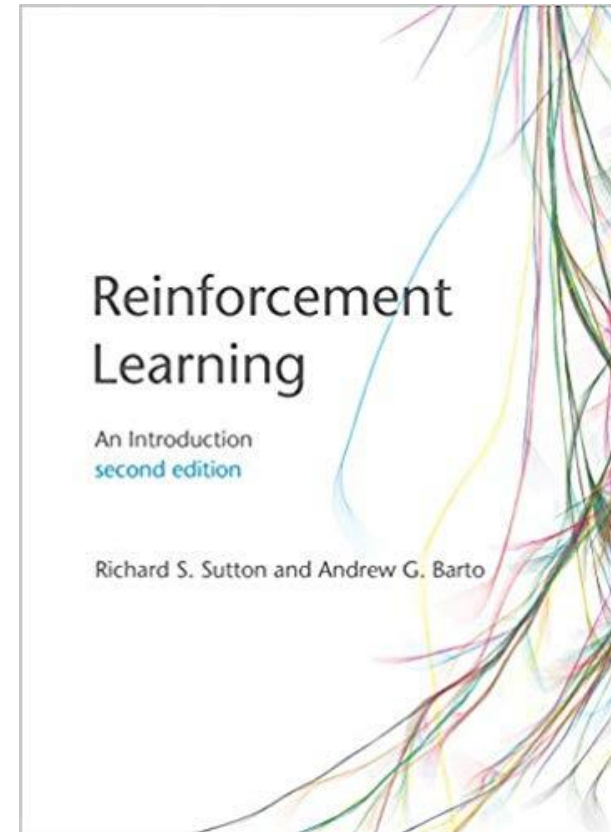
Outline

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The very short history



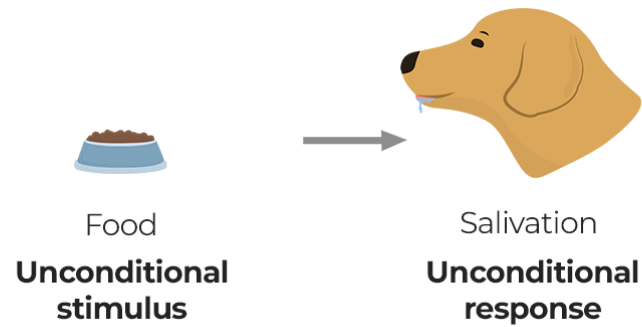
1998



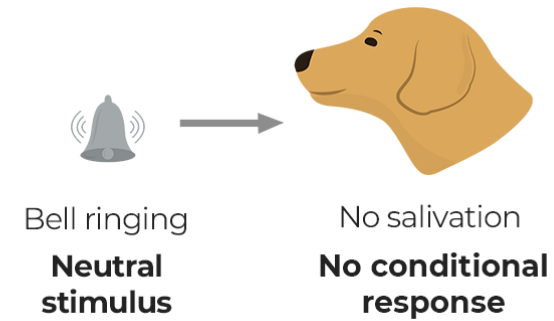
2018

why is it relevant?

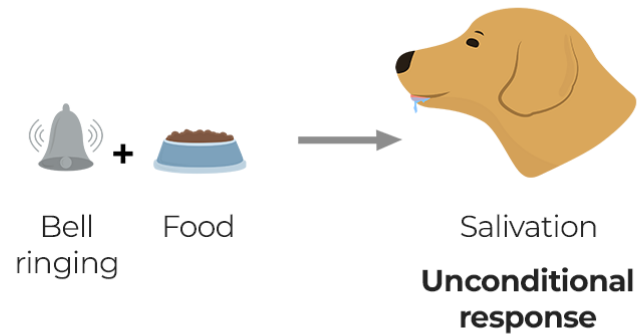
1. Before conditioning



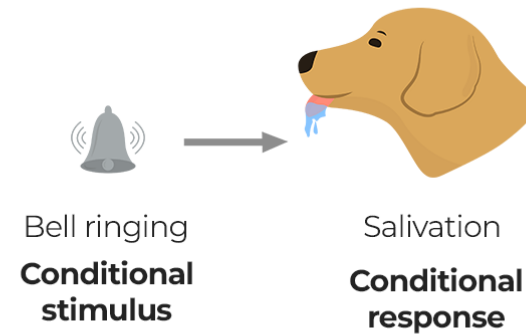
2. Before conditioning



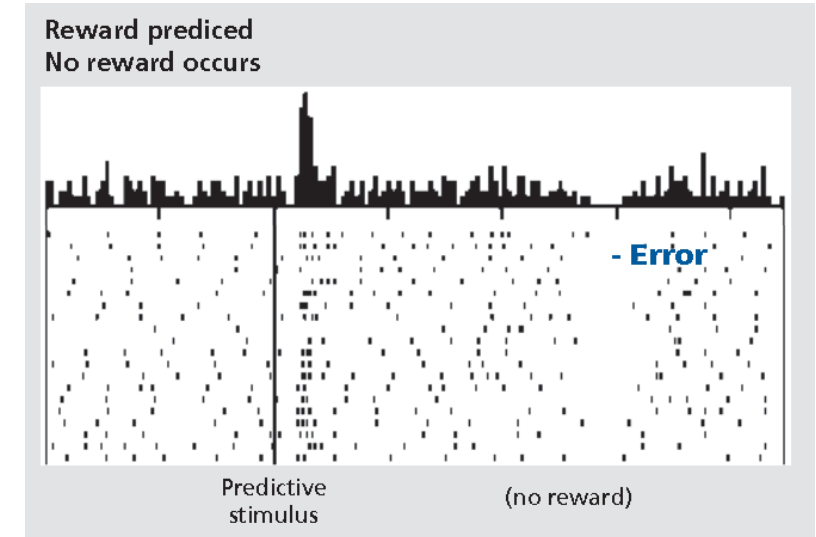
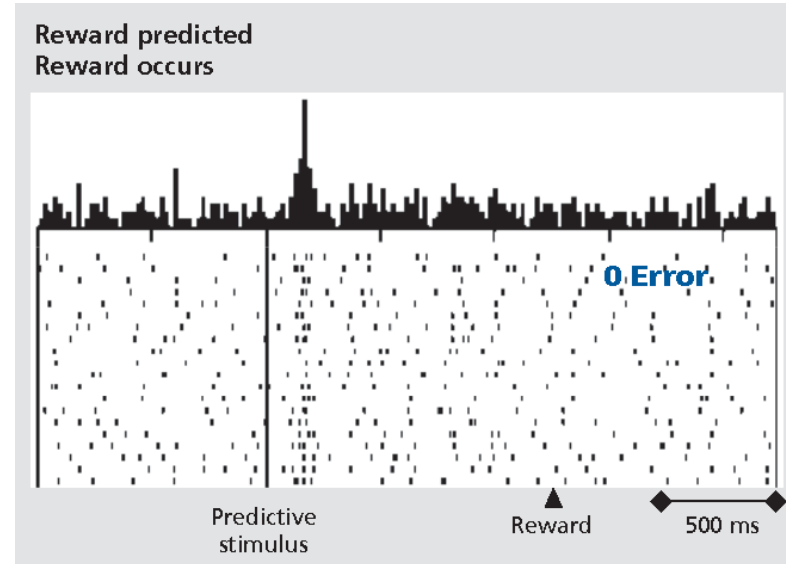
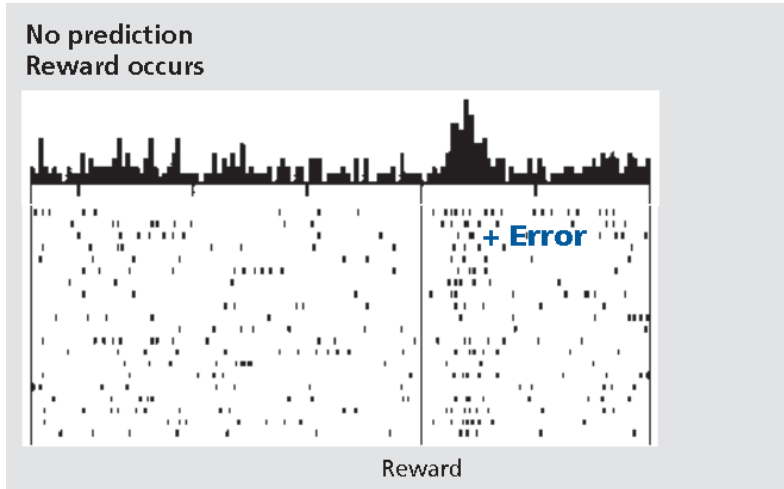
3. During conditioning



4. After conditioning



Reward Prediction error



Predicted reward trials



A neural substrate of prediction and reward

[W Schultz](#), [P Dayan](#), [PR Montague](#) - Science, 1997 - science.org

The capacity to predict future events permits a creature to detect, model, and manipulate the causal structure of its interactions with its environment. Behavioral experiments suggest that ...

☆ Speichern Zitieren Zitiert von: 12021 Ähnliche Artikel Alle 46 Versionen

Simple reinforcement learning: 2-armed bandit task



a simple task often used in the laboratory:

- repeated choice between N options (N -armed bandit)
- ...whose properties (reward amounts, probabilities) are learned through trial-and-error
- ...with a goal in mind: maximize the overall reward

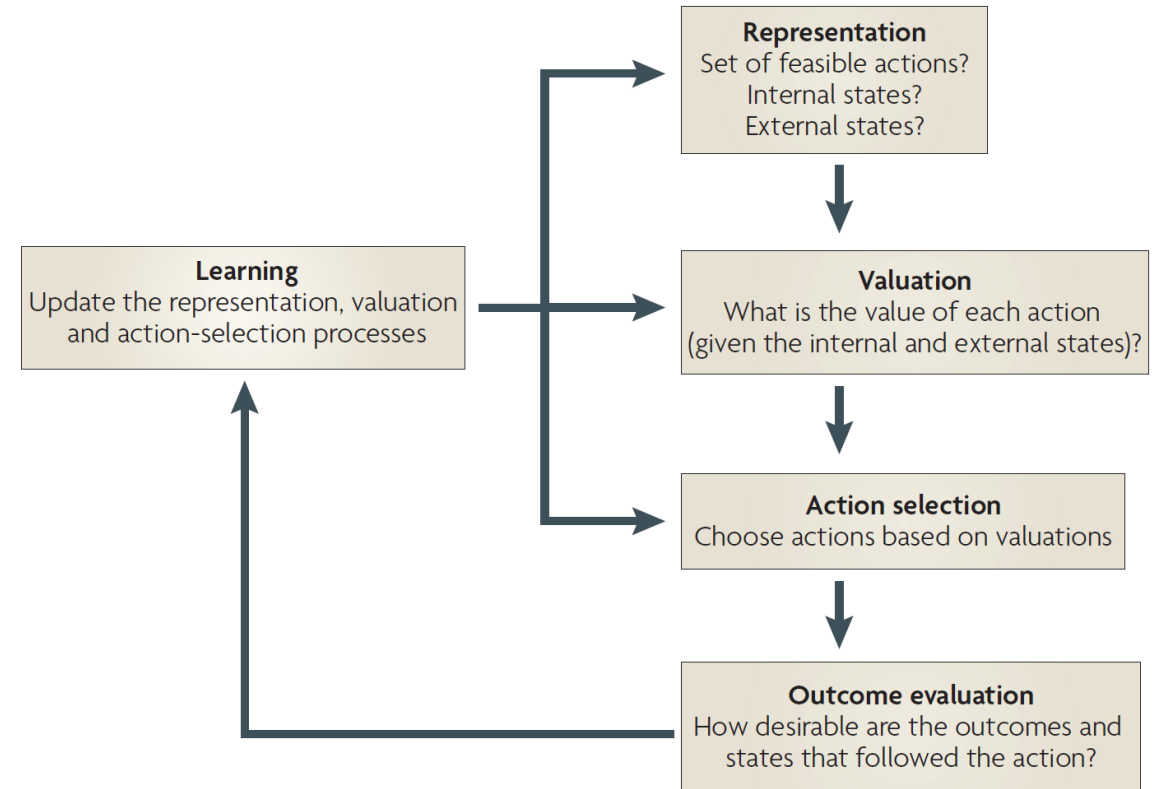
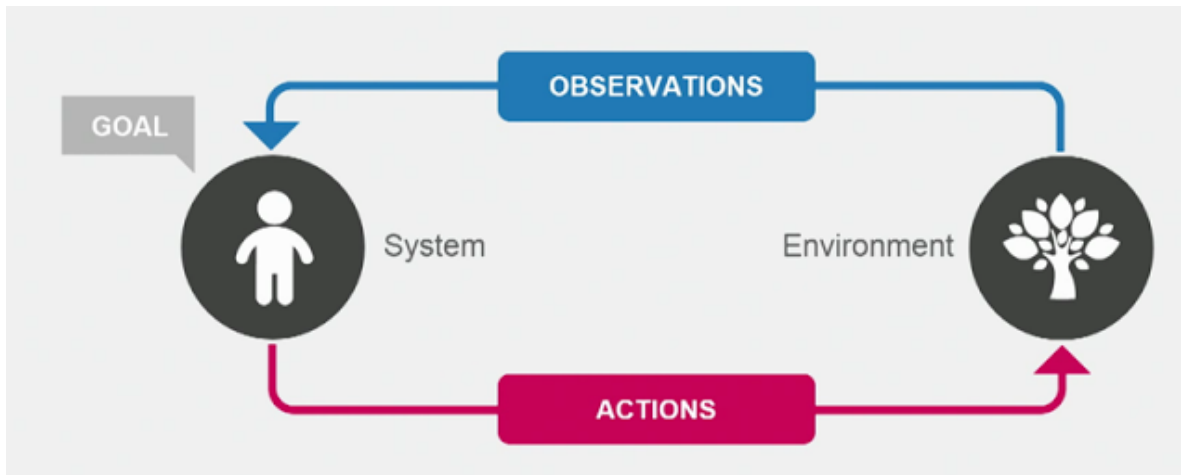
2-armed bandit task



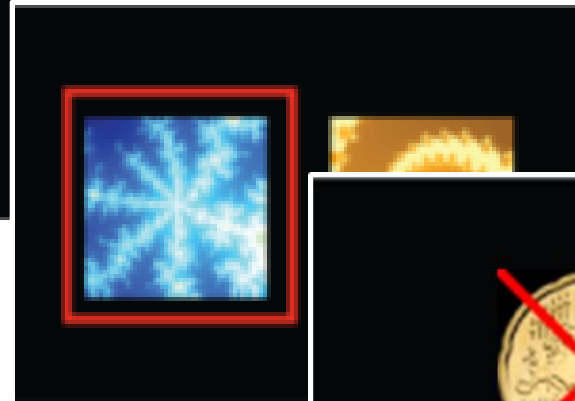
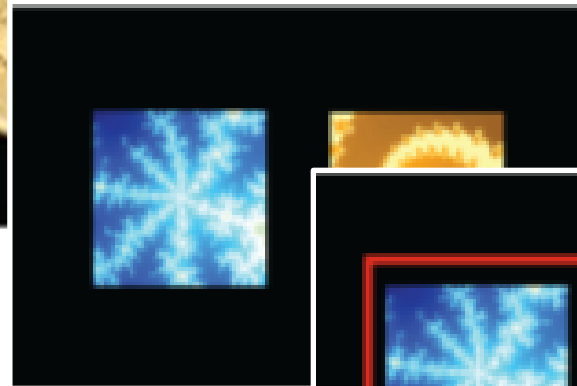
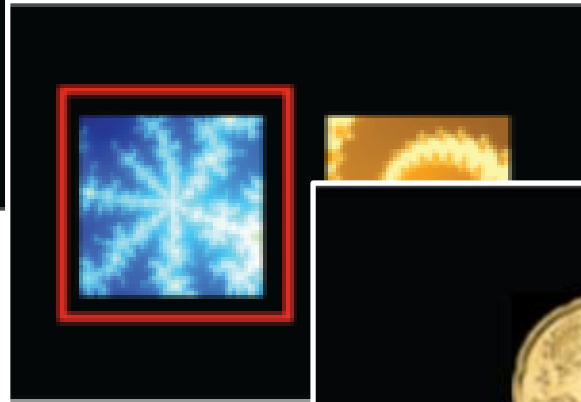
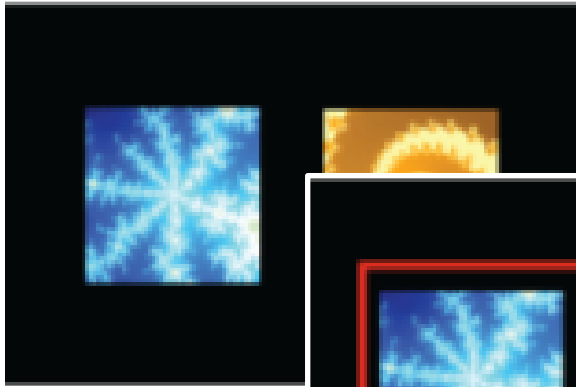
What can be your **strategies**:

1. **predict** the value of each deck
2. **choose** the best
3. **learn** from outcome to update predictions
(repeat)

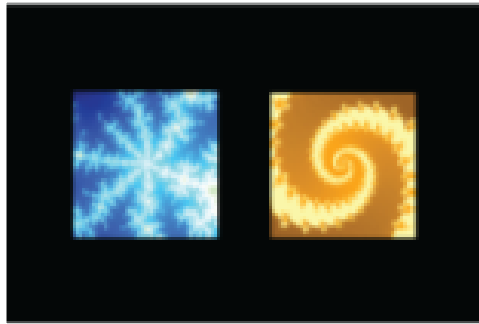
How is prediction shaped by learning?



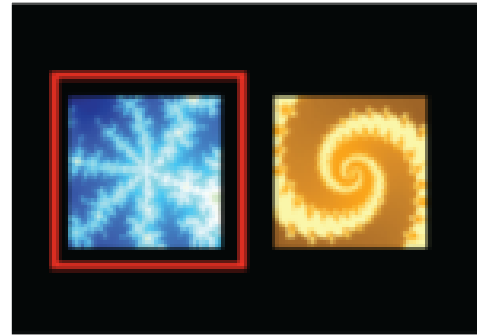
reward contingency 80:20



One simple experiment: two choice task



choice presentation



action selection



outcome

what do we know?

choice & outcome

what can we measure?

choice accuracy

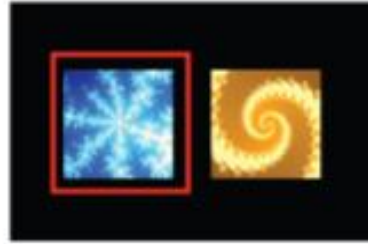
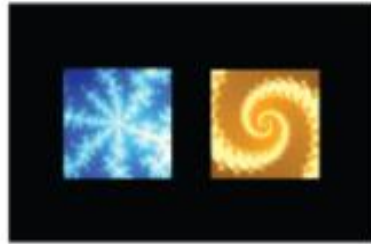
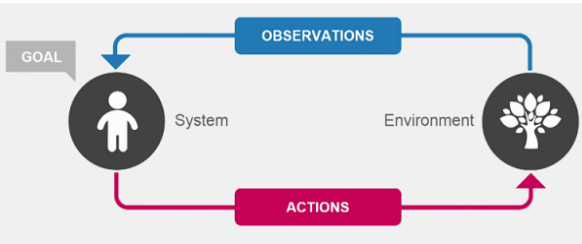
what do we not know?

RL update

$p(\text{choosing the better option})$

	subjID	trialID	choice	outcome
1	1	1	1	1
2	1	2	1	1
3	1	3	1	1

Q-learning Value Update



Cognitive Model

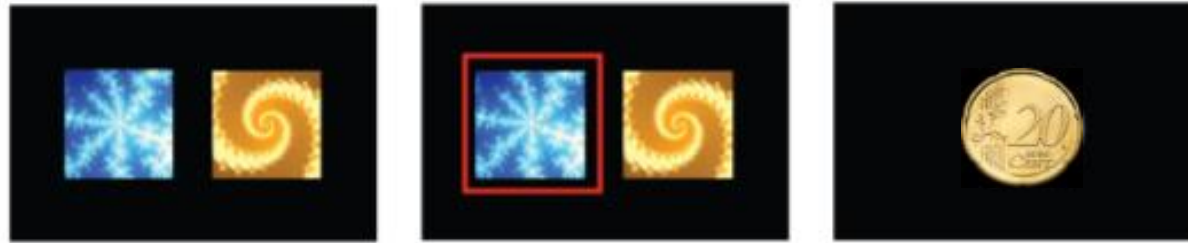
- cognitive process
- using internal variables and free parameters

Observation Model (Data Model)

- relate model to observed data
- has to account for noise

Q-Learning

- The idea: **error-driven** learning
- Change in value is proportional to the difference between actual and predicted outcome



Value update: $V_t = V_{t-1} + a * PE_{t-1}$

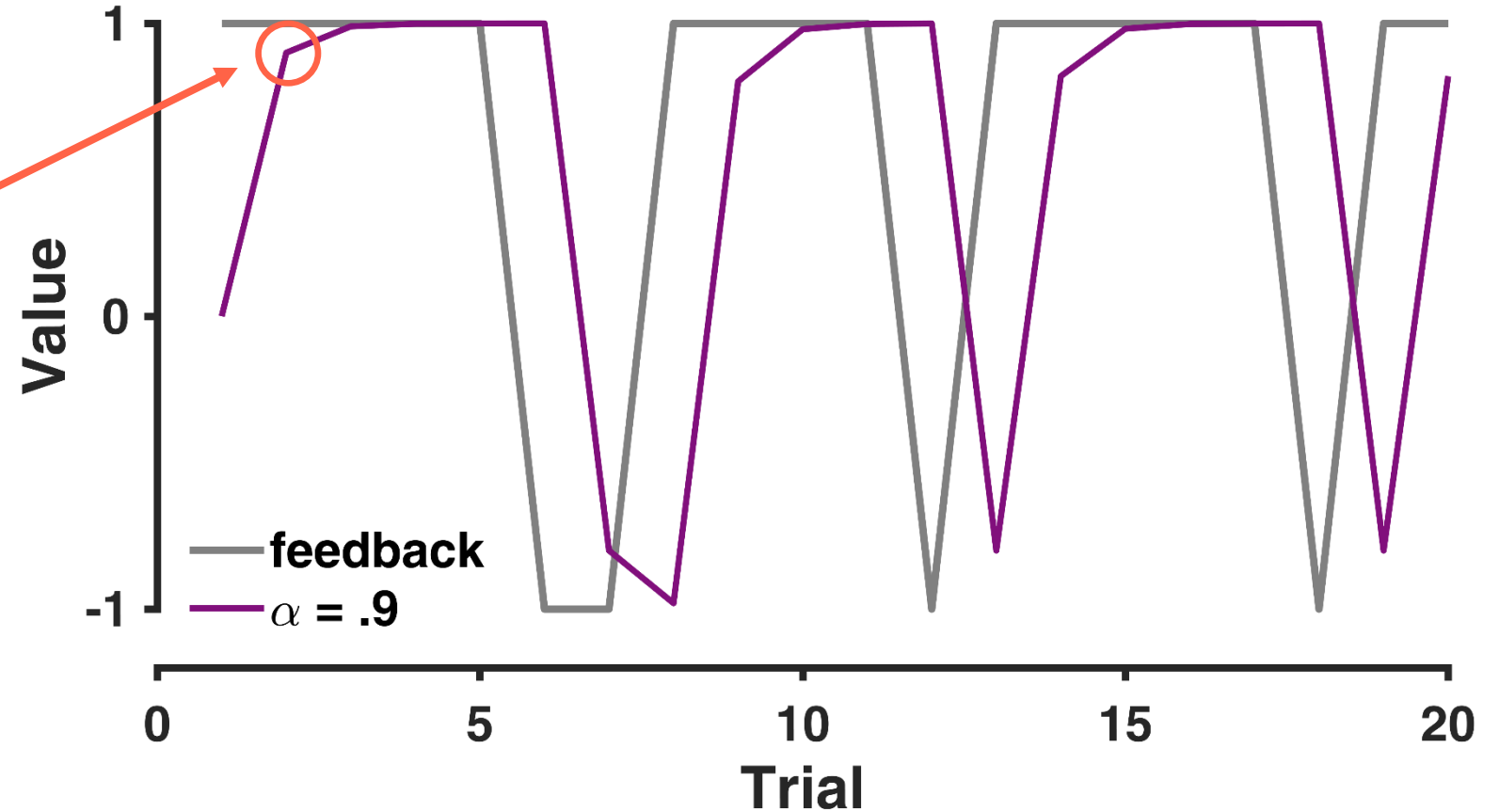
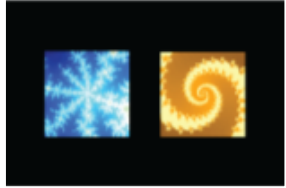
Prediction error: $PE_{t-1} = R_{t-1} - V_{t-1}$

*Expectations on the next trial = the expectation on the current trial + learning rate *
prediction error (reward – current expectation)*

Understand the learning rate

Value update: $V_t = V_{t-1} + a * PE_{t-1}$

Prediction error: $PE_{t-1} = R_{t-1} - V_{t-1}$



if $\alpha = 0.9$

$$V_1 = 0$$

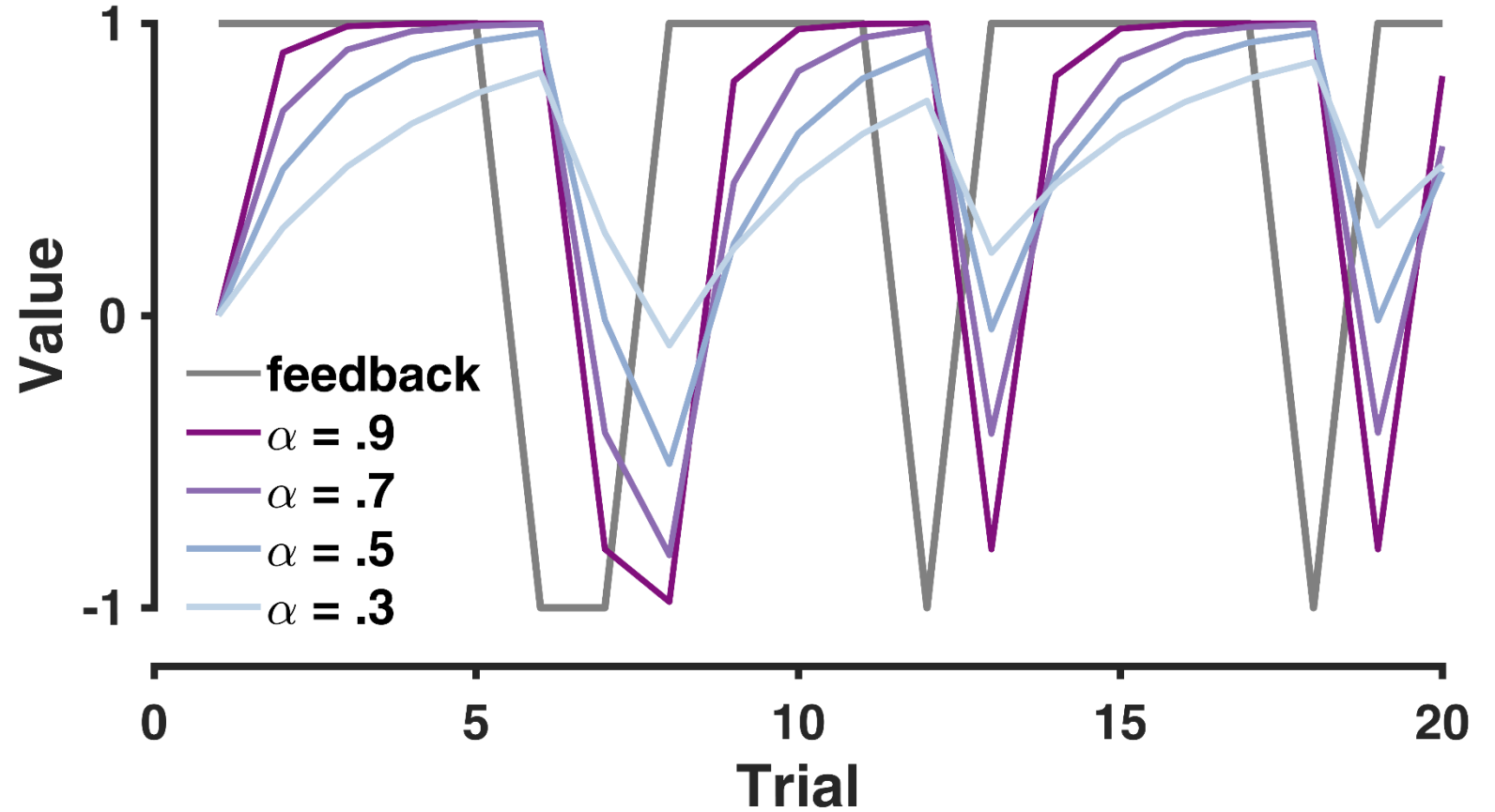
$$\begin{aligned} V_2 &= V_1 + 0.9 * (1 - V_1) \\ &= 0 + 0.9 * (1 - 0) \\ &= 0.9 \end{aligned}$$

reward contingency – 80:20

Understand the learning rate

Value update: $V_t = V_{t-1} + a * PE_{t-1}$

Prediction error: $PE_{t-1} = R_{t-1} - V_{t-1}$



reward contingency – 80:20

Understand the learning rate

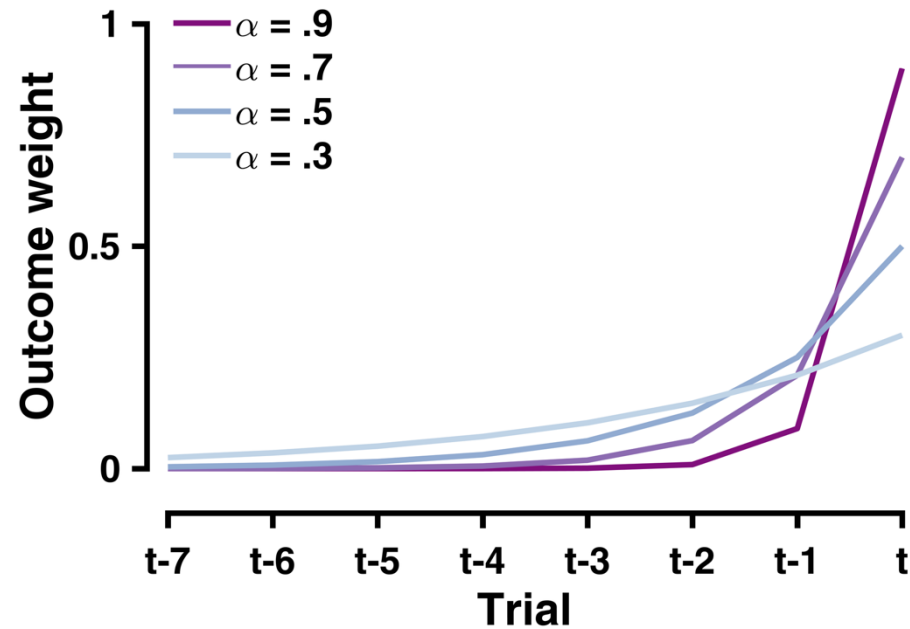
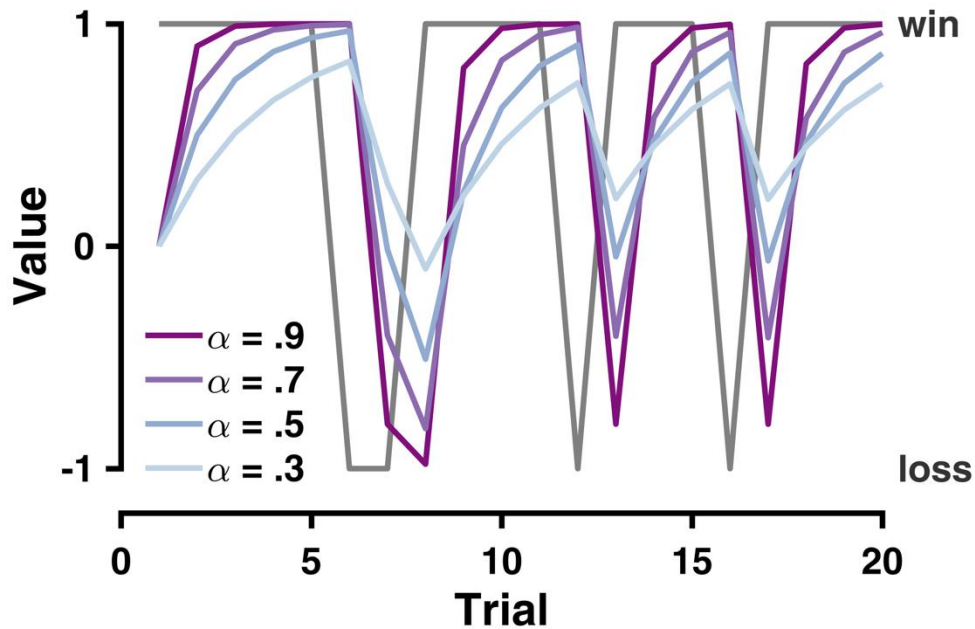
Value update: $V_t = V_{t-1} + a * PE_{t-1}$

Prediction error: $PE_{t-1} = R_{t-1} - V_{t-1}$

$$V_t = (1 - a) V_{t-1} + a R_{t-1}$$

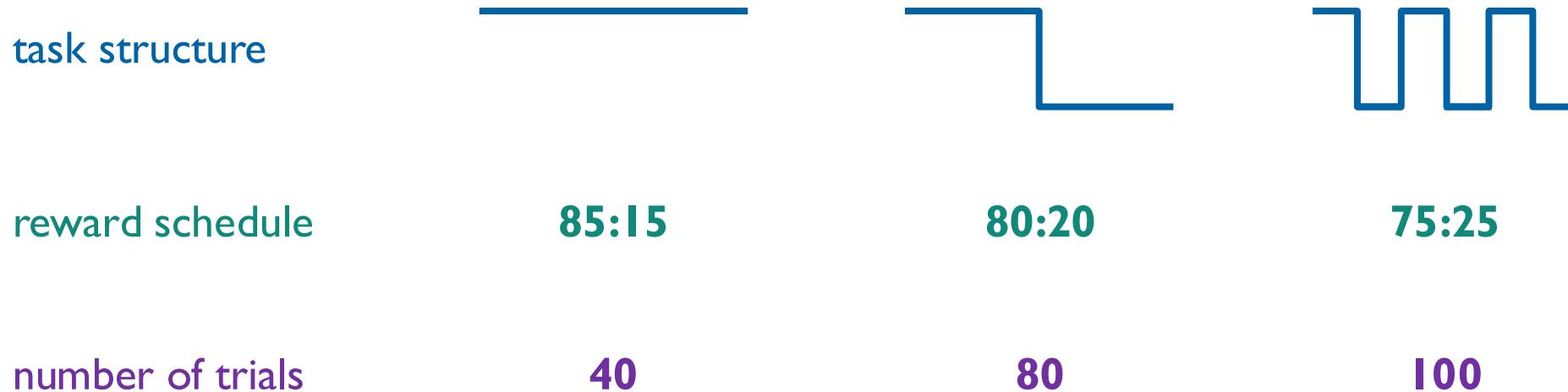
$$= (1 - a) (V_{t-2} + a (R_{t-2} - V_{t-2})) + a R_{t-1}$$

$$= (1 - a)^t V_0 + \sum_{i=1}^t (1 - a)^{t-i} a R_i$$



Is there an **optimal** learning rate?

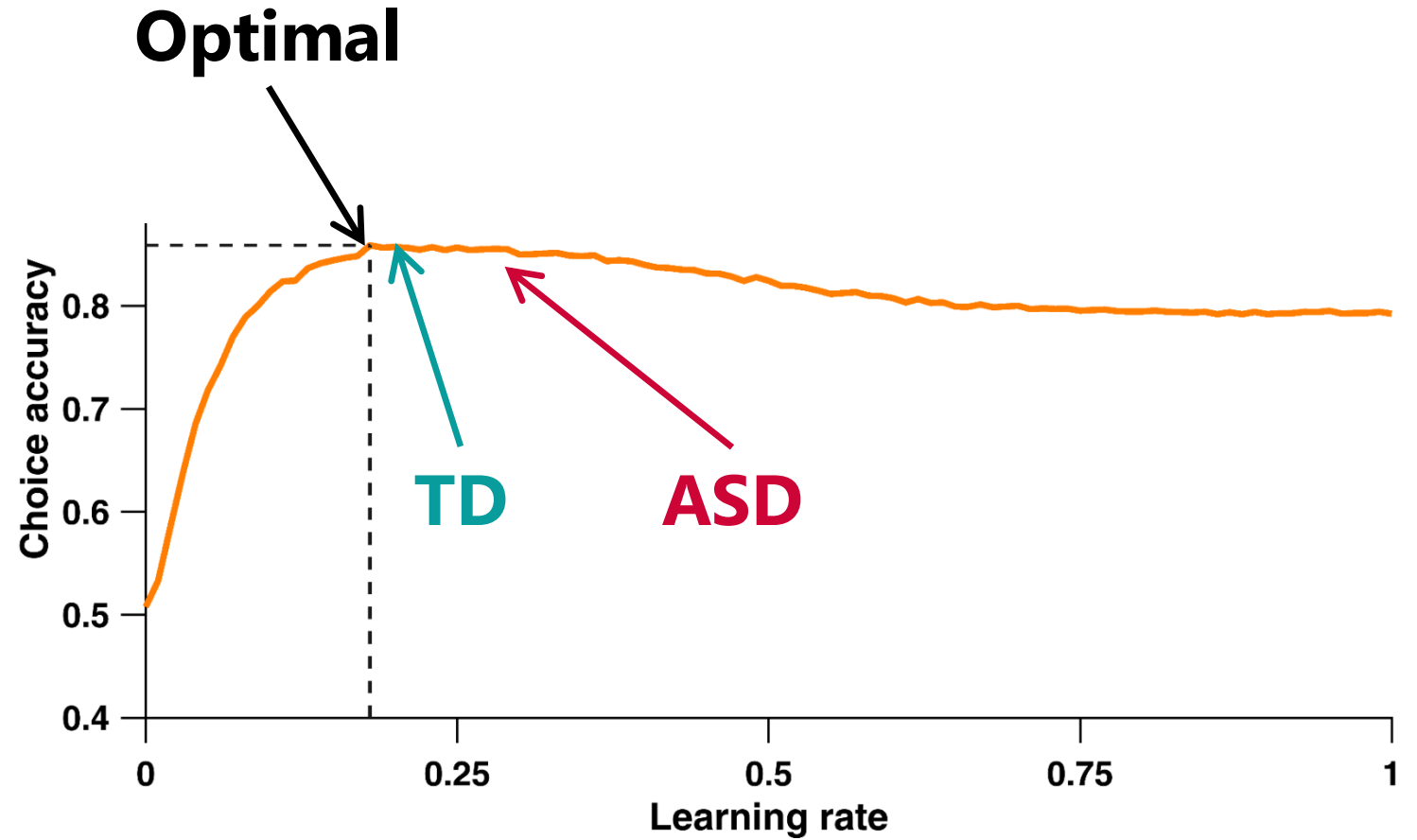
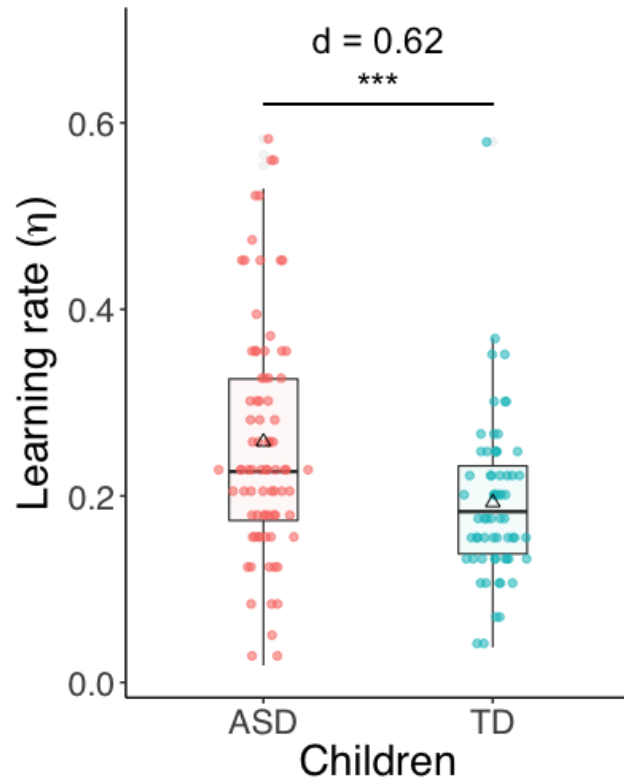
- it hugely depends on the task!



a task = task structure + reward schedule + number of trials

An optimal learning rate exists only when the task is settled.

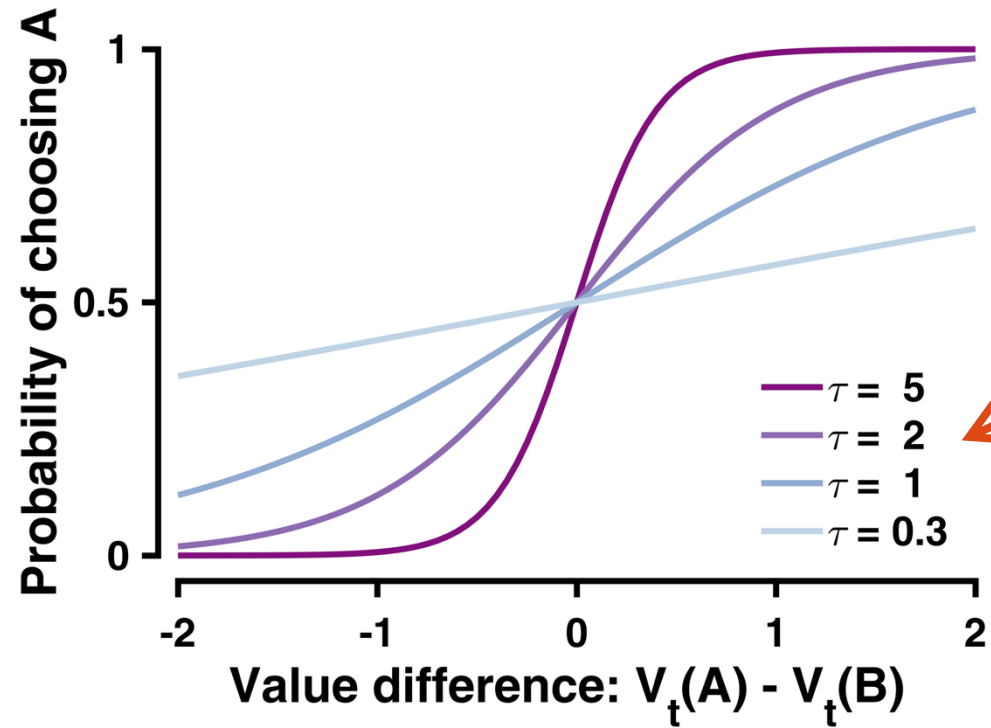
Optimal learning rate?



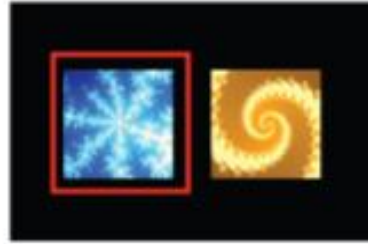
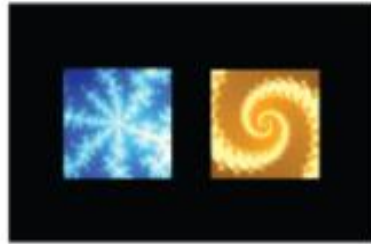
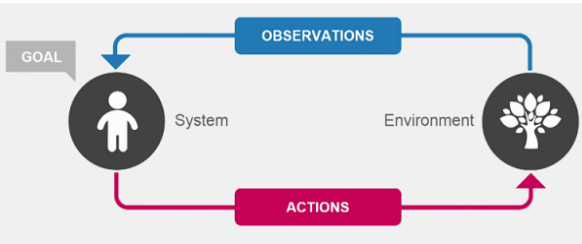
Choice rule: softmax



$$p_t(A) = \frac{e^{x^* V_t(A)}}{e^{x^* V_t(A)} + e^{x^* V_t(B)}}$$
$$= \frac{1}{1 + e^{-x^* (V_t(A) - V_t(B))}}$$



Q-Learning



Value update:

$$V_{t+1} = V_t + \alpha * PE_t$$

Prediction error:

$$PE_t = R_t - V_t$$

choice rule (sigmoid /softmax):

$$p(C=a) = \frac{1}{1 + e^{\tau * (v(b) - v(a))}}$$

α - learning rate
PE - reward prediction error
V - value
R - reward
 τ - softmax temperature

Some terminologies

simple Q-Learning

$$V_{t+1} = V_t + \alpha (R_t - V_t)$$

$$p_t(A) = \frac{1}{1 + e^{-\tau * (v(A) - v(B))}}$$

$$LL = \text{categorical_lpmf}(C_t | p_t)$$

simple linear regression

$$\mu = a + b X$$

$$LL = \text{normal_lpdf}(Y | \mu, \delta)$$

Some terminologies

simple Q-Learning

$$V_{t+1} = V_t + \alpha (R_t - V_t)$$

$$p_t(A) = \frac{1}{1 + e^{-\tau * (v(A) - v(B))}}$$

$$LL = \text{categorical_lpmf}(C_t | p_t)$$

simple linear regression

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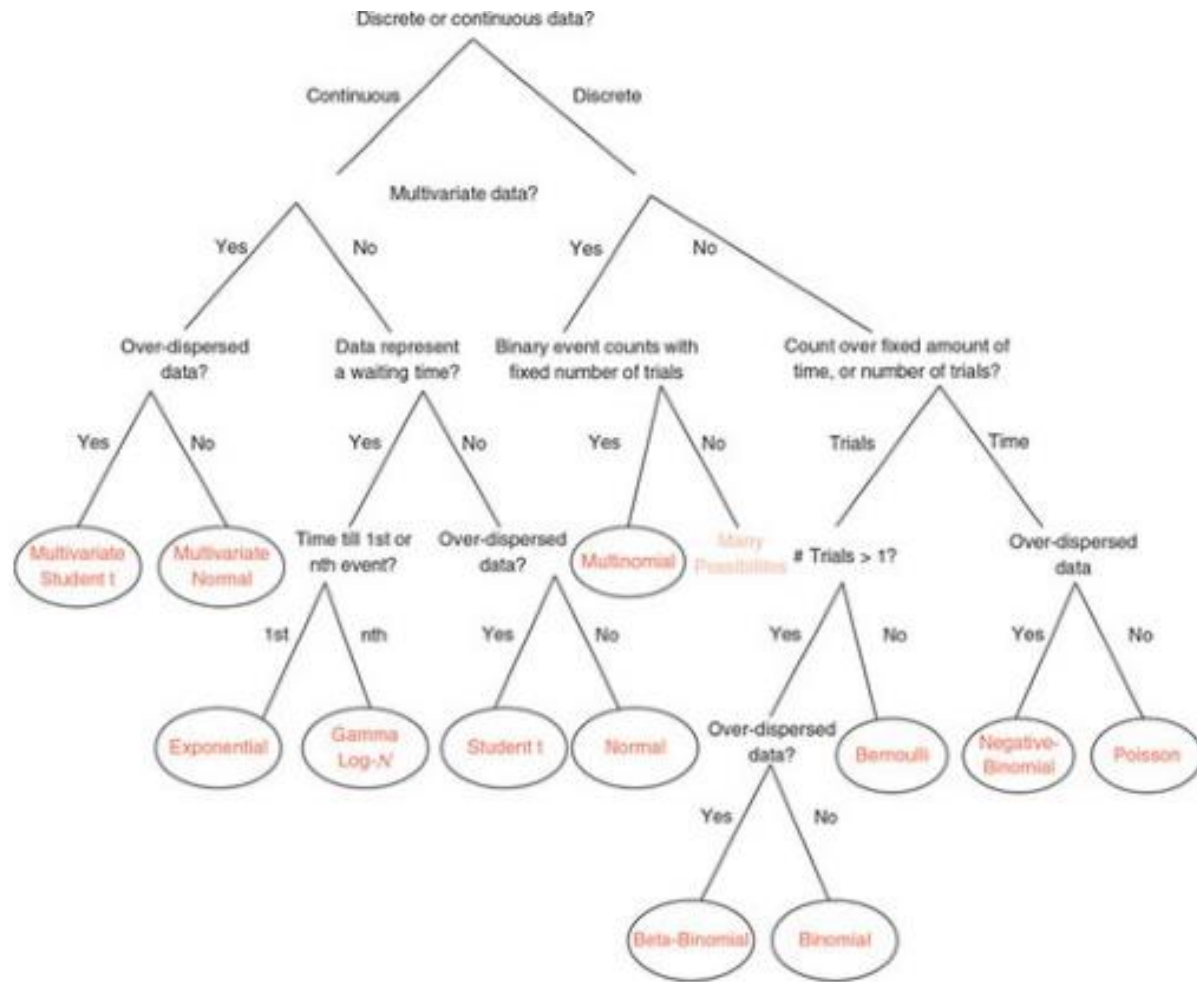
data / observation

free parameter / model parameter

internal/latent (model) variable

likelihood function

How do I know which likelihood to use?



The distribution zoo

by

Ben Lambert and Fergus Cooper

Last month: used by 285 people over 451 sessions in 41 countries

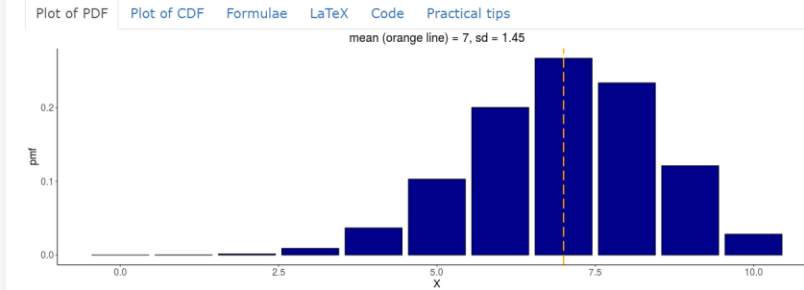
Since created: used by 4072 people over 6785 sessions in 107 countries

Category of distribution: Discrete Univariate

Distribution type: Binomial

size: 10

probability: 0.7



Category of distribution: Multivariate

Distribution type: Multivariate Normal

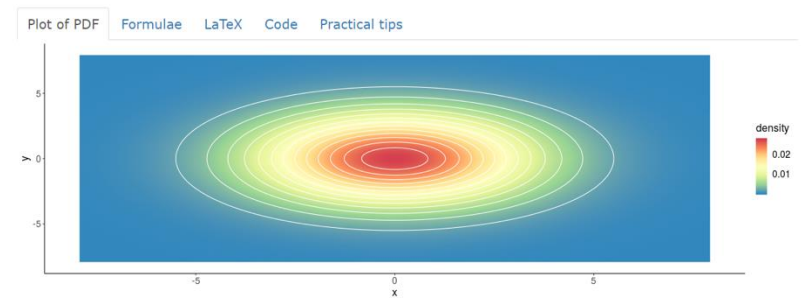
Mean of X: 0

Mean of y: 0

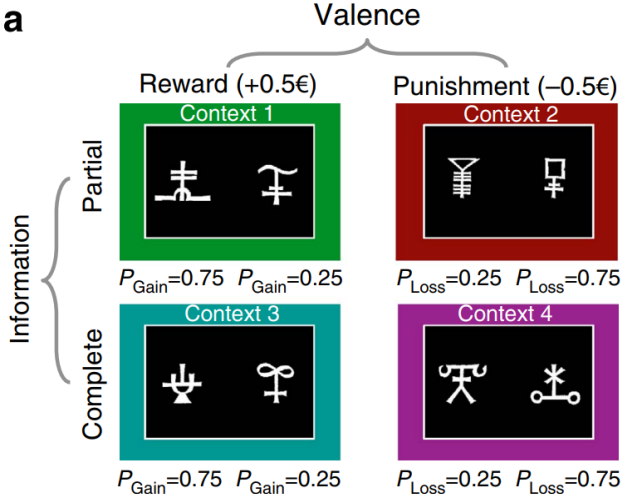
Standard deviation of x: 2.0

Standard deviation of y: 2.0

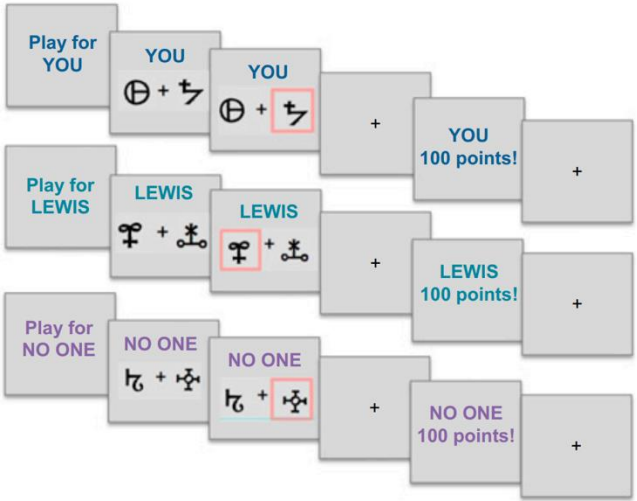
Correlation between x and y: 0



Generalizing RL framework

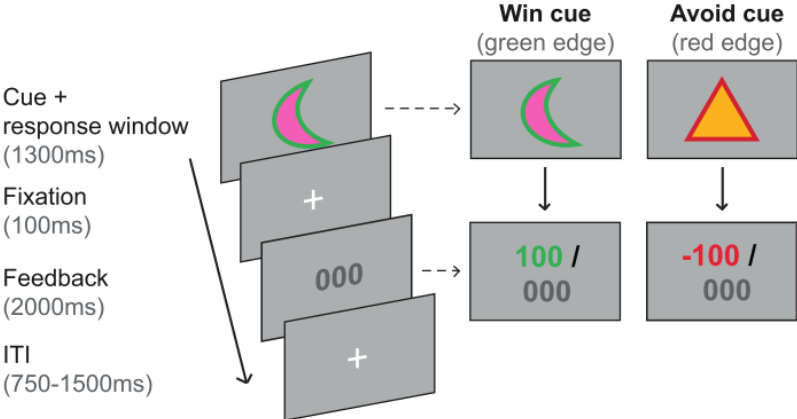


[Palminteri et al. \(2015\)](#)

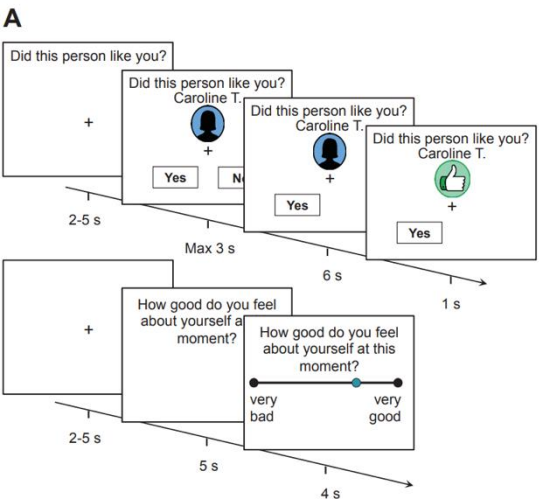


[Lockwood et al. \(2016\)](#)

A. Trial details



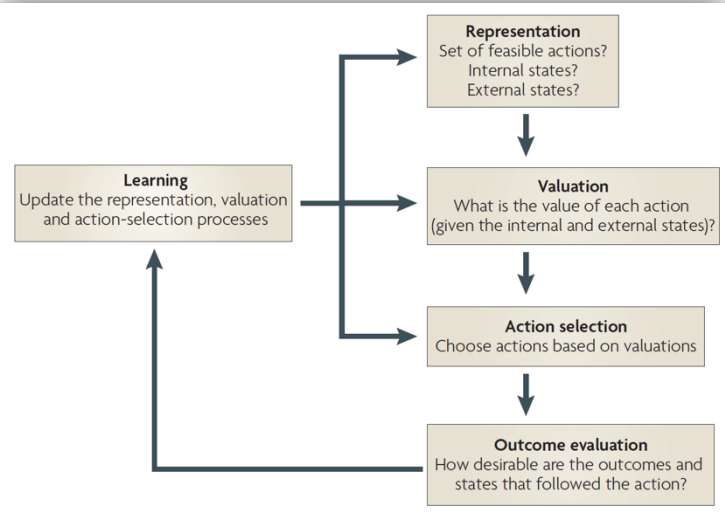
[Swart et al. \(2017\)](#)



[Will et al. \(2017\)](#)

B

	👍	👎
👤	85%	15%
👤	70%	30%
👤	30%	70%
👤	15%	85%

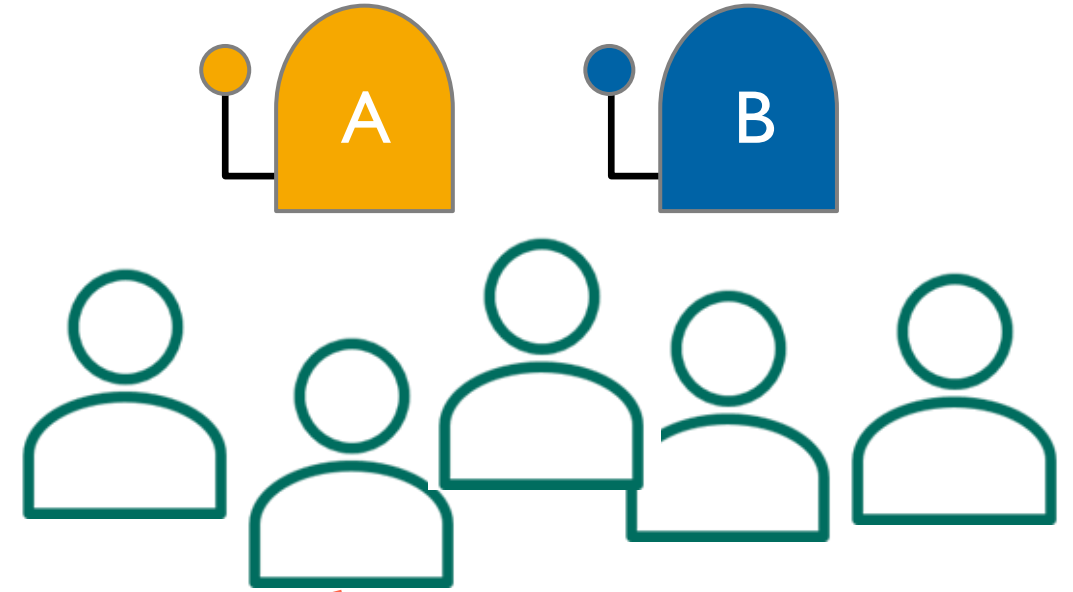
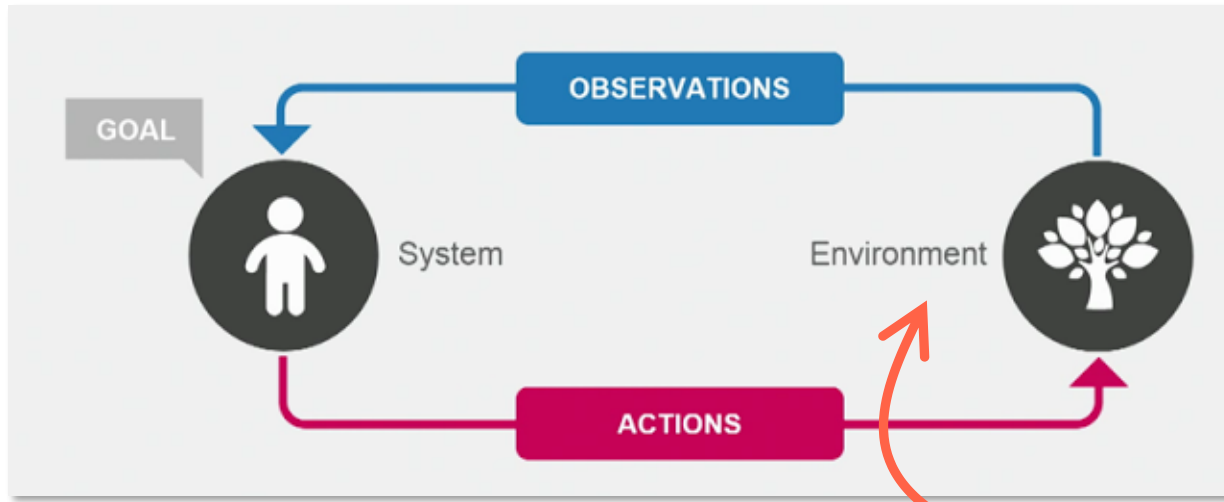


Social learning & decision-making

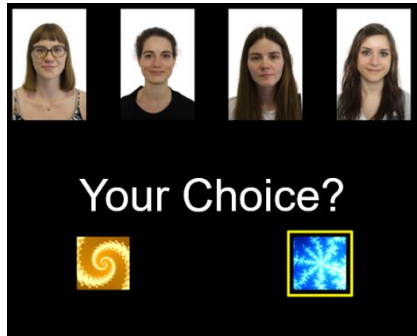
Decision-making stage (value type)	Target (and reference frame) of valuation		
	Value of other people (for oneself)	Value of other people's experiences (for them)	Value of social constellation (for normative social principle)
Choice (decision value and anticipated value)	Deciding whether to marry someone	Choosing a school for your child	Altruistic punishment of norm violations
Outcome (experienced value)	Being smiled at	Empathy with someone's pain	Enjoying fair distributions
Learning (anticipated value)	Learning about someone's trustworthiness	Learning about someone else's preferences	Changing your opinion to increase social conformity



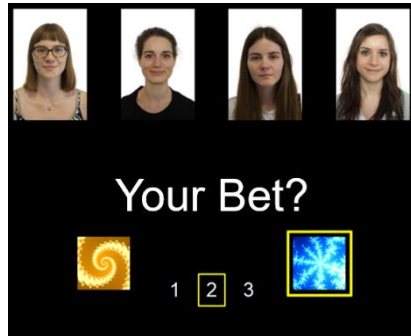
learn from others (to benefit oneself)



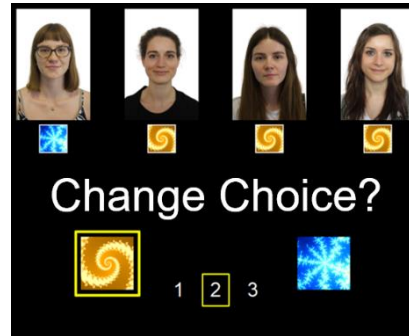
Social environment



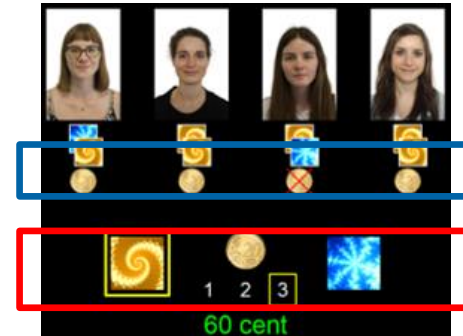
1st choice



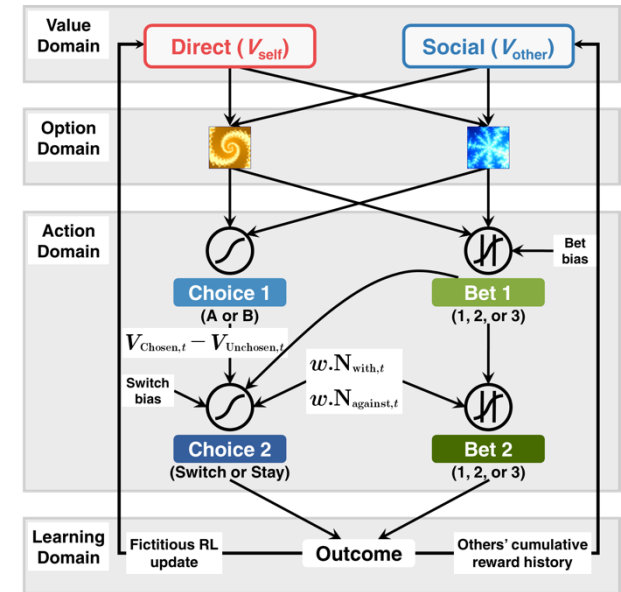
1st bet



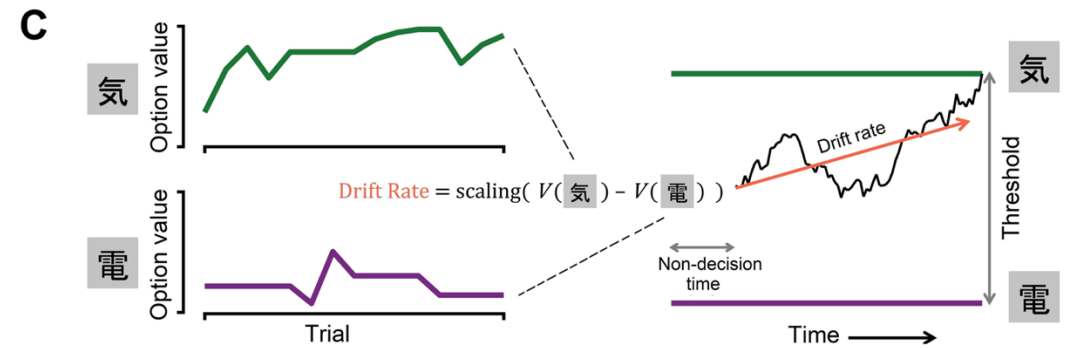
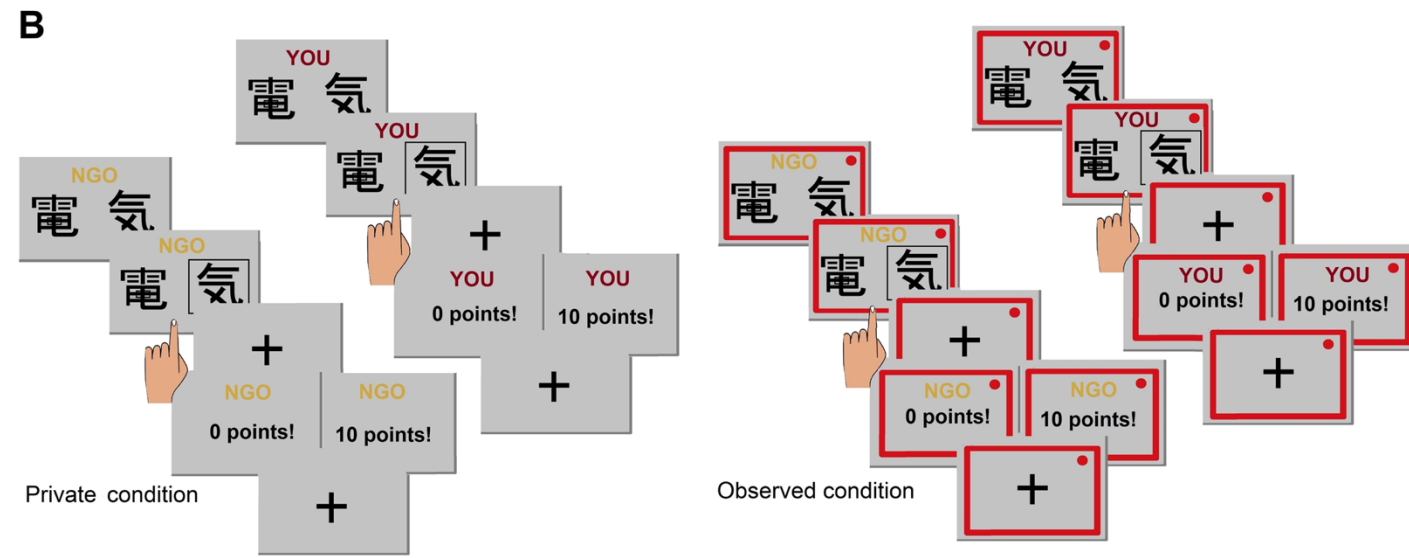
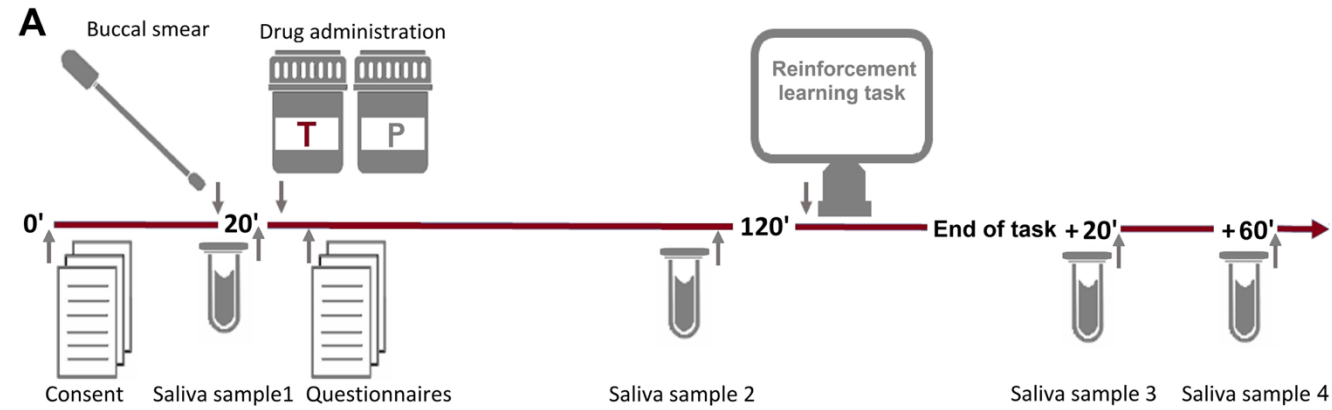
2nd choice



2nd bet



learn for others (pro-social)



Some more food for thought

What is a social RL task?

There are commonly two variations: either reward learning in social contexts (e.g. learn to expect monetary reward for a social partner) or social feedback learning (e.g. learn to expect social status or social evaluation). When the goal is to compare different types of feedback (e.g. social vs. non-social feedback), we suggest matching the feedback as closely as possible on ‘domain general’ properties, such as salience or preference

[Lockwood et al., 2016](#); [Will et al., 2017](#)

Is the learning rate static across trials, or dynamically adapting along the course of the experiment?

The learning rate does not necessarily have to be constant. But in the case of the Rescorla–Wagner model (and related models), the learning rate is indeed static. A dynamic learning rate, however, is possible when other types of models are applied. Note that the interpretation of the learning rate we discussed in the main text is independent of this constant vs dynamic property

[Li et al., 2011](#); [Mathys et al., 2011](#)

Is it possible to use RL models in the absence of choice data?

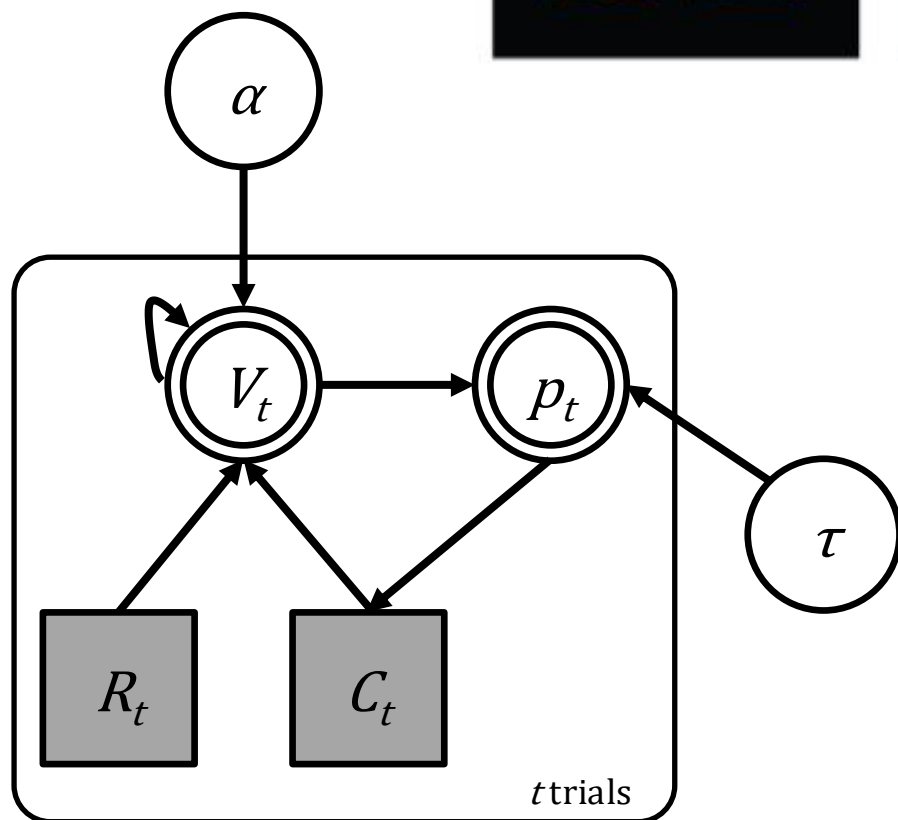
At least some sort of data is needed to perform model estimation. For example, skin conductance response (SCR) or pupil size response (PSR) have been used to fit RL models in associative fear learning tasks, where choice data was not available

[Li et al., 2011](#); [Tzovara et al., 2018](#)

Outline

- Recap on Bayesian modeling
- Why Stan?
- What is (computational) cognitive modeling?
- The idea of the simple Q-Learning model
- **Implementing Q-Learning model for one subject in Stan**
- Fitting multiple subjects with the hBayesDM package
- Q&A

RL – Implementation



$$\alpha \sim \text{Uniform}(0, 1)$$

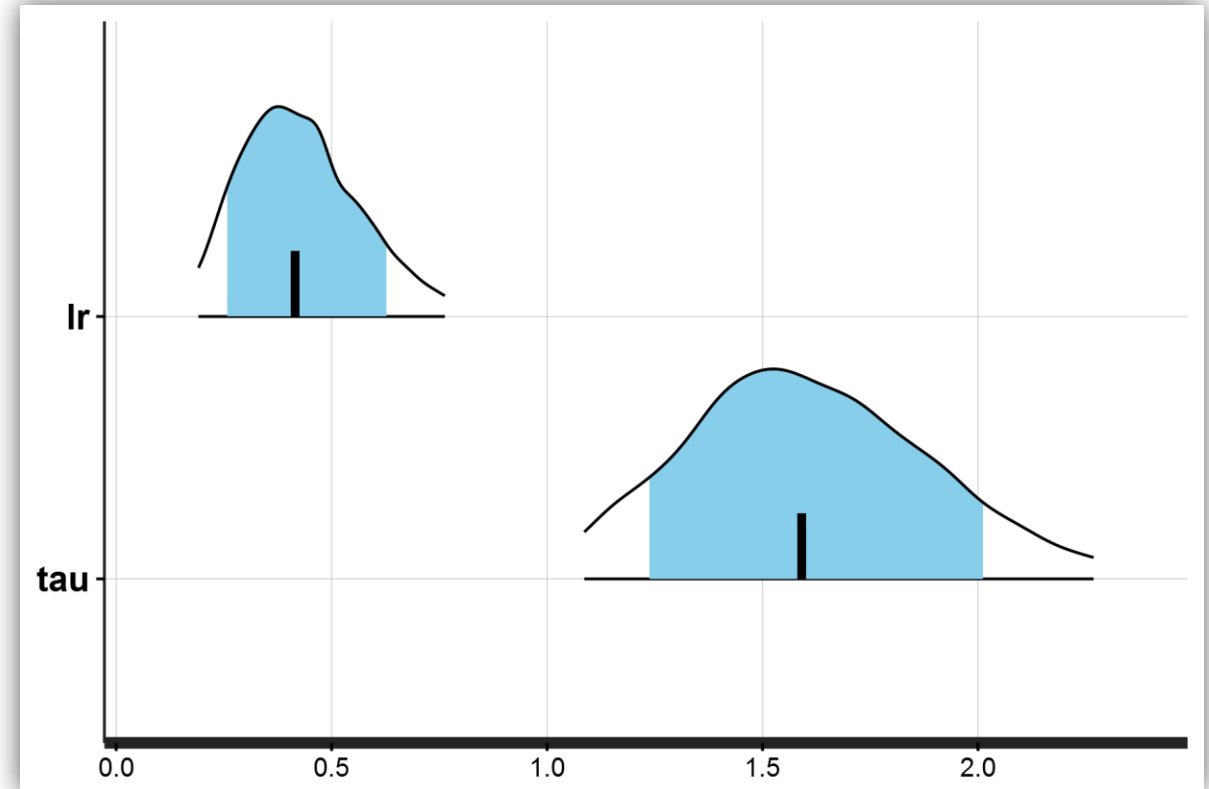
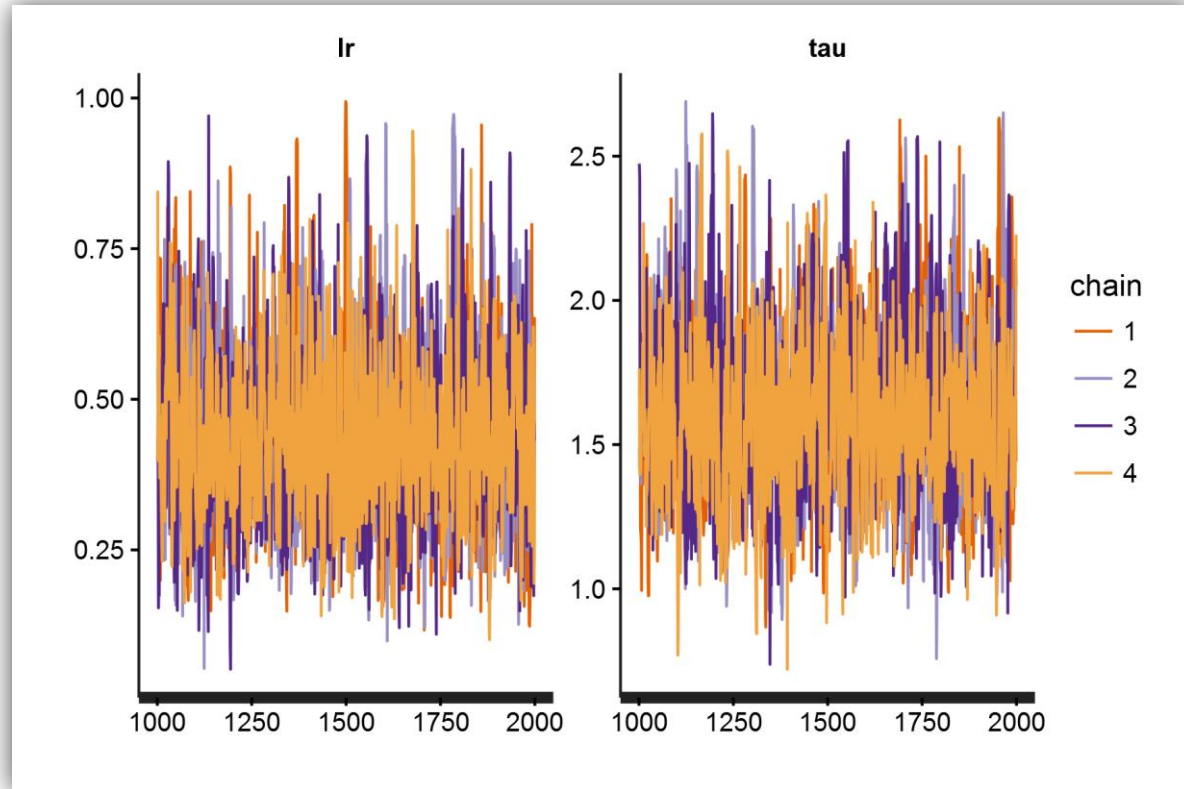
$$\tau \sim \text{Uniform}(0, 3)$$

$$p_t(C = A) = \frac{1}{1 + e^{\tau(V_t(B) - V_t(A))}}$$

$$V_{t+1}^c = V_t^c + \alpha(R_t - V_t^c)$$

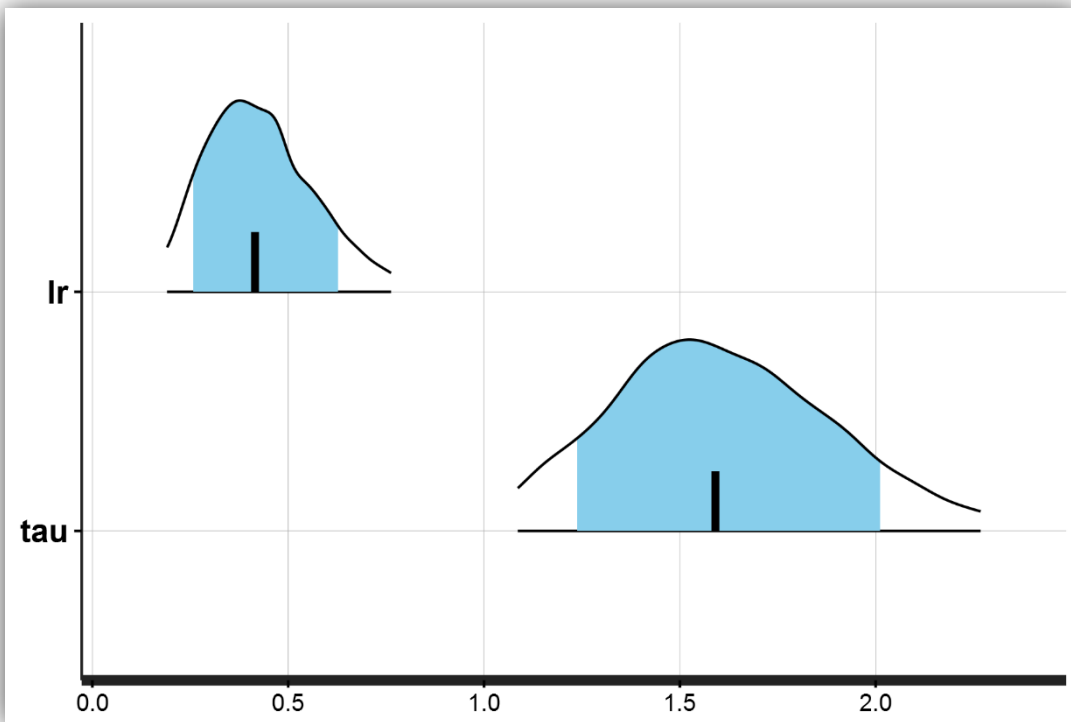
Live demo of building the Q-Learning model
model in Stan from scratch

RL – MCMC Output

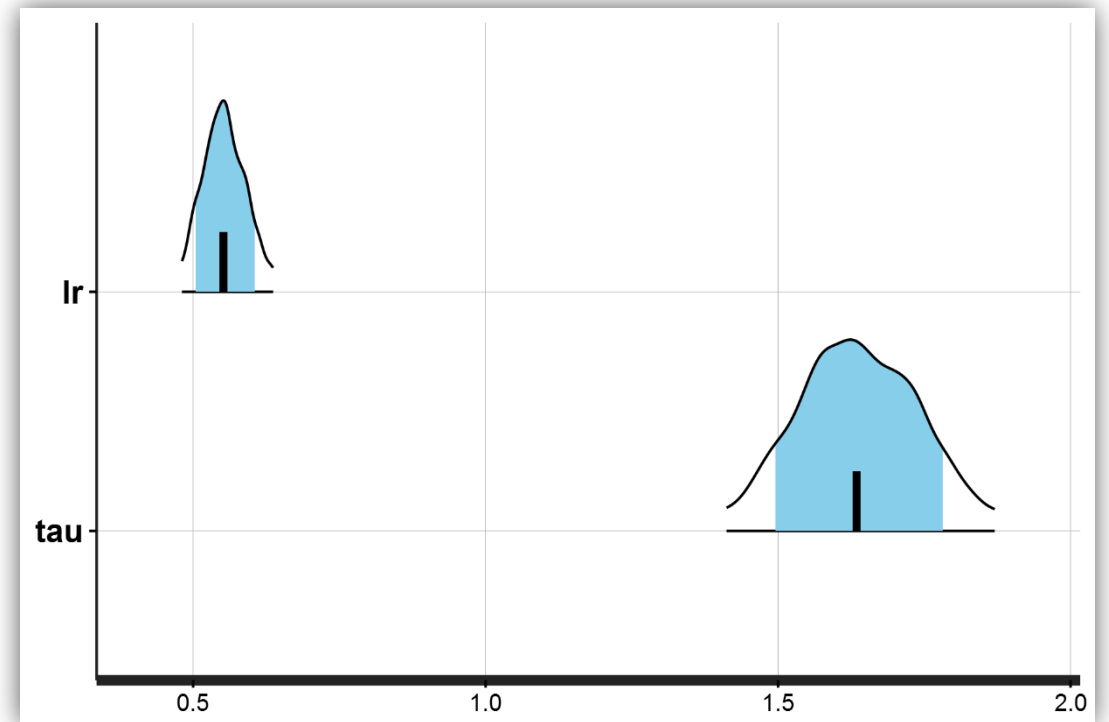


What if we have more data?

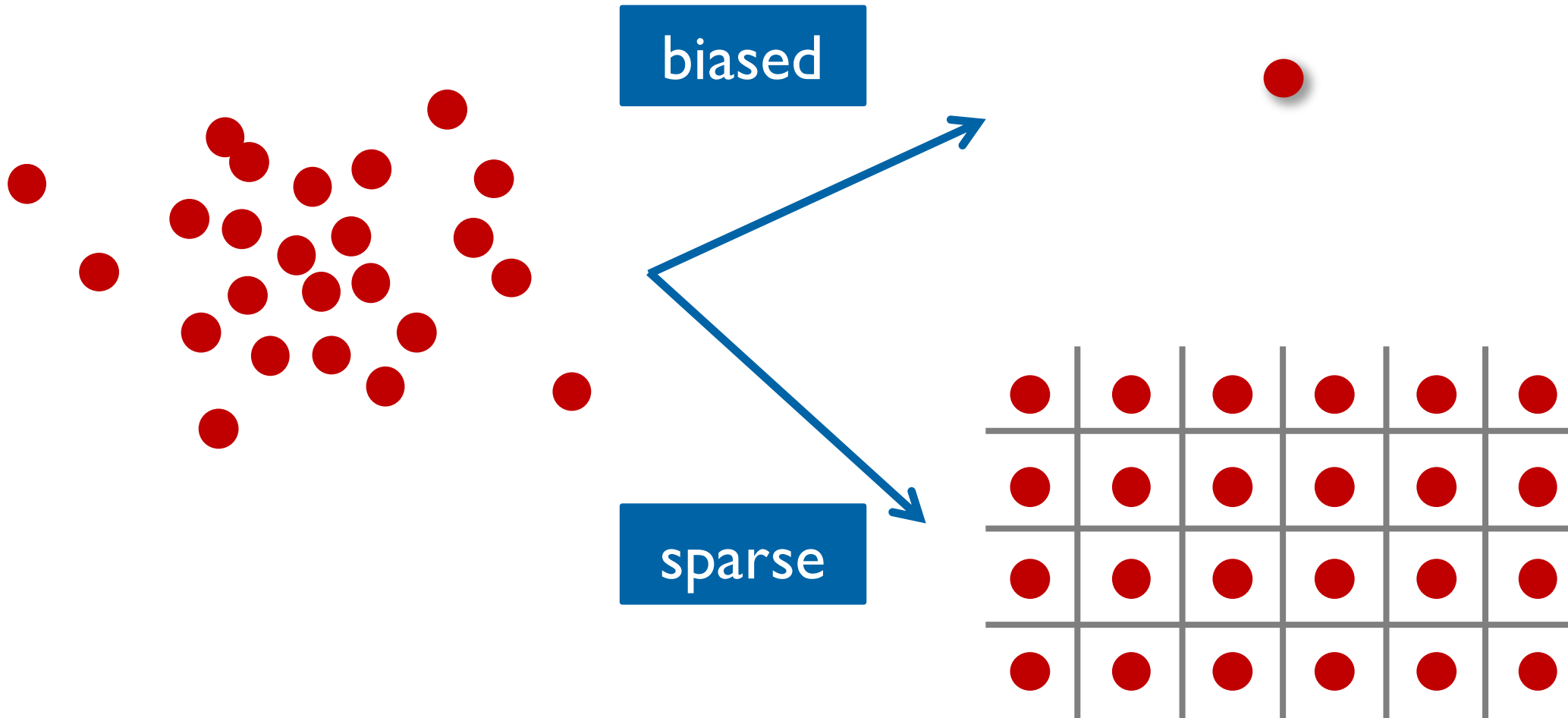
$N = 1$



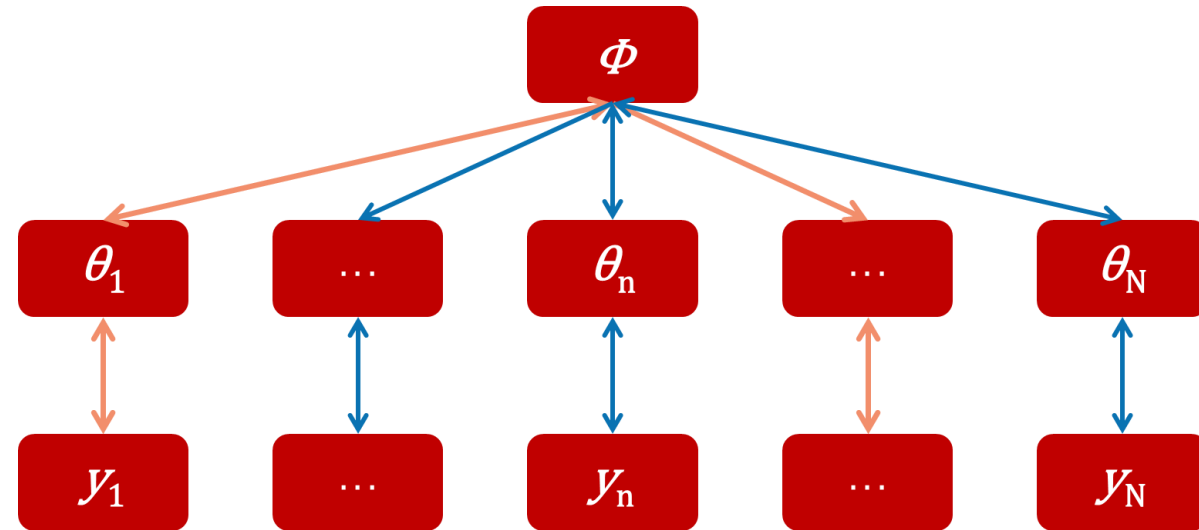
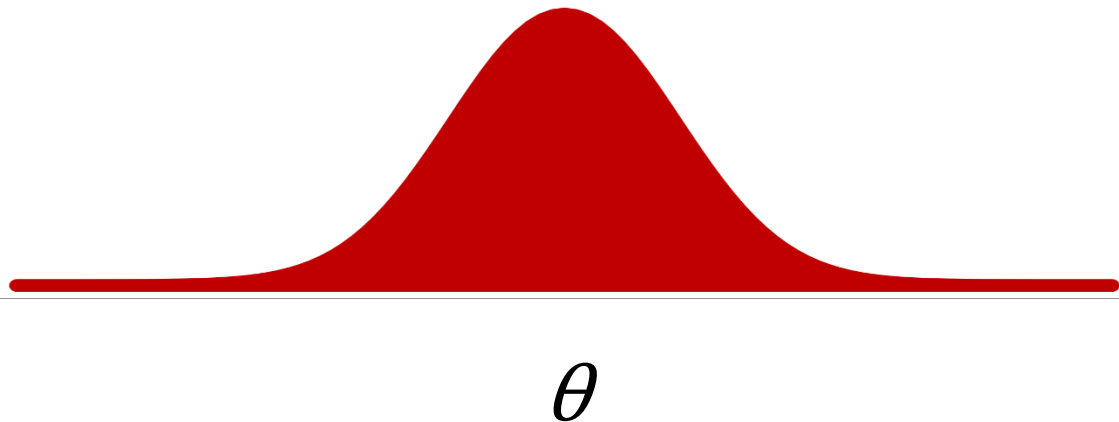
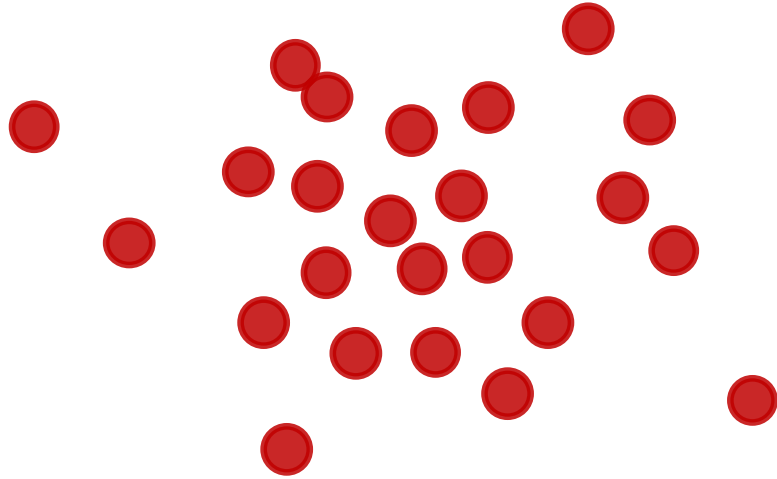
$N = 10$



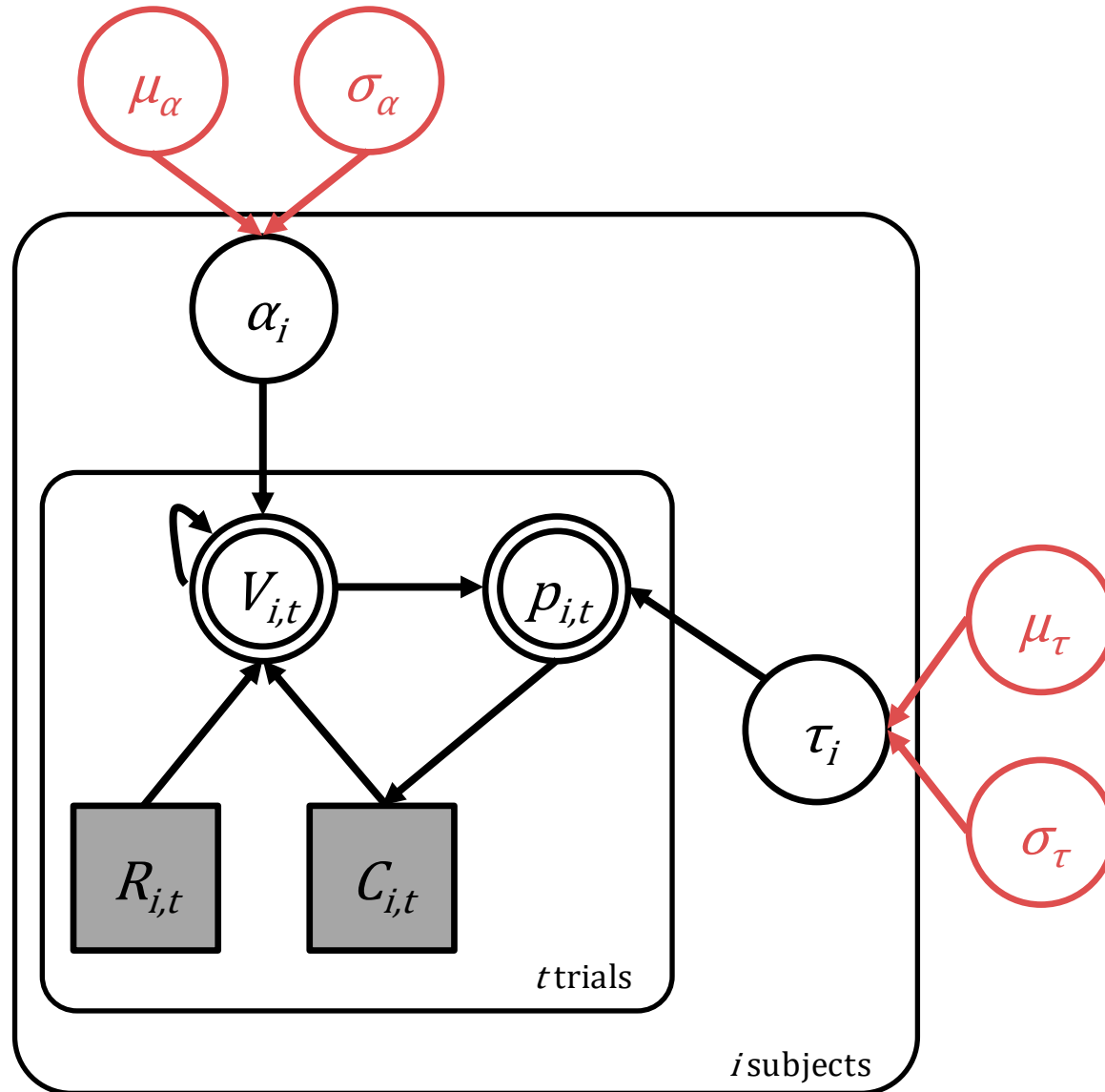
Fitting Multiple Participants



Fitting Multiple Participants with hierarchical Bayesian analysis (HBA)



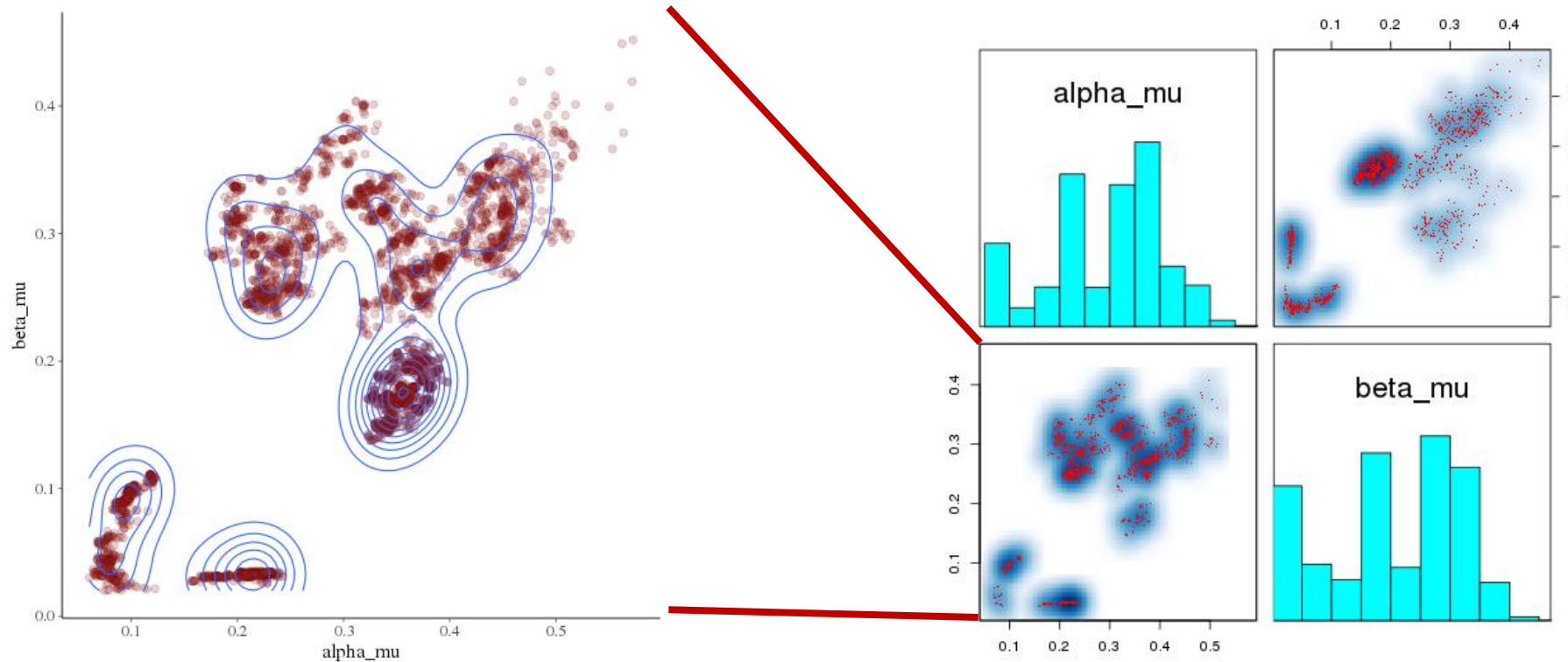
Hierarchical RL Model



Satisfied with the results?

Warning messages:

1: There were 3998 divergent transitions after warmup. Increasing adapt_delta above 0.8 may help. See <http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup>
2: Examine the pairs() plot to diagnose sampling problems



Non-centered Reparameterization*

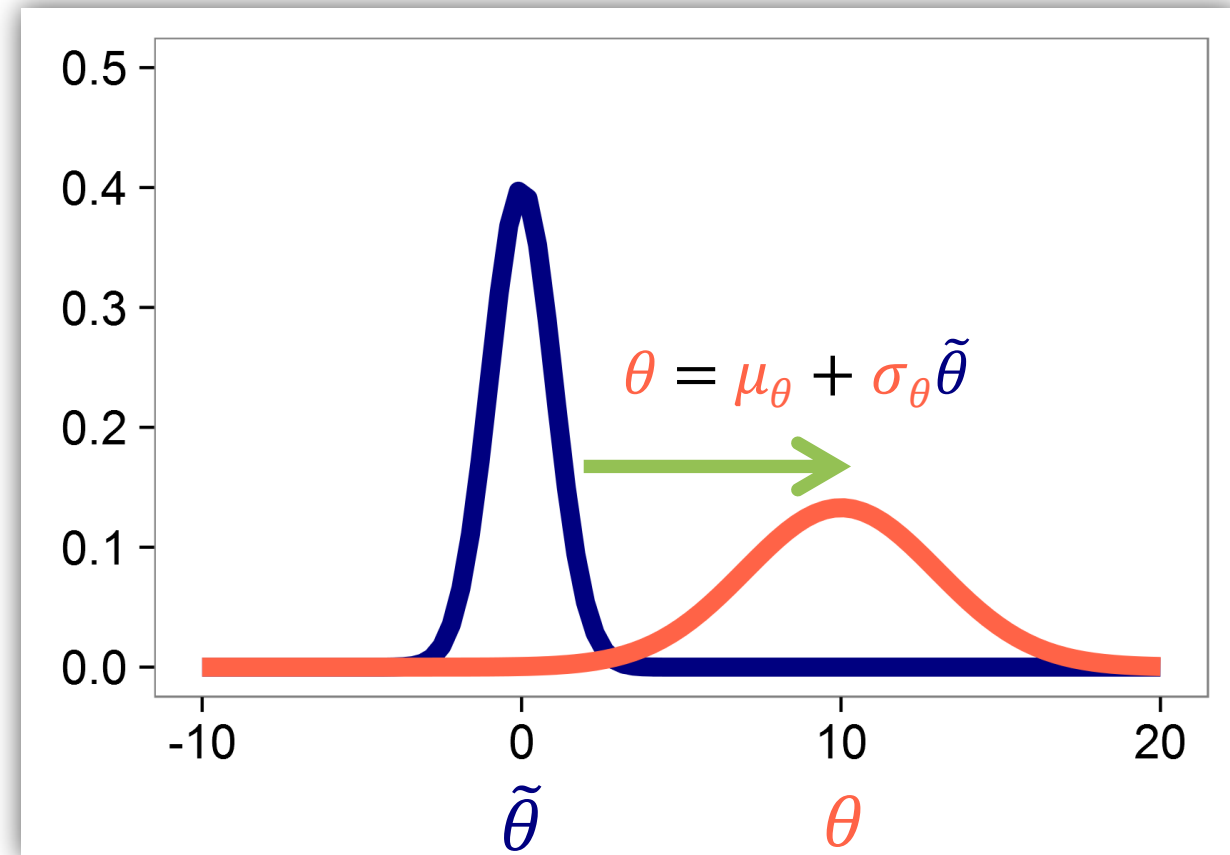
$$\theta \sim \text{Normal}(\mu_\theta, \sigma_\theta)$$



$$\tilde{\theta} \sim \text{Normal}(0, 1)$$

$$\theta = \mu_\theta + \sigma_\theta \tilde{\theta}$$

Stan likes **simple**
distributions!

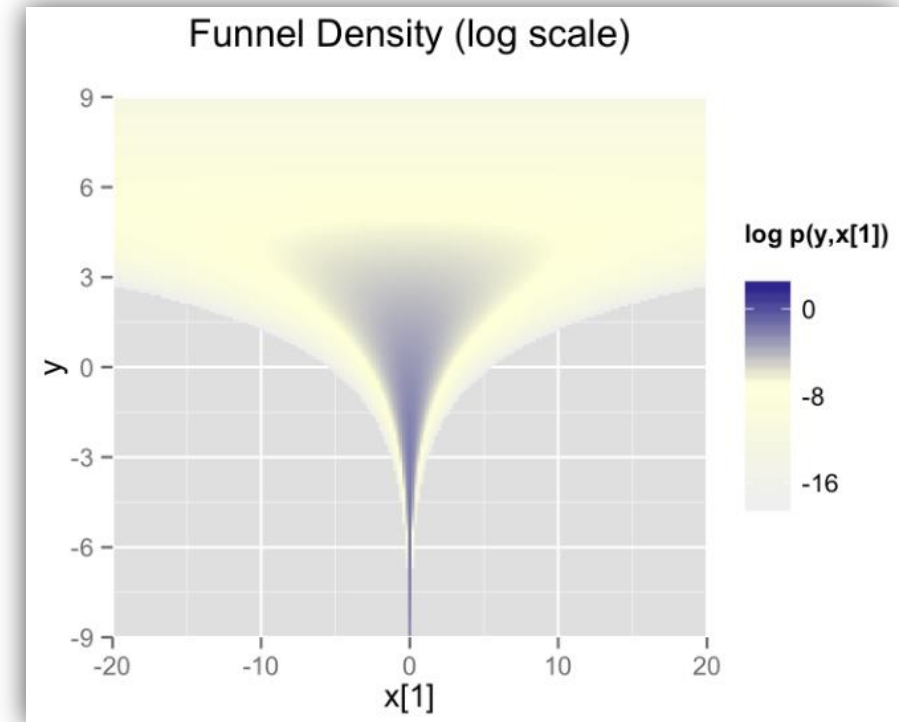


Reparameterization

Neal's Funnel

$$p(y, x) = \text{Normal}(y|0, 3) \times \prod_{n=1}^9 \text{Normal}(x_n|0, \exp(y/2))$$

```
parameters {  
  real y;  
  vector[9] x;  
}  
model {  
  y ~ normal(0, 3);  
  x ~ normal(0, exp(y/2));  
}
```



Reparameterization

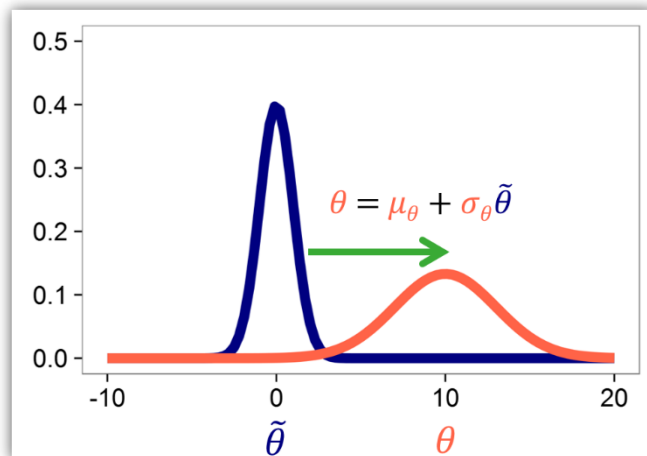
Neal's Funnel

$$p(y, x) = \text{Normal}(y|0, 3) \times \prod_{n=1}^9 \text{Normal}(x_n|0, \exp(y/2))$$

```
parameters {  
  real y;  
  vector[9] x;  
}  
model {  
  y ~ normal(0, 3);  
  x ~ normal(0, exp(y/2));  
}
```



```
parameters {  
  real y_raw;  
  vector[9] x_raw;  
}  
transformed parameters {  
  real y;  
  vector[9] x;  
  
  y = 3.0 * y_raw;  
  x = exp(y/2) * x_raw;  
}  
model {  
  y_raw ~ normal(0, 1);  
  x_raw ~ normal(0, 1);  
}
```



Stan Sampling Parameters

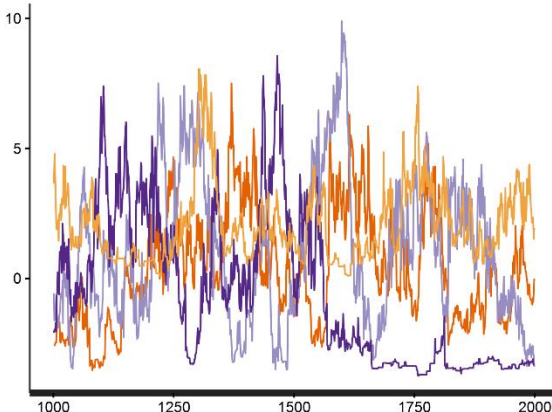
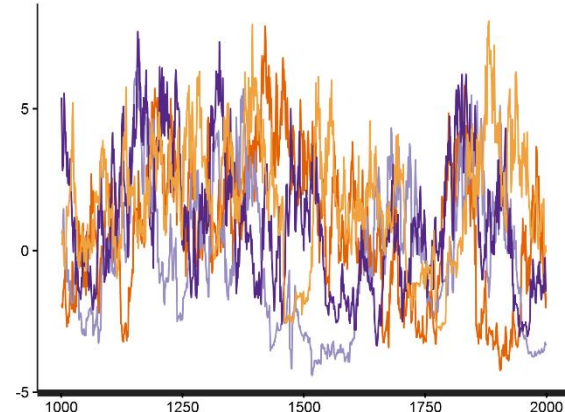
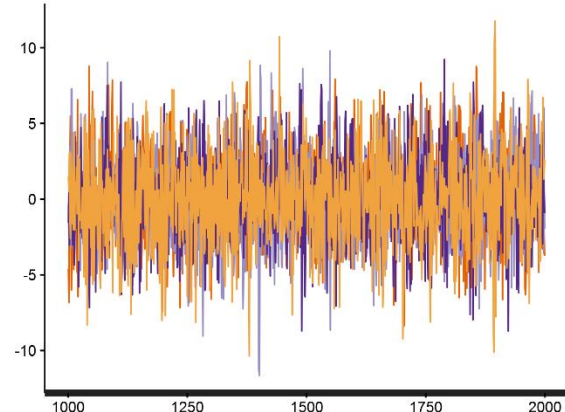
parameter	description	constraint	default
iterations	number of MCMC samples (per chain)	int, > 0	2000
delta: δ	target Metropolis acceptance rate	$\delta \in [0, 1]$	0.80
stepsize: ε	initial HMC step size	real, $\varepsilon > 0$	2.0
max_treedepth: L	maximum HMC steps per iteration	int, $L > 0$	10

Typical adjustments

- Increase iterations
- Increase delta
- Decrease stepsize
- Might have to increase max_treedepth

```
funnel_fit2 <- stan("_scripts/funnel.stan",  
  iter = 4000,  
  control = list(adapt_delta = 0.999,  
    stepsize = 1.0,  
    max_treedepth = 20))
```

Neal's Funnel: Comparing Performance

	direct model	adjusted direct model	reparameterized model
Rhat (y)	1.22	1.1	1.0
n_eff (y)	18	42	3886
runtime*	48.50 sec	50.76 sec	50.12 sec
n_eff (y) / runtime	0.37 / sec	0.82 / sec	77.53 / sec
n_divergent	53	0	0
traceplot (y)			

*: 2 cores in parallel, including compiling time

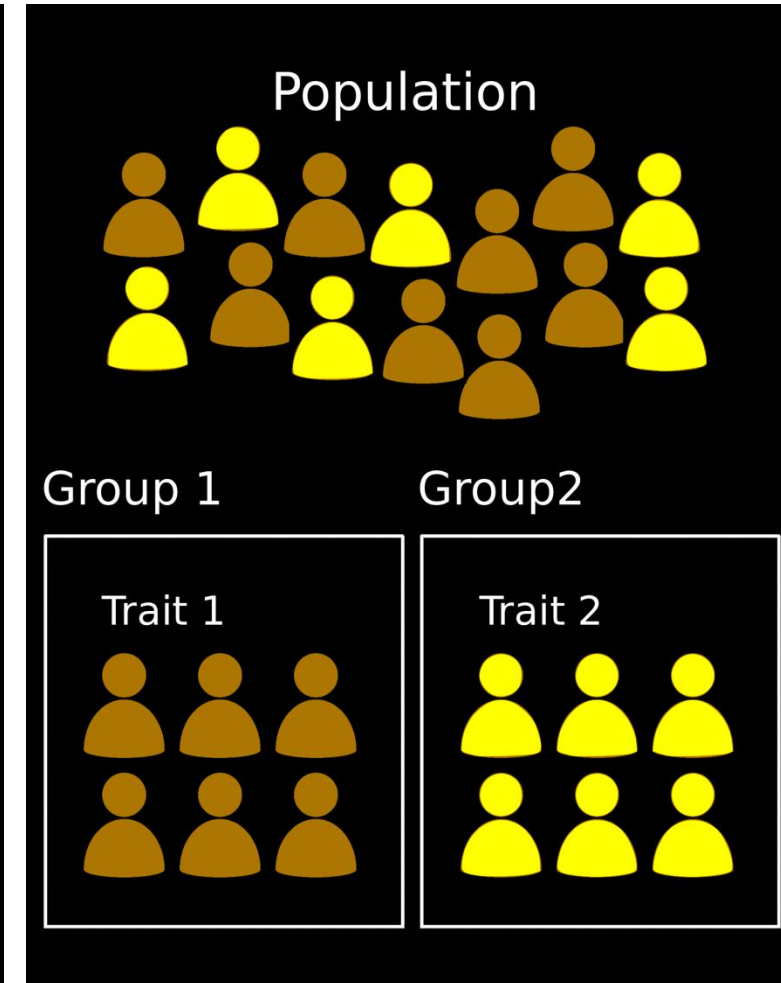
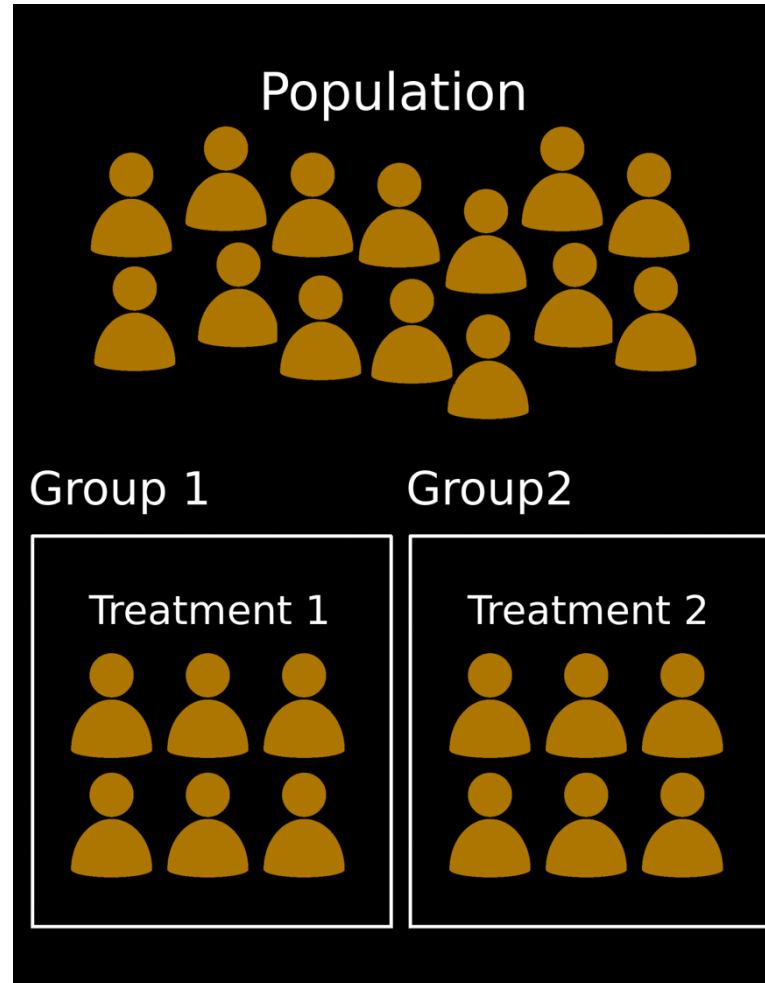
How about **Bounded** Parameters?

$$\begin{array}{ccc}
 \tilde{\theta} \sim Normal(0, 1) & \xrightarrow{\text{Sigmoid}} & \tilde{\theta} \sim Normal(0, 1) \\
 \theta = \mu_{\theta} + \sigma_{\theta} \tilde{\theta} & & \theta = Probit^{-1}(\mu_{\theta} + \sigma_{\theta} \tilde{\theta}) \\
 \theta \in (-\infty, +\infty) & & \theta \in [0, 1]
 \end{array}$$

constraint	reparameterization
$\theta \in (-\infty, +\infty)$	$\theta = \mu_{\theta} + \sigma_{\theta} \tilde{\theta}$
$\theta \in [0, N]$	$\theta = Probit^{-1}(\mu_{\theta} + \sigma_{\theta} \tilde{\theta}) \times N$
$\theta \in [M, N]$	$\theta = Probit^{-1}(\mu_{\theta} + \sigma_{\theta} \tilde{\theta}) \times (N-M) + M$
$\theta \in (0, +\infty)$	$\theta = \exp(\mu_{\theta} + \sigma_{\theta} \tilde{\theta}); \theta = \log(1 + \exp(\mu_{\theta} + \sigma_{\theta} \tilde{\theta}))$

Between-subject design

- each participant is tested in only one condition
- e.g., mood → IQ performance in the previous example



$$\theta_{\text{patient},i} \sim \text{Normal}(\mu_{\theta,\text{patient}}, \sigma_{\theta,\text{patient}})$$

$$\theta_{\text{control},i} \sim \text{Normal}(\mu_{\theta,\text{control}}, \sigma_{\theta,\text{control}}),$$

$$\mu_{\theta,\text{patient}} \sim \text{Normal}(\mu_{\theta}, \sigma_{\theta})$$

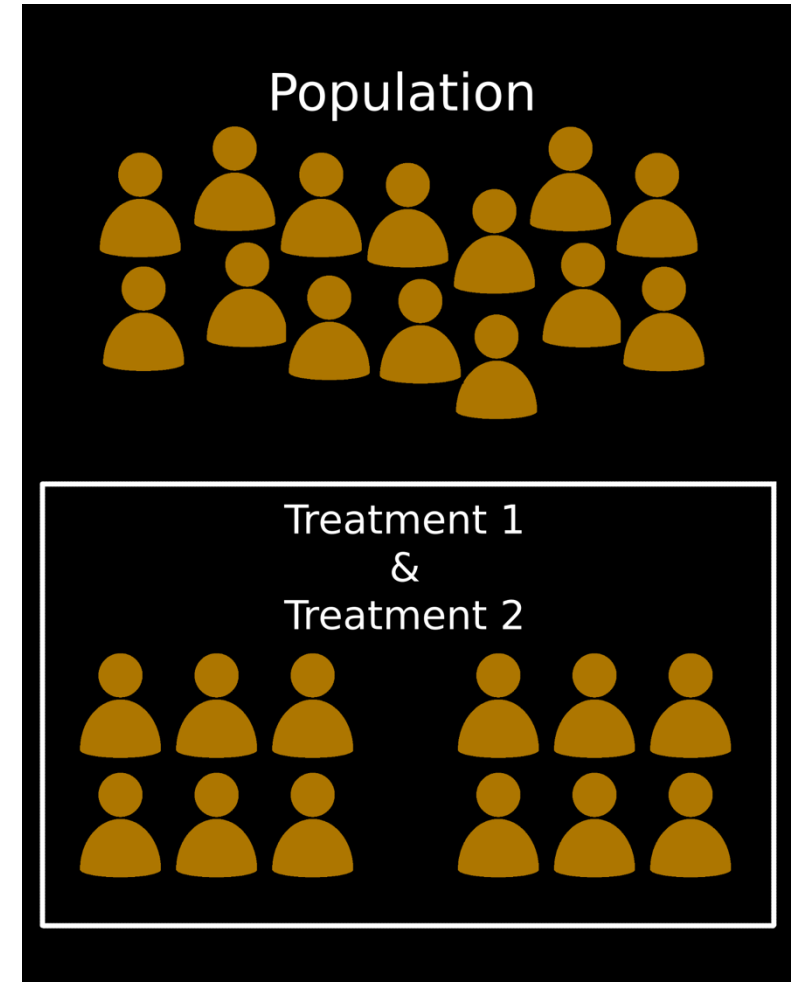
$$\mu_{\theta,\text{control}} \sim \text{Normal}(\mu_{\theta}, \sigma_{\theta})$$

$$\theta_{\text{patient},i} \sim \text{Normal}(\mu_{\theta,\text{patient}}, \sigma_{\theta,\text{patient}})$$

$$\theta_{\text{control},i} \sim \text{Normal}(\mu_{\theta,\text{control}}, \sigma_{\theta,\text{control}})$$

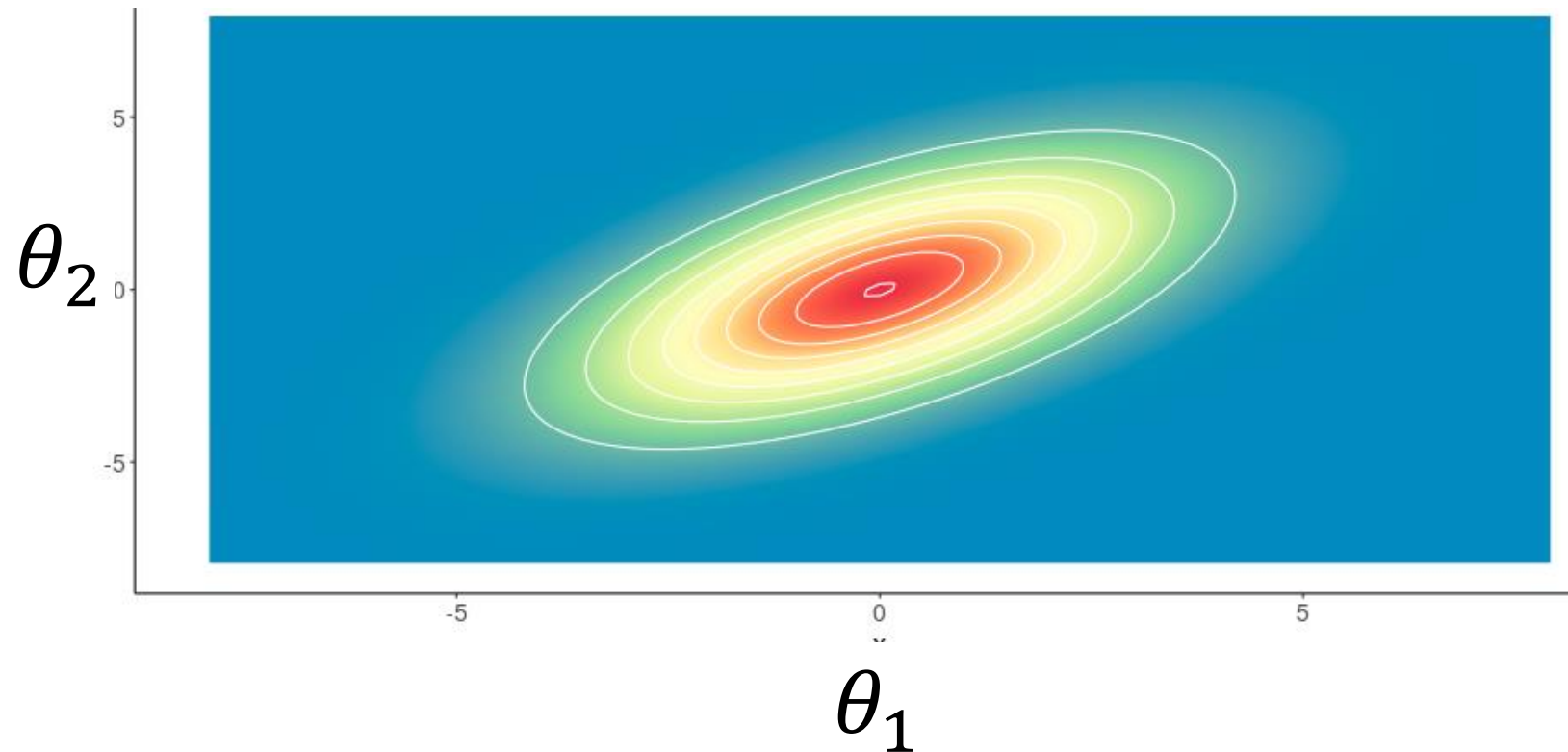
Within-subject design

- each participant is tested under all conditions
- e.g., learning performance at the beginning vs. at the end of the semester



Effect coding

$$\theta_2 = \theta_1 + \Delta\theta$$



Effect coding

	θ_1	$\theta_2 - \theta_1$
θ_1	1	0
θ_2	1	1

Effect coding

	θ_1	$\theta_2 - \theta_1$	$\theta_3 - \theta_1$
θ_1	1	0	0
θ_2	1	1	0
θ_3	1	0	1

Effect coding

$$\theta_i \sim \text{Normal}(\mu, \Sigma)$$

$$\mu = (\mu_1, \mu_2, \mu_3) = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} (\mu_1, \mu_2 - \mu_1, \mu_3 - \mu_1)^T$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 \end{pmatrix} = \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix} \Omega \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix}$$

Multiple IV: Factorial design

		FACTOR A (TV VIOLENCE)	
		Level 1 (A1) (Violent)	Level 2 (A2) (Nonviolent)
FACTOR B (REALISM)	Level 1 (B1) (Real)	A1B1 (Real/Violent)	A2B1 (Real/Nonviolent)
	Level 2 (B2) (Cartoon)	A1B2 (Cartoon/Violent)	A2B2 (Cartoon/Nonviolent)

Experimental Procedures

1. **Screening:** Online Questionnaire

2. **Baseline session** 
Reward tasks

3. **Iso sessions** (counterbalanced)

Iso total



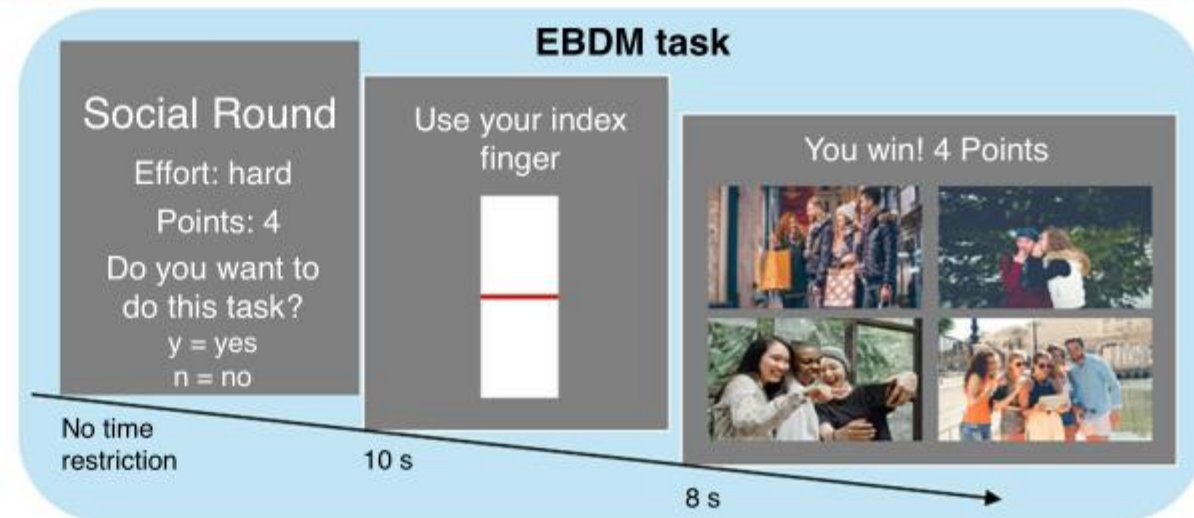
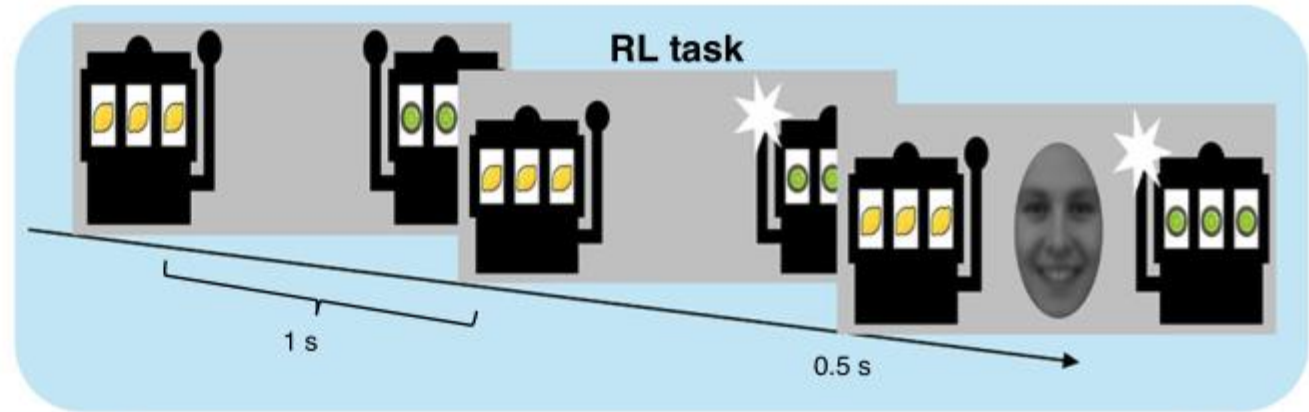
Reward tasks

Iso w media



Reward tasks

Reward tasks: EBDM task, RL task (counterbalanced)



	baseline	iso w media	iso total	baseline	iso w media	iso total
non-social	1	0	0	0	0	0
	1	1	0	0	0	0
	1	0	1	0	0	0
social	1	0	0	1	0	0
	1	0	0	1	1	0
	1	0	0	1	0	1

Information Criteria

AIC – Akaike information criterion

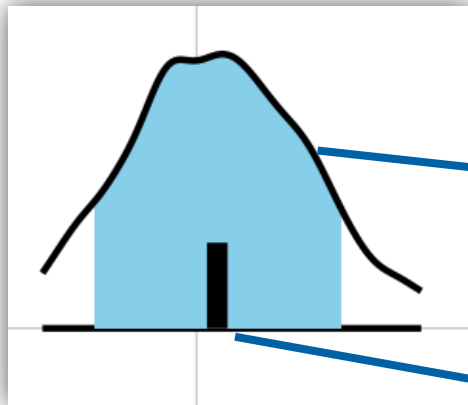
DIC – Deviance Information Criterion

WAIC – Widely Applicable Information Criterion

(Watanabe–Akaike information criterion)

finding the model that has
the highest out-of-sample
predictive accuracy

approximation to LOO



WAIC: using **entire posterior** distribution

AIC/DIC: using **point** estimation

Compute WAIC from Likelihood

$$\text{WAIC} = -2 \widehat{\text{elpd}}_{\text{waic}}$$

expected log pointwise predictive density

$$\widehat{\text{elpd}}_{\text{waic}} = \widehat{\text{lpd}} - \hat{p}_{\text{waic}}$$

$$\begin{aligned} \widehat{\text{lpd}} &= \text{computed log pointwise predictive density} \\ &= \sum_{i=1}^n \log \left(\frac{1}{S} \sum_{s=1}^S p(y_i | \theta^s) \right). \end{aligned}$$

```
lpd <- log(colMeans(exp(log_lik)))
```

$$\hat{p}_{\text{waic}} = \sum_{i=1}^n V_{s=1}^S (\log p(y_i | \theta^s))$$

estimated effective number of parameters

```
p_waic <- colVars(log_lik)
```

The {loo} Package

```
> library(loo)
> LL1      <- extract_log_lik(stanfit)
> loo1     <- loo(LL1)      # PSIS leave-one-out
> waic1    <- waic(LL1)    # WAIC
```

Computed from 4000 by 20 log-likelihood matrix

	Estimate	SE
elpd_loo	-29.5	3.3
p_loo	2.7	1.0
looic	58.9	6.7

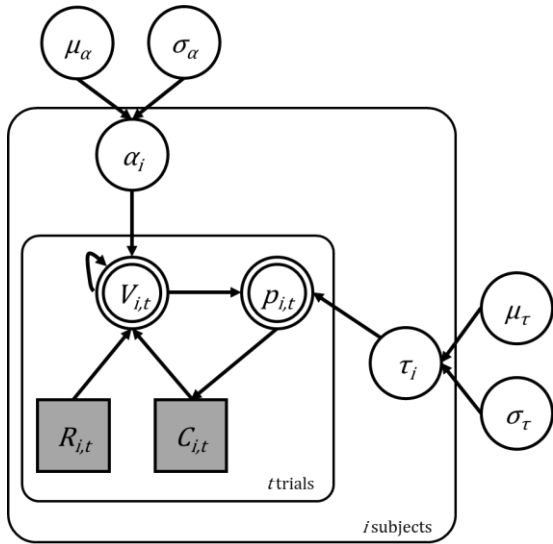
Pareto Smoothed Importance Sampling



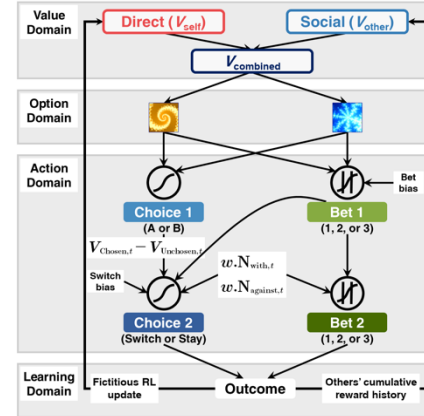
Recording the Log-Likelihood in Stan

```
generated quantities {  
  ...  
  real log_lik[nSubjects];  
  ...  
  
  { # local section, this saves time and space  
    for (s in 1:nSubjects) {  
      vector[2] v;  
      real pe;  
  
      log_lik[s] = 0;  
      v = initV;  
  
      for (t in 1:nTrials) {  
        log_lik[s] = log_lik[s] + categorical_logit_lpmf(choice[s,t] | tau[s] * v);  
  
        pe = reward[s,t] - v[choice[s,t]];  
        v[choice[s,t]] = v[choice[s,t]] + lr[s] * pe;  
      }  
    }  
  }  
}
```

HBA sounds good, but...



$$\begin{aligned}\mu_\alpha &\sim \text{Uniform}(0,1) \\ \sigma_\alpha &\sim \text{halfCauchy}(0,1) \\ \mu_\tau &\sim \text{Uniform}(0,3) \\ \sigma_\tau &\sim \text{halfCauchy}(0,3) \\ \alpha_i &\sim \text{Normal}(\mu_\alpha, \sigma_\alpha) \tau(0,1) \\ \tau_i &\sim \text{Normal}(\mu_\tau, \sigma_\tau) \tau(0,3) \\ p_{i,t}(C=A) &= \frac{1}{1 + e^{\tau_i(V_{i,t}(B) - V_{i,t}(A))}} \\ V_{i,t+1}^C &= V_{i,t}^C + \alpha_i(R_{i,t} - V_{i,t}^C)\end{aligned}$$



$$\begin{aligned}V_{\text{self},t} &= [V_{\text{self},t}(A), V_{\text{self},t}(B)] \\ V_{\text{other},t} &= [V_{\text{other},t}(A), V_{\text{other},t}(B)] \\ V_t &= \beta_{\text{self}} V_{\text{self},t} + \beta_{\text{other}} V_{\text{other},t} \\ C1_t &\sim \text{Categorical}(\text{Softmax}(V_t)) \\ U_{\text{bet},t} &= \beta_{\text{bias}} + \beta_{\text{diff}} (V_{\text{chosen},C1,t} - V_{\text{unchosen},C1,t}) \\ B1_t &\sim \text{OrderedLogistic}(U_{\text{bet},t} | \theta) \\ w.N_{\text{against},t} &= \frac{\sum_{s=1}^K w_{s,t}}{\sum_{s=1}^{s=1} w_{s,t}}, K=0,1,...,4 \\ w.N_{\text{with},t} &= \frac{\sum_{s=1}^{s=1} w_{s,t}}{\sum_{s=1}^{s=1} w_{s,t}} \\ V_t(\text{switch}) &= \beta_{\text{bias}} + \beta_{\text{diff}} (V_{\text{chosen},C1,t} - V_{\text{unchosen},C1,t}) + \beta_{\text{against}} w.N_{\text{against},t} \\ C2 &\sim \text{Bernoulli}(V_t(\text{switch})) \\ U_{\text{bet},t} &= \begin{cases} U_{\text{bet},t} + \beta_{\text{with}} w.N_{\text{with},t} + \beta_{\text{against}} w.N_{\text{against},t}, & \text{if } C1 = C2 \\ U_{\text{bet},t} + \beta_{\text{with}} w.N_{\text{with},t} + \beta_{\text{against}} w.N_{\text{against},t}, & \text{if } C1 \neq C2 \end{cases} \\ B2_t &\sim \text{OrderedLogistic}(U_{\text{bet},t} | \theta) \\ \Phi(x) &= \frac{1}{1 + e^{-x}} \\ \delta_{\text{self},\text{chosen},C2,t} &= R_{\text{self},t} - V_{\text{self},\text{chosen},C2,t} \\ \delta_{\text{self},\text{unchosen},C2,t} &= -R_{\text{self},t} - V_{\text{self},\text{unchosen},C2,t} \\ V_{\text{self},\text{chosen},C2,t+1} &= V_{\text{self},\text{chosen},C2,t} + \alpha \delta_{\text{self},\text{chosen},C2,t} \\ V_{\text{self},\text{unchosen},C2,t+1} &= V_{\text{self},\text{unchosen},C2,t} + \alpha \delta_{\text{self},\text{unchosen},C2,t}\end{aligned}$$

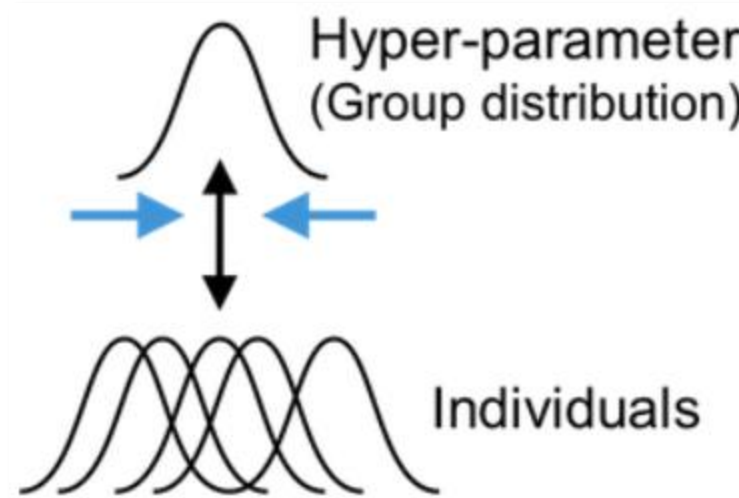
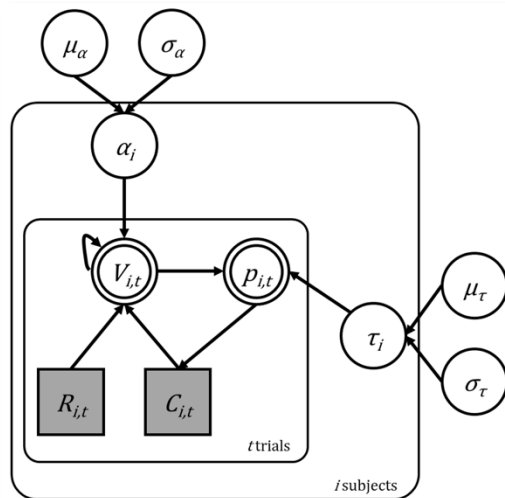
hBayesDM package

hBayesDM

repo status **Active** build **passing** CRAN **1.0.2 – 2019-11-13** downloads **33K**

DOI [10.1162/CPSY_a_00002](https://doi.org/10.1162/CPSY_a_00002)

hBayesDM (hierarchical Bayesian modeling of Decision-Making tasks) is a user-friendly package that offers hierarchical Bayesian analysis of various computational models on an array of decision-making tasks. hBayesDM uses [Stan](#) for Bayesian inference.



Bayesian Statistics and Hierarchical Bayesian Modeling for Psychological Science

Lecture 01

Lei Zhang

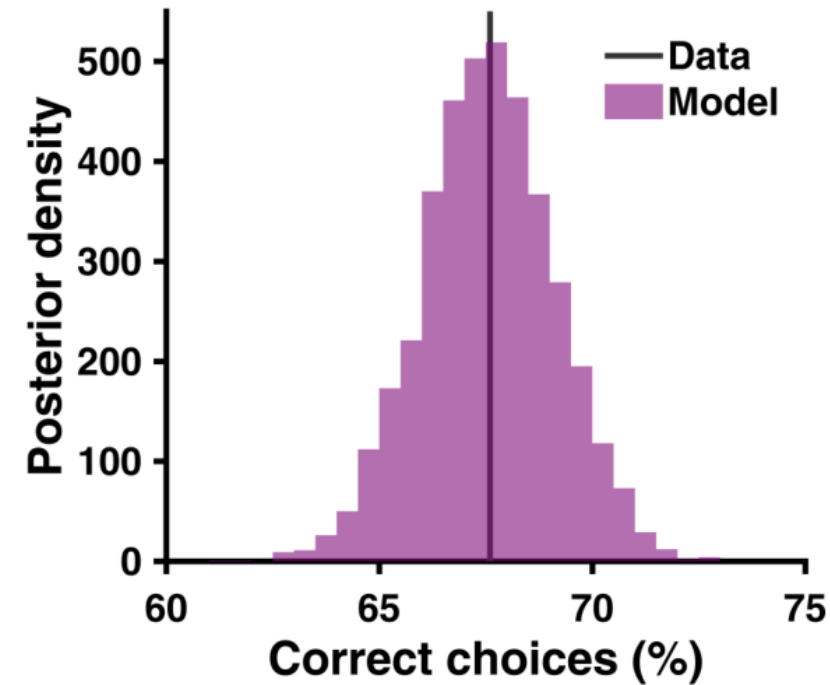
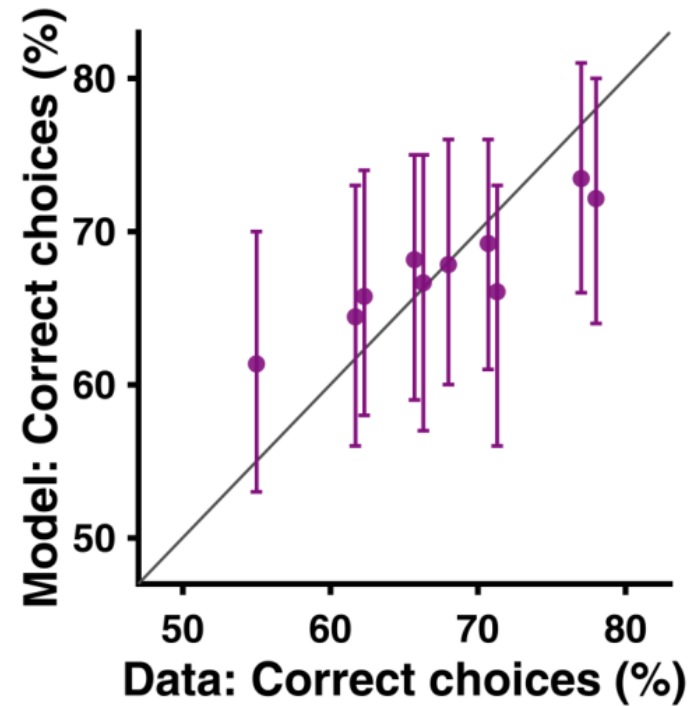
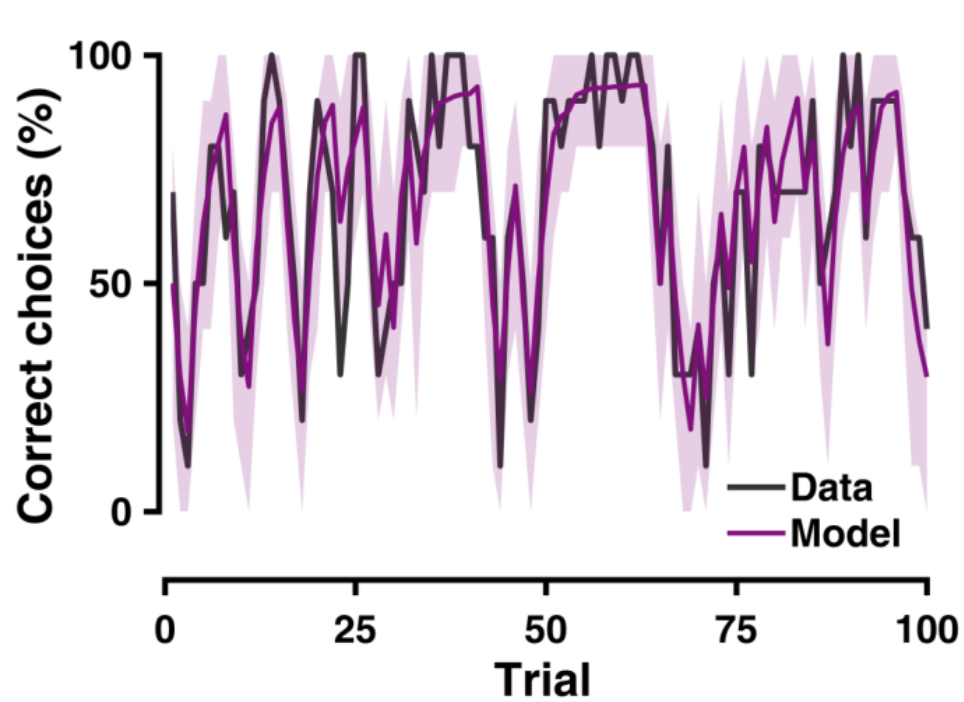
Social, Cognitive and Affective Neuroscience Unit (SCAN-Unit)
Department of Cognition, Emotion, and Methods in Psychology

lei.zhang@univie.ac.at
https://github.com/lei-zhang/BayesCog_Wien

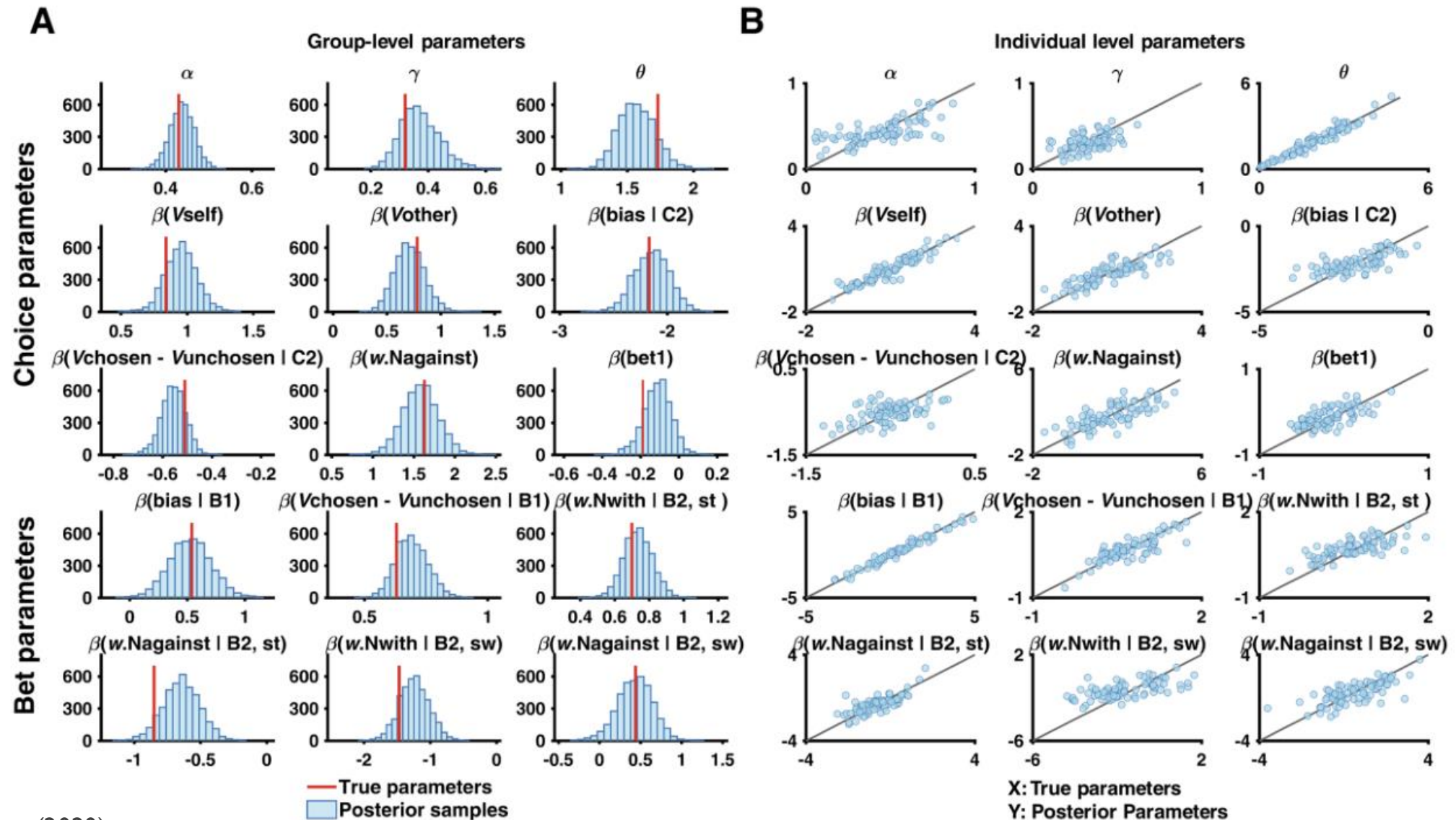
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Posterior predictive check: *can the model resonate with data?*



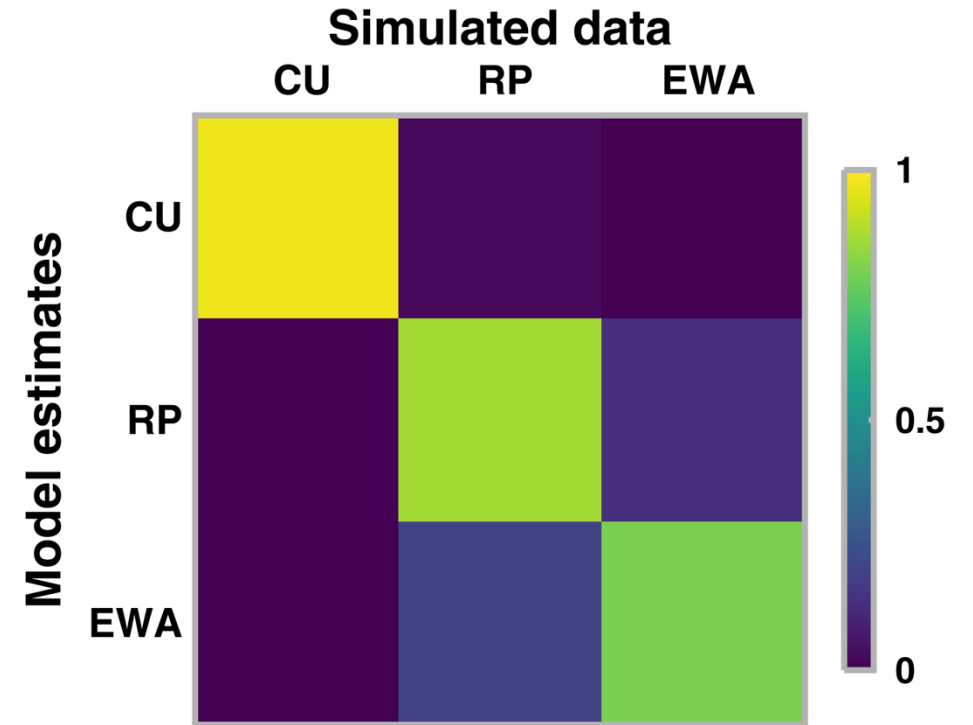
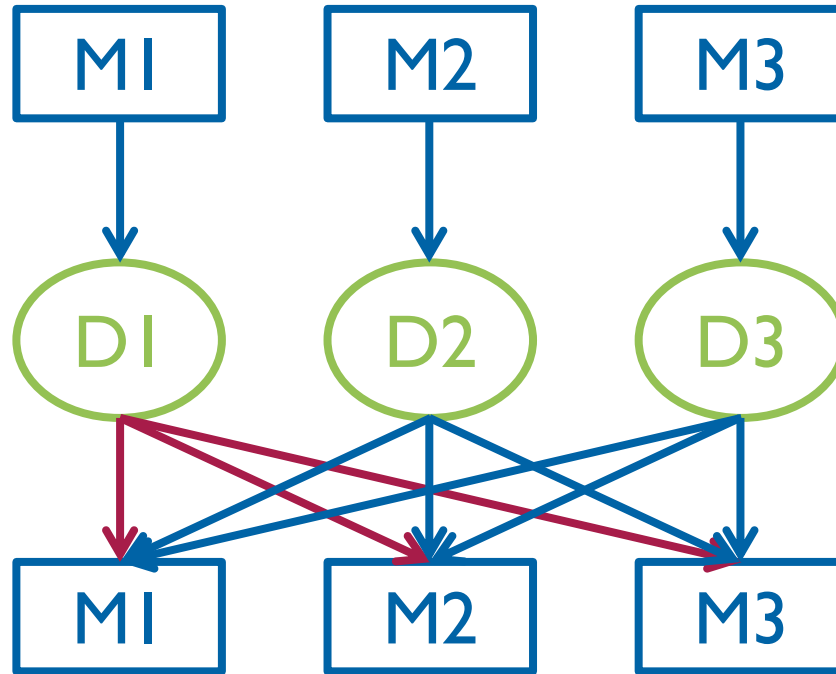
Parameter recovery: *are parameters identifiable?*



Model recovery: *are models identifiable?*

generative
process

fitting
process



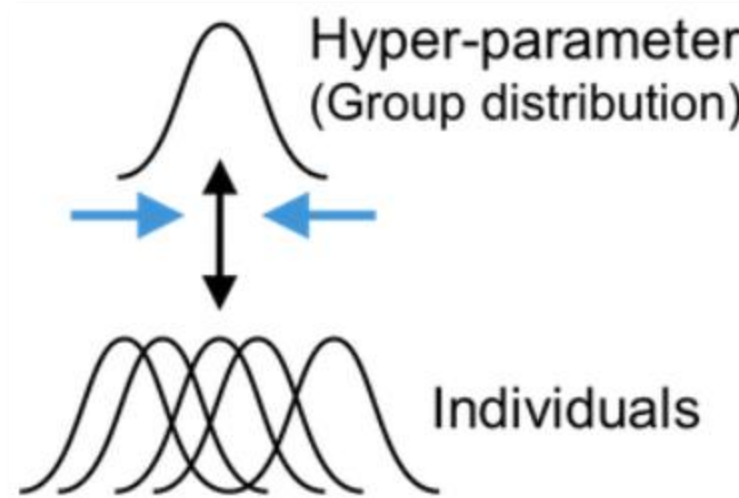
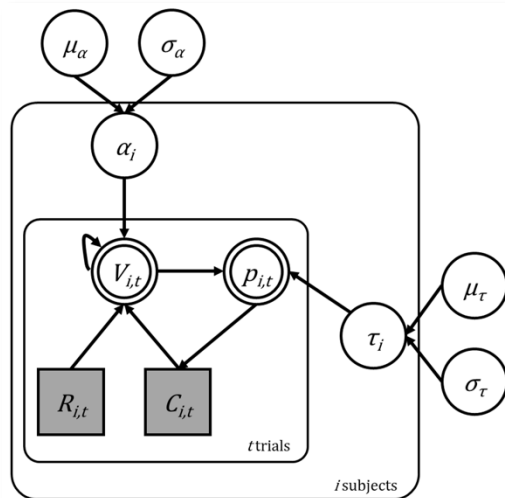
hBayesDM package

hBayesDM

repo status **Active** build **passing** CRAN **1.0.2 – 2019-11-13** downloads **33K**

DOI [10.1162/CPSY_a_00002](https://doi.org/10.1162/CPSY_a_00002)

hBayesDM (hierarchical Bayesian modeling of Decision-Making tasks) is a user-friendly package that offers hierarchical Bayesian analysis of various computational models on an array of decision-making tasks. hBayesDM uses [Stan](#) for Bayesian inference.



Bayesian Statistics and Hierarchical Bayesian Modeling for Psychological Science

Lecture 01

Lei Zhang

Social, Cognitive and Affective Neuroscience Unit (SCAN-Unit)
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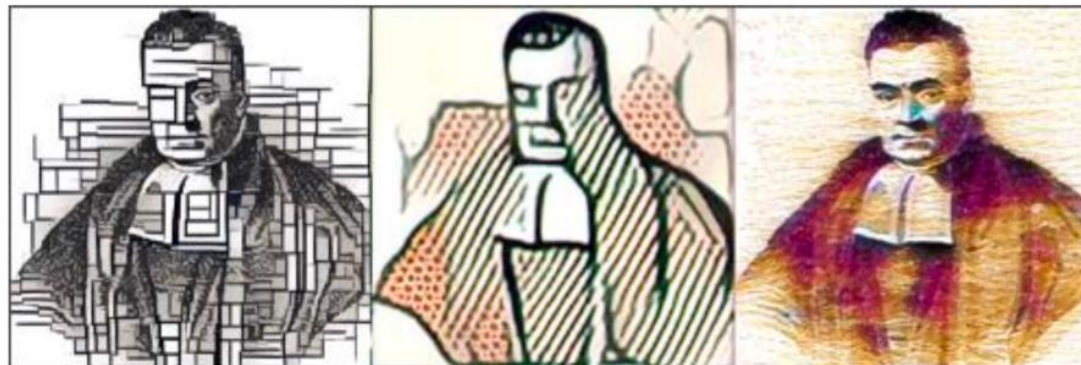
Adaptive Learning Psychology and Neuroscience





Welcome to BayesCog!

- Course overview
- Workshop 1: R Basics
- Workshop 2: Probability and an introduction to Bayes' theorem
- Workshop 3: Building simple models conceptually
- Workshop 4: Introduction to building models in Stan
- Workshop 5: Bernoulli and linear regression models
- Workshop 6: Reinforcement learning models
- Workshop 7: Hierarchical Bayesian modeling
- Workshop 8: Model comparison
- Workshop 9: Debugging in Stan
- Bonus workshop: Introduction to model-based fMRI
- Acknowledgements
- Resources



Visitors 1 / 119 contributors 2 last commit july repo size 360.2 MiB

Welcome

This website is an adapted version of teaching materials originally made for the [award winning*](#) **BayesCog** seminar at the [Faculty of Psychology, University of Vienna](#), as part of the Advanced Seminar for master's students (Mind and Brain track; recorded during Summer Term 2020/2021). Further content from the [BayesCog workshop at UKE Hamburg](#) (2023) have also been added¹.

Recording: Recordings from the original version of the course are available on [YouTube](#) (also see below). The most recent recording from the 2021 summer semester is also available on [YouTube](#).

Outreach: [Twitter thread](#) (being **liked 700+ times** on Twitter) summarizing the contents of the course.

Award/Recognition: The original course received a commendation award from the [Society for the Improvement of Psychological Science \(SIPS\)](#) (also see a [tweet](#)), as well as an ECR Teaching Award from the [Faculty of Psychology, University of Vienna](#).



1 BayesCog: A freely available course in Bayesian statistics and hierarchical Bayesian modeling for psychological science

4 Lei Zhang ^{1,2,3} and Aamir Sohail ^{1,2}

5 1 Centre for Human Brain Health, School of Psychology, University of Birmingham, Birmingham, UK 2
 6 Institute for Mental Health, School of Psychology, University of Birmingham, Birmingham, UK 3 Centre for
 7 Developmental Science, School of Psychology, University of Birmingham, Birmingham, UK

<https://alpn-lab.github.io/BayesCog/>

Recommended reading: tutorial

Using reinforcement learning models in social neuroscience: frameworks, pitfalls and suggestions of best practices

Lei Zhang , Lukas Lengersdorff, Nace Mikus, Jan Gläscher, Claus Lamm  Author Notes

Social Cognitive and Affective Neuroscience, Volume 15, Issue 6, June 2020, Pages 695–707, <https://doi.org/10.1093/scan/nsaa089>

<https://doi.org/10.1093/scan/nsaa089>

ACCEPTED MANUSCRIPT

Computational modelling of social cognition and behaviour—a reinforcement learning primer

Patricia L Lockwood , Miriam Klein-Flügge 

Social Cognitive and Affective Neuroscience, nsaa040, <https://doi.org/10.1093/scan/nsaa040>

Published: 30 March 2020 [Article history](#) 

<https://doi.org/10.1093/scan/nsaa040>

Ten simple rules for the computational modeling of behavioral data



Robert C Wilson , Anne GE Collins 

University of Arizona, United States; University of California, Berkeley, United States

<https://elifesciences.org/articles/49547>

The Importance of Falsification in Computational Cognitive Modeling

Stefano Palminteri,^{1,2,*,‡} Valentin Wyart,^{1,2,*,‡} and Etienne Koechlin^{1,2,*}

<https://doi.org/10.1016/j.tics.2017.03.011>

Recommended reading: empirical work

Catecholaminergic challenge uncovers distinct Pavlovian and instrumental mechanisms of motivated (in)action



Jennifer C Swart^a, Monja I Froböse, Jennifer L Cook, Dirk EM Geurts, Michael J Frank, Roshan Cools, Hanneke EM den Ouden^a
Radboud University, The Netherlands; University of Birmingham, United Kingdom; Radboud University Medical Center, The Netherlands; Linguistic and Psychological Sciences, Brown University, United States; Brown University, United States

<https://elifesciences.org/articles/22169>

COGNITIVE NEUROSCIENCE

A brain network supporting social influences in human decision-making

Lei Zhang^{1,2*} and Jan Gläscher^{1*†}

<https://advances.sciencemag.org/content/6/34/eabb4159>

Social threat learning transfers to decision making in humans

Björn Lindström^{a,b,c,1}, Armita Golkar^{c,d}, Simon Jangard^c, Philippe N. Tobler^b, and Andreas Olsson^c

^aDepartment of Social Psychology, University of Amsterdam, 1018 WT Amsterdam, The Netherlands; ^bLaboratory for Social and Neural Systems Research, Department of Economics, University of Zürich, 8001 Zürich, Switzerland; ^cSection for Psychology, Department of Clinical Neuroscience, Karolinska Institutet, 171 77 Stockholm, Sweden; and ^dDepartment of Clinical Psychology, University of Amsterdam, 1018 WT Amsterdam, The Netherlands

<https://www.pnas.org/content/116/10/4732.abstract>

Modeling flexible behavior in childhood to adulthood shows age-dependent learning mechanisms and less optimal learning in autism in each age group

Daisy Crawley^{1‡}, Lei Zhang^{2,3,4‡}, Emily J. H. Jones⁵, Jumana Ahmad^{1,6}, Bethany Oakley¹, Antonia San José Cáceres^{1,7}, Tony Charman^{8,9}, Jan K. Buitelaar^{10,11,12}, Declan G. M. Murphy^{1,9,13}, Christopher Chatham⁴, Hanneke den Ouden^{10‡}, Eva Loth^{1,13‡}, the EU-AIMS LEAP group[†]

<https://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.3000908>

Recommended reading: new direction

nature

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Article | [Open access](#) | Published: 02 July 2025

Discovering cognitive strategies with tiny recurrent neural networks

[Li Ji-An](#), [Marcus K. Benna](#) & [Marcelo G. Mattar](#) 

[Nature](#) (2025) | [Cite this article](#)

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A foundation model to predict and capture human cognition

[Marcel Binz](#) , [Elif Akata](#), [Matthias Bethge](#), [Franziska Brändle](#), [Fred Callaway](#), [Julian Coda-Forno](#), [Peter Dayan](#), [Can Demircan](#), [Maria K. Eckstein](#), [Noémi Éltető](#), [Thomas L. Griffiths](#), [Susanne Haridi](#), [Akshay K. Jagadish](#), [Li Ji-An](#), [Alexander Kipnis](#), [Sreejan Kumar](#), [Tobias Ludwig](#), [Marvin Mathony](#), [Marcelo Mattar](#), [Alireza Modirshanechi](#), [Surabhi S. Nath](#), [Joshua C. Peterson](#), [Milena Rmus](#), [Evan M. Russek](#), ... [Eric Schulz](#)  Show authors

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<https://www.nature.com/articles/s41586-025-09215-4>



Aamir
Sohail

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Using large language models to facilitate academic work in the psychological sciences

[Open access](#) | Published: 28 January 2025

Volume 44, pages 7910–7918, (2025) [Cite this article](#)

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[Aamir Sohail](#)  & [Lei Zhang](#) 

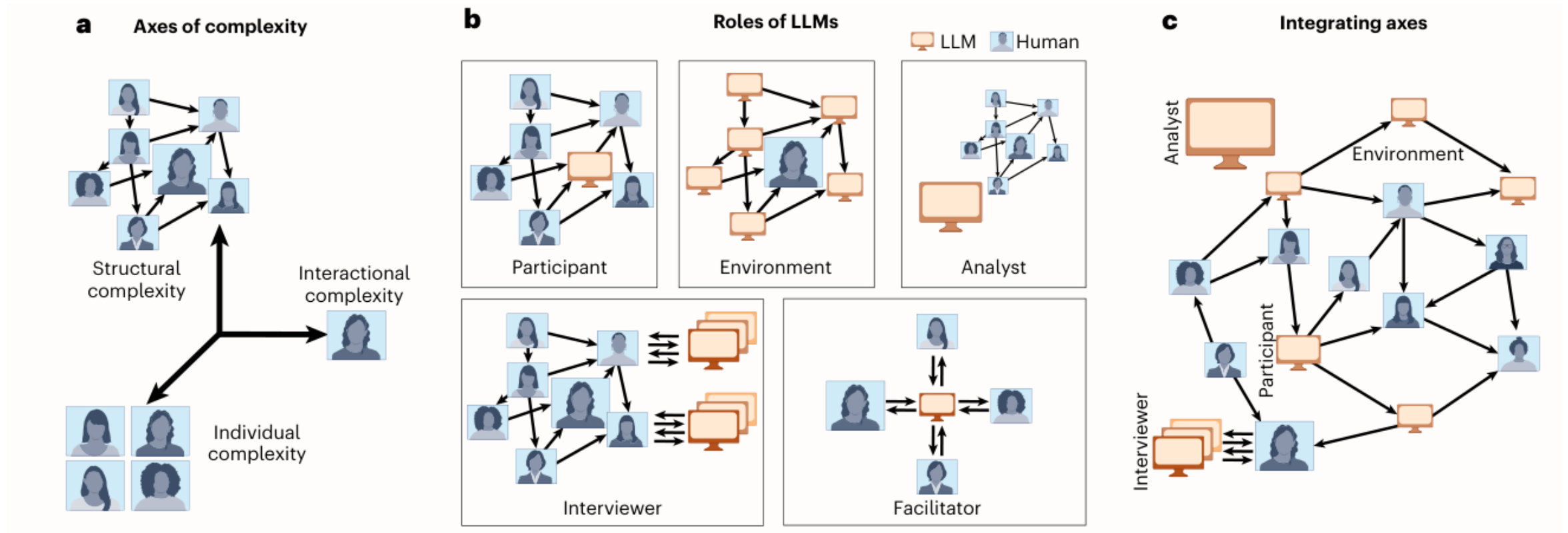
 **4412** Accesses  **10** Citations  **7** Altmetric [Explore all metrics](#) →

Simulating human participants with large language models

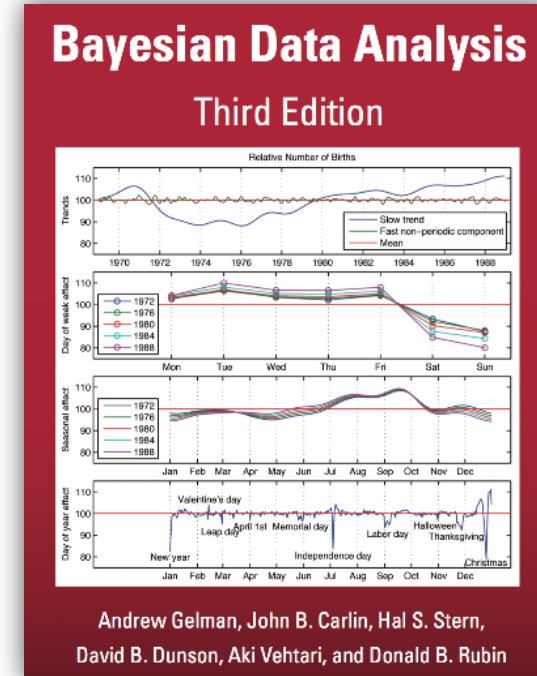
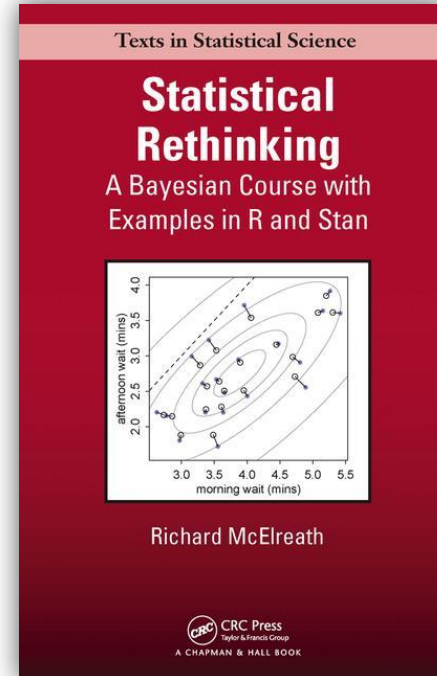
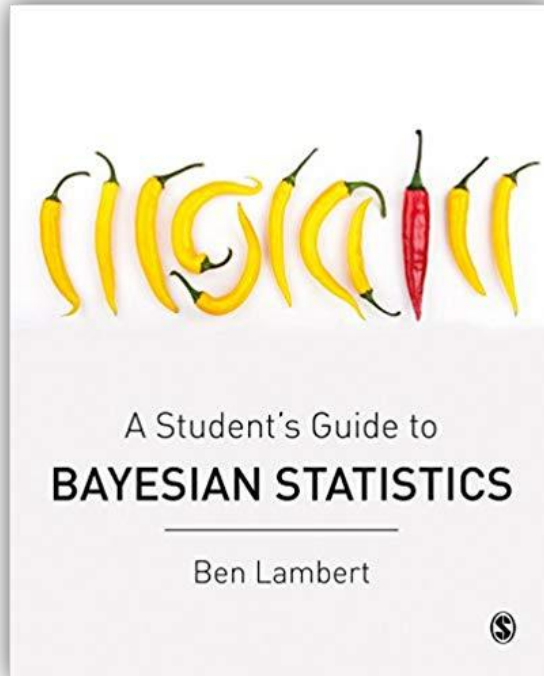
Multiple fields of research including psychology, sociology, economics, and neuroscience utilize experiments to assess behaviors as part of their research methodology repertoire. However, despite its importance and usefulness, this process has several challenges and potential limitations, including high financial costs and data quality concerns.

Using LLMs to advance the cognitive science of collectives

Ilia Sucholutsky, Katherine M. Collins, Nori Jacoby, Bill D. Thompson & Robert D. Hawkins



Recommended reading: book



Want to learn more about Stan?

Workshops

- [StanCon 2019 Hierarchical Models](#)
- [PyData NYC 2019](#)
- [StanCon 2018 Intro Stan](#)
- [StanCon 2018 Hierarchical Models](#)

<https://mc-stan.org/workshops/>

Stan forum

🚩 StanCon 2020. August 11-14 at Oregon State University

■ Announcements ■ stancon

StanCon 2020 will be at Oregon State University! There will be two days of tutorials followed by two days of talks, open discussions, and statistical modeling. Up-to-date details at <https://mc-stan.org/events/stancon2020...> [read more](#)



🔒 🚩 Welcome to the Stan Forums!

The Stan Forums provide a community for asking and answering questions about all aspects of Stan. Before creating a new topic please search the Forums to see if your questions has already been answered, or check out the... [read more](#)



Unable to retrieve parameters from a dynamic model

■ Modeling ■ fitting-issues ■ specification



<https://discourse.mc-stan.org/>

Twitter



Richard McElreath
@rhmcelreath



Quiche Lorraine,
@dan_p_simpson



\mathfrak{Michael}
@betanalpha



EJ Wagenmakers
@EJWagenmakers

Summary

- Computational modeling is never new → don't let it scare you!
- Learn some statistics (e.g., different statistical distributions)
- Learn some math (e.g., linear algebra)
- Learn some programming (e.g., R/Python/Julia/Matlab)
- Learn to seek external help (e.g., existing packages)
- Learn in pairs; practice makes perfect!



Richard McElreath
@rilmcelreath



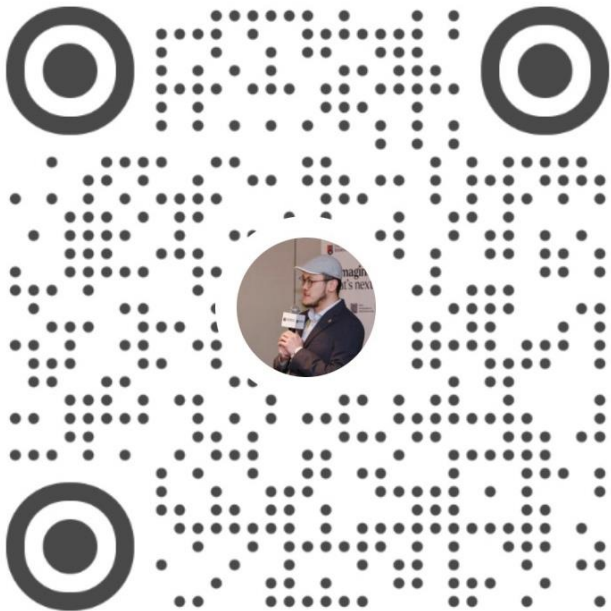
I say this a lot, bc I am also confused quite often.



Anna Jacobson @AnnaChingChing · Feb 21

"If you are confused, it is only because you are trying to understand." -
@rilmcelreath in Statistical Rethinking

Research opportunities (PhD) @ UoB



Acknowledgement

Funding:



Adaptive
Learning
Psychology and
Neuroscience

Thank you!



ANY
QUESTIONS
?

Happy Computing!